

UNDERSTANDING THE AGGRESSIVE BRAIN

High-Definition transcranial Direct Current Stimulation
(HD-tDCS) in reducing aggression and as a treatment
intervention in forensic patients.

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Colophon

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Understanding the Aggressive Brain

High-Definition transcranial Direct Current Stimulation (HD-tDCS) in reducing aggression and as a treatment intervention in forensic patients.

Het Agressieve Brein Ontrafeld

Het gebruik van High-Definition transcranial Direct Current Stimulation (HD-tDCS) in het reduceren van agressie en als behandel interventie voor forensische patiënten.

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“ALLES VAN WAARDE IS WEERLOOS”



Dit proefschrift is opgedragen aan jou Papa.

Dr. Marius Dijkhoorn

Winnaar proefschrift prijs 1986

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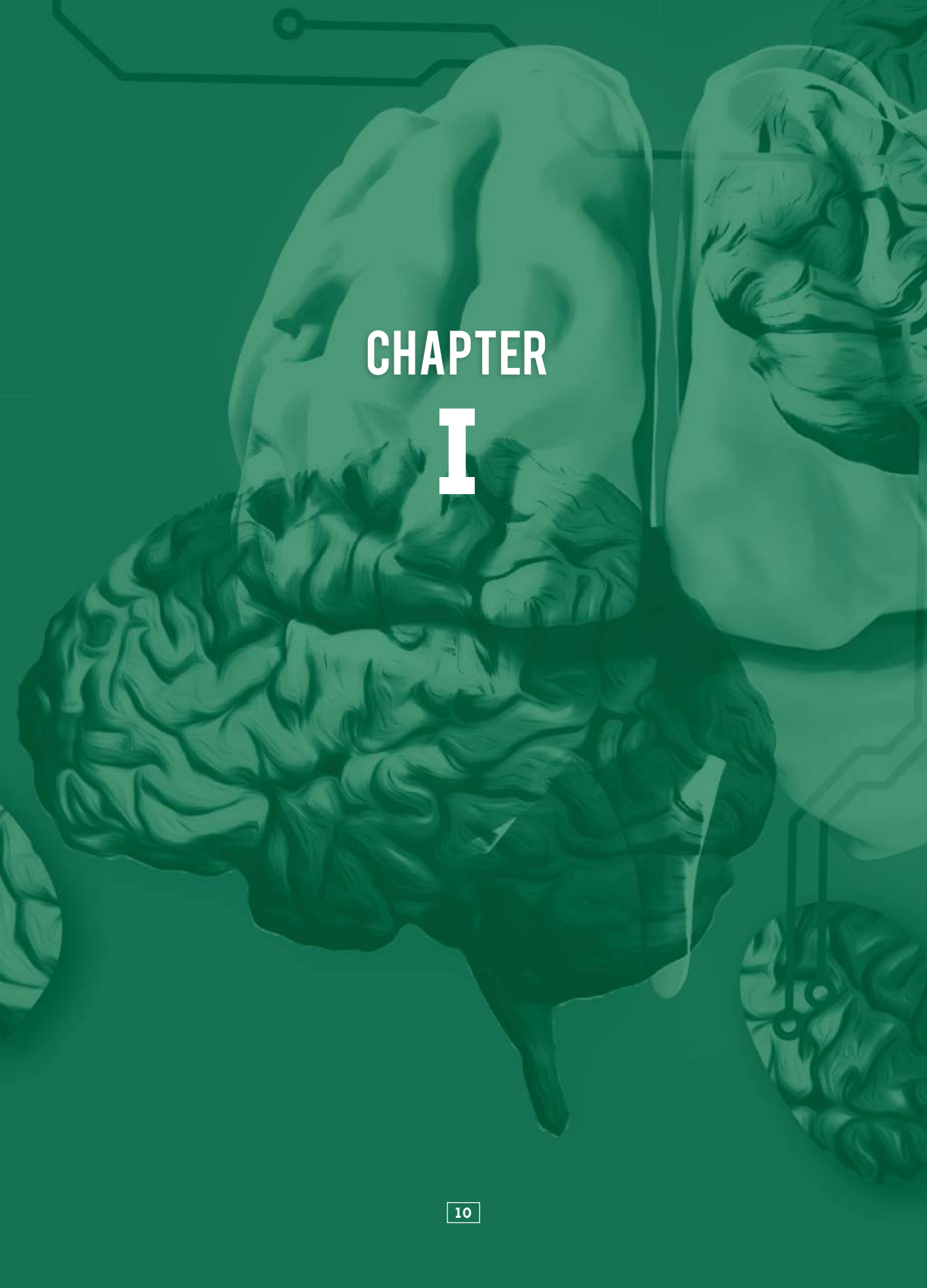
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ACC	Anterior Cingulate Cortex	FVA	Forensische Verslavings Afdeling (Forensic Addiction Department)
AFZ	Afdeling Forensische Zorg (Forensic Care Department)	GCP	Good Clinical Practice
AIC	Anterior Insular Cortex	GLM	General Linear Model
ASPD	Antisocial Personality Disorder	HD-tDCS	High-Definition transcranial Direct Current Stimulation
AUDIT	Alcohol Use Disorder Identification Test	HKT-R	Dutch Risk Assessment Tool
BCM	Bienenstock-Cooper-Munroe rule	Hz	Hertz
BISII	Behavioral Impulsivity Scale	IB	Investigator's Brochure
CCMO	Central Committee on Research Involving Human Subjects; in Dutch: Central Commissie Mensgebonden Onderzoek	IC	Informed Consent
CE	European Certified	ICA	Independent Component Analysis
CI	Confidence Interval	IMP	Investigational Medicinal Product
CT	European drug regulatory affairs Clinical Trails	IMPD	Investigational Medicinal Product Dossier
DMPFC	Dorsomedial Prefrontal Cortex	IRI	Interpersonal Reactivity Index
DLPFC	Dorsolateral Prefrontal Cortex	LPP	Late Positive Potential
DSM-5	Diagnostic and Statistical Manual of mental disorders	LTD	Long Term Depression
DUDIT	Drug Use Disorder Identification Test	LTP	Long Term Potentiation
EEG	Electroencephalography	MET	Multifaceted Empathy Task
ERP	Event-Related Potential	METC	Medical Research Ethics Committee (MREC); in Dutch: Medische Ethische Toetsings Commissie (METC)
EU	European Union	NIC	Neuroelectric Instrument Controller
FFT	Fast Fourier Transform	NIBS	Non Invasive Brain Stimulation
fMRI	functional Magnetic Resonance Imaging	NIDA	National Institute of Drug Abuse

OCDS	Obsessive Compulsive Drinking Scale
OCDUS	Obsessive Compulsive Drug Use Scale
P ₃	P300 (event-related potential)
PFC	Prefrontal Cortex
PLV	Phase Locking Value
PSAP	Point Subtraction Aggression Paradigm
RISQ	Risky, Impulsive, and Self-Destructive Behavior Questionnaire
RPQ	Reactive and Proactive Aggression Questionnaire
rsEEG	Resting-State Electroencephalography
rMET	Reading the Mind in the Eyes Test
(S)AE	(Serious) Adverse Event
SPC	Summary of Product Characteristics; in Dutch: Officiële Productinformatie Sponsor): The sponsor is the party that commissions the organization or performance of the research, for example a pharmaceutical company, academic hospital, scientific organization or investigator. A party that provides funding for a study but does not commission it is not regarded as the sponsor, but referred to as a subsidizing party.
SPSS	Statistical Package for Social Sciences

SRP-SF	Self-Report Psychopathy Scale Short Form
SUD	Substance Use Disorder
SUSAR	Suspected Unsuspected Serious Adverse Reaction Wbp Personal Data Protection Act (in Dutch: Wet Bescherming Persoonsgegevens).
TAP	Taylor Aggression Paradigm
TAU	Treatment as Usual
tCS	transcranial Current Stimulation
tDCS	transcranial Direct Current Stimulation
tMS	transcranial Magnetic Stimulation
ToM	Theory of Mind
TPJ	Temporo-Parietal Junction
VIM	Violence Inhibition Mechanism
vlPFC	ventrolateral Prefrontal Cortex
vmPFC	ventromedial Prefrontal Cortex
WMO	Medical Research Involving Human Subjects act; in Dutch: Wet Medisch-wetenschappelijk Onderzoek met mensen



CHAPTER I



I

INTRODUCTION



INTRODUCTION

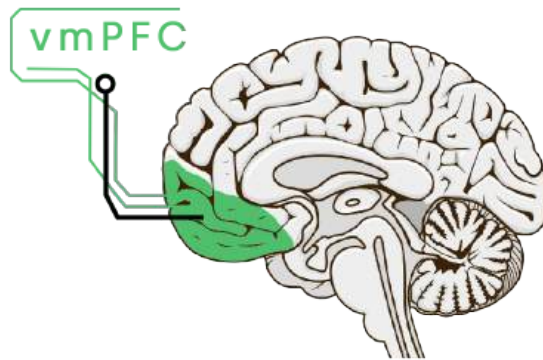
Violence is a major problem in our society. While violence can occur in many contexts, individual acts of aggression account for the majority of the cases. Therefore, research into the neural underpinnings of aggression has grown exponentially. Predicting aggressive behavior in forensic patients is crucial for treatment in forensic care. Nevertheless, there is still a large gap between neuroscience and the forensic practice, and the useful insights of the neurological mechanism underlying aggressive behavior are not well understood or communicated towards forensic care. Current treatment interventions to reduce aggression in forensic patients are not found to be effective and are not leading to a reduction in violent reoffending (Greeven & Bosma, 2012; Koeter et al., 2009). One underlying cause to the ineffectiveness of the current treatments may be that a large part of these forensic patients also suffers from a substance use disorder. This group of forensic patients with a substance dependence is highly unmotivated to treatment due to side effects of medication and the repeated use of substances that has led to structural neuroadaptations in the brain. Substance use can enhance violent behavior through the psychopharmacological properties of drugs and alcohol, which can lead towards more aggression and with that an increase in crime (Collins, 1981; De Looff et al., 2017; Virkkunen & Linnoila, 1993; White, & Gorman, 2000). Therefore, the treatment of forensic patients with a substance use dependence is of significant importance to society.



Therefore, the need for a new add-on treatment in this difficult to treat patient group is needed. Studies have started investigating the neurological underpinnings of violent behavior, aiming towards neuropsychology in the treatment of violent forensic patients. A promising new treatment option to investigate this is transcranial Direct Current Stimulation (tDCS), which can modulate brain regions of interest and regulate deficits in the brain.

Deficits in the Prefrontal Cortex (PFC) are associated with aggression and violent behavior (Anderson et al., 1999b; Blair, 2004; Calzada-Reyes et al., 2013, 2016; Krämer et al., 2009; Raine, 1993; Raine et al., 2000). An important part of the PFC in regulating emotion and aggression is the Ventromedial Prefrontal Cortex (vmPFC). This is an area located in the lower (ventral) central (medial) region of the PFC and is known for regulating emotions, inhibition, impulsivity and aggression (See Figure 1). In particular, impairments in the vmPFC in alcohol and cocaine use patients play a crucial role in these predictions towards violent behavior (Gizewski et al., 2012; Preller et al., 2014). Therefore, changes in neural substrates through substance use, the increased odds of aggression in these patients and the prediction of violent behavior can be associated to abnormalities in these areas of the brain. Due to this, investigating the brain abnormalities of forensic patients could be of substantial importance to gain knowledge on the neurobiological underpinnings of aggressive behavior.

Hence, this dissertation focusses on the effectiveness of modulating the vmPFC as a treatment to increase empathic abilities and decrease aggression and with that gaining knowledge in understanding the brain of forensic patients with a violence offence. To give a better insight in the background of this dissertation an overview of the constructs will be described first.

**Figure 1.**

the Ventromedial Prefrontal Cortex (vmPFC)

Forensic Patients

Forensic patients are delinquents with a mental disorder who are also given mental health care. The forensic patients studied in this dissertation mostly have a personality disorder from the DSM-IV Cluster B disorders (histrionic personality disorder, borderline personality disorder, narcissistic personality disorder and antisocial personality disorder). The main focus of this dissertation is on antisocial personality disorder (ASPD). ASPD is associated with dysfunctional social and behavioral processes with exploitive, delinquent, and criminal behavior with no remorse (Black, 2015; Farrington, 1993; Moran, 1999). Therefore, this specific group of offenders has increased odds of posing violent behavior partly caused by the underlying mechanisms of aggression.

Aggression

Aggression is conceptualized here as behavior that is mostly defined by any behavior that is intended to harm someone who is motivated to avoid being harmed (Baron et al., 1994). Aggression can be subdivided in reactive and proactive aggression (Dodge & Coie, 1987). Reactive aggression (i.e. impulsive aggression) is 'hot' and emotional, causes an increase in arousal, is associated with hostile attributions and is expressed in reaction to provocation (Crick & Dodge, 1996; Dodge & Coie, 1987; Dodge et al., 1990). On the other hand, proactive aggression (i.e. instrumental disorder) is a 'cold', instrumental and premeditated form of aggression, which is often motivated by a desire to dominate another person or to obtain a personal gain (Blair 2001; Blair et al., 2006; Dodge & Coie, 1987).

Empathy

One important aspect in understanding aggression and violent offending is empathy. Empathy is a multi-dimensional concept that can be described as the ability to



understand the emotions of others and to share the mental states of other individuals in social contexts, to facilitate their motivation, action and affections (Bahji et al., 2021; Hein et al., 2008; Hétu et al., 2012; Singer et al., 2006; van Dongen, 2020). Therefore, empathy is crucial for our emotional and social life, social interactions and social enhancement (Bernhardt & Singer, 2021; Geer et al., 2016). Due to the complexity of the term empathy the construct will be further explained in its different components. Social psychology and neuroscience (Decety, 2007; Kanske et al., 2017; Winter et al., 2017) have identified two crucial routes in understanding others: (i) an affective route that encompasses the sharing for others' emotions and feeling for them (including empathy, and compassion) and (ii) a cognitive route that allows for the reasoning about others' mental states (Theory of Mind [ToM] and perspective-taking). Generally, researchers have postulated that empathy includes both affective and cognitive components (Decety and Jackson, 2004; Eisenberg and Eggum, 2009; Decety, 2010; van Dongen, 2020; Zaki and Ochsner, 2012). The affective component entails the bottom-up process that derives from perception-action coupling, and emotion perception (Preston and de Waal, 2002) and can be labeled as affective emotional contagion (van Dongen, 2020). The affective component is developed early in life, automatic and could be seen as the mimicry of the self and other (van Dongen et al., 2020). The cognitive component entails perspective taking (i.e., Theory of mind; ToM) or mentalizing in making a distinction between self and other mental states. The cognitive component includes executive functioning such as attention, self-regulation and inhibitory control (van Dongen, 2020).

Association Between Empathy and Aggression

According to the models of Blair (Blair 2001; 2005; Blair et al., 2006) individuals with less empathic abilities may be less susceptible and motivated to inhibit violent behavior, which therefore increases the risk for violent behavior. In addition, inhibiting empathic responding (McGrady et al., 2008) is regarded as a cause for antisocial behavior. Studies demonstrated that a deficit in empathic abilities is associated with antisocial and deviant behavior (Preller et al., 2014). Furthermore, Galan and colleagues (2017) reported that empathy predicted moral disengagement and juvenile petitions for violent crimes; lower empathy predicted increased violent behavior and more arrests. Also, Winter et al., (2017) demonstrated that men with a history of aggressive behavior showed diminished empathic responses and Heilbrun (1982) showed that male offenders had lower empathy than male controls. The above-mentioned findings are confirmed in a meta-analysis of Jolliffe and Farrington (2004), that revealed a negative association between empathy and offending. Hence, empathic abilities have shown to have an association with aggressive behavior. To gain more insight in these crucial roles it is important to investigate the neurobiological processes underlying empathy and aggression.

Neurobiological insights in Empathy and Aggression

Antisocial behavior, especially aggression and impulsive behavior, is associated with dysfunctions in the prefrontal cortex (PFC; Anderson et al., 1999b; Blair, 2004; Calzada-Reyes et al., 2013, 2016; Krämer et al., 2009; Raine, 1993; Raine et al., 2000). Most notably are the impairments in the vmPFC (e.g. emotional regulation, moral decision making) which are linked to empathic abilities, and the Dorsolateral Prefrontal cortex (DLPFC; e.g. disinhibition and impulsiveness), which is associated with aggression and violent behavior (Blair et al., 2006; Giancola, 2004; Hoaken et al., 2003; Morgan & Lillienfeld, 2000). Based on the aforementioned theory of Blair linking impaired empathic abilities to violence, and the model demonstrated (Fig. 2), it could be proposed that the vmPFC plays a crucial role in modulating both empathic abilities and thereby also in violent behavior. Furthermore, one of the most substantial causes of aggressive behavior on a biological level is a lesion of the frontal lobe of the brain (Tateno et al., 2013). A meta- analysis (Yang & Raine, 2009) of 43 imaging studies also demonstrated that impairments of the DLPFC are related to antisocial behavior.

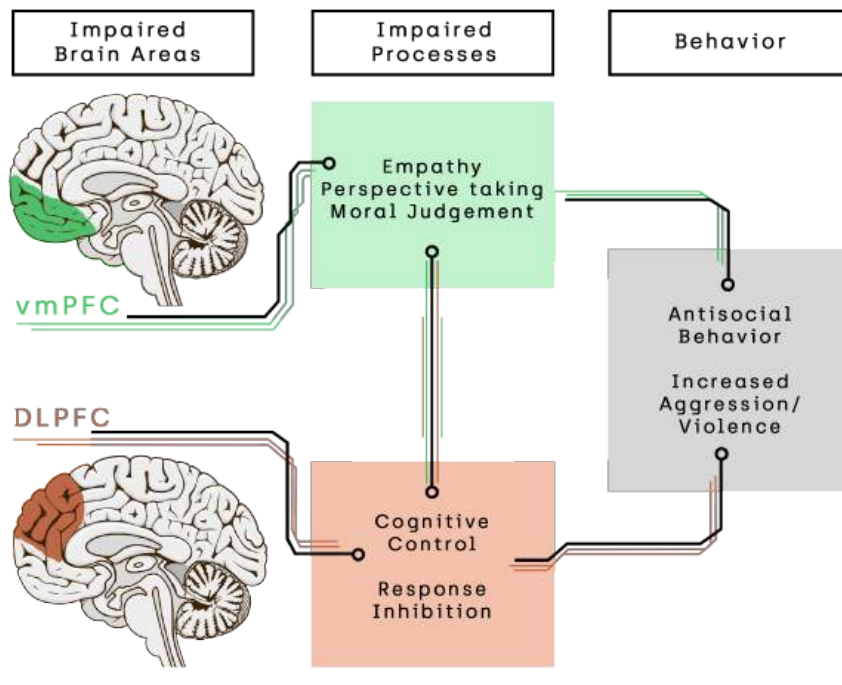


Figure 2.

Relationship between the impaired brain areas, impaired processes and behavior. The highlighted brain area, impaired process and behavior are discussed in this dissertation.
 *DLPFC= Dorsolateral Prefrontal Cortex, vmPFC= Ventromedial Prefrontal Cortex



Although, the DLPFC modulates cognitive control and response inhibition which are associated with aggression and violent behavior (Blair et al., 2006; Giancola, 2004; Hoaken et al., 2003; Morgan & Lillienfeld, 2000), in this study we focus on the vmPFC. The vmPFC appears to play a crucial role in both aggressive behavior and empathy (Anderson et al., 2006; Blair 2001; Zheng et al., 2016). Neuroimaging studies showed that the structures that are relevant for empathy and the modulation of aggressive behavior, namely the vmPFC and medial frontal regions (Anderson et al., 2006; Blair 2001; Blair et al., 2006; Covell & Scalora, 2002; Eisenberg et al., 2010; Giancola 2004; Hoaken & Stewart, 2003; Preller et al., 2014; Marshall, 1996; Mathur et al., 2010; Miller & Eisenberg, 1988; Morgan & Lillienfeld, 2000; Waytz et al., 2012; ; Zheng et al., 2016), are impaired in individuals with antisocial traits (Decety et al., 2013; Grafman, 1996; Koenings et al., 2007; Lockwood, 2016; Shamay-Tsoory et al., 2010; van Dongen, 2020). Damage to the vmPFC is associated with poor decision-making in antisocial behavior (Bechara et al., 1997; Damasio et al., 1990). In the same vein, studies of Raine et al. (1993; 1998; 2003) demonstrated that abnormal brain structures of the medial and ventral PFC correlate with the lack of empathy, impaired moral judgment (Raine et al., 2003) and hyper metabolism (Raine et al., 1998), which are all contributing to increased aggressive impulses.

One study of Skibsted et al. (2017) used a laboratory aggression task, the Point Subtraction Aggression Paradigm (PSAP; Cherek, 1992), and also indicated a link between brain activity in the vmPFC and aggressive behavior, proposing the vmPFC to be a neural substrate related to aggressive provocation (Skibsted et al., 2017). Other studies (Blair, 2018; Blair, 2016; Gilam et al., 2015; Gilam et al., 2017; White et al., 2014) demonstrated that the vmPFC is related to increasing costs and retaliation of the aggressor and greater punishment choices. Furthermore, Grafman and colleagues (1996) demonstrated that war veterans with lesions of the vmPFC exhibited increased aggressive behavior compared to veterans with no lesion. Another study of Bufkin and Luttrell (2005) showed that dysfunctions in the vmPFC could differentiate between violent and non-violent offenders. In addition, the study of Coccaro et al., (2007) found a reduced activity of the vmPFC and vmPFC-amgdala connectivity in patients with explosive disorders when watching angry faces.

Furthermore, studies show that changes in the brain areas related to less empathic ability (i.e., vmPFC) in abusers of alcohol and cocaine plays an important role in violent behavior (Gizewski et al., 2012; Preller et al., 2014). Moreover, repeatedly using substances has been found to be associated with functional neuroadaptations in the PFC (Goldstein & Volkow, 2011; Volkow et al., 2009).

Substance-dependent Offenders

Substance abuse is a major concern in forensic patients and poses an additional risk for violence in these individuals (Pickhard & Fazel, 2013). Forensic patients with a substance dependence are found to have increased odds of violent behavior (Cristello et al., 2020; Friedman, 1998; Goldstein & Volkow, 2011; Gizewski et al., 2012; Ghossoub et al., 2019; Preller et al., 2014; Volkow et al., 2009; Wagner, 1996). Impaired functioning of the DLPFC in patients with substance use disorders (SUD), are underlying to a diminished cognitive and inhibitory control leading to increased tendency to relapse and maintain in addictive behaviors (Capriles et al., 2003; Garavan, 2007; Janes et al., 2010; Kosten et al., 2006; van Holst & Schilt, 2011; Volkow et al., 2001; Volkow et al., 2007). Especially the vmPFC is found to be associated with SUD in regulating reward processing and motivational salience (Engelman et al., 2012; Filbey et al., 2008; Goldstein & Volkow, 2011; Kearney-Ramos et al., 2018; Kilts et al., 2001; Li et al., 2015; Schacht et al., 2013). In addition, Preller and colleagues (2014) found in their study that cocaine users show deficits in emotional empathy and that substance abuse patients are less emotionally responsive to the emotions of other individuals and their mental state. To measure the deficits in emotional processing and changes in brain activity related to aggression electroencephalography (EEG) can be used.

Electroencephalography

Electroencephalography (EEG) is a method used to record and analyze the electrical activity of the brain and is an excellent instrument to measure neurobiological dysregulation (Sur & Sinha, 2019). Using EEG, cognitive processing can be investigated in using different methods. One method is to explore event-related potentials (ERPs) to measure a certain moment in time linked to a certain stimuli or task, another method is to investigate ongoing activity in a particular frequency band (power) and the coherence between different electrodes (connectivity). In this dissertation I have investigated all the above-mentioned methods.

ERPs are small changes in the EEG signal linked to a certain stimuli or event in time and can be used to assess cognitive processing and have been linked to different neurological disorders (Duncan et al., 2009; Kappenman & Luck, 2016; Kremláček et al., 2016; Michie et al., 2016; Morlet & Fischer, 2014; Seer et al., 2016). Therefore, analyzing ERPs can be a useful technique for investigating empathy. Studies have shown that both early and late ERP components are related to perceived pain in others (Cheng & Decety, 2012; Decety et al., 2010; Meng et al., 2012) and related to self-reported trait empathy (Corbera et al., 2014; Fabi & Leuthold, 2017; van Dongen et al., 2018; Vaess et al., 2016). Two ERPs that have been studied extensively in the context of empathy and aggression are the P3, and the late positive potential (LPP).



The P3 is an ERP that has a positive voltage in the latency of 300-650 milliseconds (ms; Hajcak et al., 2010). The P3 is typically associated with controlling sustained attention towards salient stimuli (Hajcak et al., 2010) and has been linked to empathy for pain (Cheng & Decety, 2012; Coll, 2018; Cuthbert et al., 2000; Decety et al., 2010; Fan & Han, 2008; Ibáñez et al., 2011; Li & Han, 2010). The LPP is a positively sustained ERP that occurs around 400-1000 ms and is found to be maximal at the parietal locations and the central midline (Hajcak et al., 2012) and is associated with empathy and emotional cues. A study of Cuthbert et al. (2000) demonstrated that the LPP indicates an increase in amplitude towards emotional stimuli, arousal, regulation, facial expressions, and affective experience regulation. In the same vein, the study of van Dongen et al. (2018) found reduced P3 and LPP in individuals with decreased empathic abilities. Therefore, studying these positive potentials contribute to the understanding of emotional regulation (Cheng & Decety, 2012; Cuthbert et al., 2000; Fan et al., 2011; Lang & Bradley, 2010), and serve as an indication that the temporal dynamics of empathic processing in the brain can be studied using late ERPs in the EEG.

Another way to investigate electrophysiological changes is investigating the ongoing brain oscillations of the spontaneous EEG. In this way, EEG enables to investigate electrical fluctuations in neural signals (Howells et al., 2018; Jorge et al., 2014), allowing brainwave patterns during rest and thought processing to be detected (Nasser et al., 2019). These neural oscillations can be divided into different frequency bands, i.e., the delta wave (δ ; 0.3–3.5 Hz), theta wave (θ ; 4–7.5 Hz), alpha wave (α ; 8–13 Hz), beta wave (β ; 14–30 Hz) and gamma wave (γ ; >30 Hz). Although these waves have been linked to arousal and many different emotional, affective and cognitive states (Di et al., 2018), the discussion below will focus on the association between these waves and relevant aggressive and violent behaviors.

Common findings in these studies investigating EEG of antisocial and violent individuals show that excessive slow-wave activity is related to aggressive tendencies within these individuals (Hill & Waterson, 1987; Scarpa et al., 1997; Brower & Price, 2001; Moya-Albiol, 2004). These aggressive tendencies such as to aggressiveness, elevated anger and violence (Convit et al., 1991) that are linked to this slow-wave activity are all core elements of antisocial personality disorder.

Furthermore, besides the excessive slowing of the EEG activity, the reduced power in alpha frequency band (8-12 Hz) seems also one of the most robust findings in individuals with antisocial and behavioral disorders (Calzada-Reyes et al., 2013; Convit et al., 1991; Fishbein et al., 1989; Forssmann & Frey, 1953; Lindberg et al., 2005; Keune et al., 2012). Activity in the alpha frequency band has been linked to being a general marker for inhibition (Klimesch et al., 2007; Weisz et al., 2011) and proposed to represent communication between different brain regions on a long-range (von Stein et al., 2000). Taking into account

the deficits in functioning of the alpha frequency bands in violent offender samples, it is therefore plausible that specifically the disrupted brain communication is one of the neuroanatomical key factors in antisocial and violent individuals (Motzkin et al., 2011; Tillem et al., 2018; Yang & Raine, 2009).

Resting-State EEG (rsEEG)

In recent decades, rsEEG has received increased attention due to its unique advantages in investigating brain connectivity and characterizing changes in brain functions among different psychiatric disorders. Resting-state is the state in which an individual is awake, but not engaged in a specific task and therefore this could reflect the intrinsic activity in the brain (Duan et al., 2020; Deco et al., 2011; Greicus et al., 2003). The importance of using rsEEG has been proven effective in many studies. For example in providing valuable information of deficits in the brain, such as in Alzheimer's disease (Babiloni et al., 2013), schizophrenia (Siebenhuhner et al., 2013), epilepsy (Rotondi et al., 2016), neuropsychiatric disease (Lehmann et al., 2005, Tomescu et al., 2014), task performance (Koenig et al., 1998; Koenig and Lehmann, 1996) and drug effects (Michel and Lehmann, 1993). Furthermore, rsEEG appears able to detect gender (Volf et al., 2014) and age differences (Koenig et al., 2002; Tomescu et al., 2018) within healthy individuals. Temporal dynamic patterns of the brain activity in resting state measured by the EEG are suggested to change more rapidly than can be detected with using techniques such as fMRI (Baker et al., 2014; Deco & Corbetta, 2011; Siegel et al., 2012). In addition, EEG already has been used to demonstrate the organization of spontaneous brain oscillations and interactions between different networks (de Pasqualle et al., 2016; Lui et al., 2017; Siegel et al., 2012) to be found a more favorable technique to use over fMRI.

Analyzing rsEEG in forensic patients to measure activity in spectral power of the frequency bands (theta, alpha and beta) and the asymmetrical activity between the left and right cortical region can give more insights in the neurological underpinnings of aggressive behavior. Increasing evidence suggests asymmetrical involvement of the frontal hemispheres in aggressive behavior (for review see Harmon-Jones, 2003). Further support for the asymmetrical cortical activity indicates that the left-sided compared to right-sided excitability underlies aggressive tendencies, and shows an association between increasing activity with higher levels of experienced anger and aggressive behavior (Peterson et al., 2008). In the same vein, the studies of Hofman and Schutter (2009, 2012) demonstrated that the asymmetrical activity in the beta frequency band has been found to be related to the development and expression of aggressive behavior (Hofman & Schutter, 2012) and in trait aggression in healthy individuals. In further support of this assumed link between asymmetry and aggression, the theory behind this indicates that relatively higher levels of left-sided as compared to right-sided cortical excitability is associated with increased



approach-related motivational tendencies and decreased levels of avoidance-related motivational tendencies (Harmon-Jones and Allen, 1998; Harmon-Jones, 2007; Schutter et al., 2008).

In addition, an interesting method to examine the synchronicity and communication between frontal regions is to measure the intercorrelation in the phase between electrodes. We used the Phase Locking Value (PLV) to calculate the phase symmetry between couples of electrodes in the frontal region. With this PLV we can calculate 'synchronization' which refers to a process in which linear or nonlinear oscillatory components reflect a property in activity over time, demonstrating a collective behavior (Boccaletti et al., 2018).

Current Treatment Perspectives

From a clinical perspective, deficits in empathic abilities have been reported in several psychiatric disorders (Decety and Moriguchi, 2007), including psychopathy (Soderstrom, 2003), antisocial personality disorder (Dolan & Fullam, 2004), depression (Schreier & Pijnenborg, 2013), schizophrenia (Bragado-Jimenez & Taylor, 2012), autistic spectrum disorder (Nagar Shmoni et al., 2012) and Alzheimer's dementia (Laisney et al., 2013). Due to the complexity of the construct empathy, it remains difficult to indicate if the deficits in empathic abilities are solely due to brain impairments, if it is characterized by the affective component or cognitive component and how underlying personal traits play a role (Bird et al., 2010; Bora et al., 2009; Michaels et al., 2014; Shah et al., 2016). As previously described, empathic abilities have been linked to impaired social functioning and offending behavior such as aggression and violence (Blair et al., 2005; Jolliffe & Farrington, 2004, 2007; Lovett and Sheffield, 2007; Miller and Eisenberg, 1988; van Lange et al., 2014). Therefore, an intervention that is combining both increasing empathic abilities and reducing violent behavior could be a crucial implementation in forensic care. Conventional interventions have tried to enhance empathy in psychopaths (Salekin et al., 2010) and sex offenders (Dennis et al., 2012), but with limited benefits.

In addition, substance abuse problems have an important relationship with criminal behavior. As demonstrated in a meta-analysis from (O'Driscoll et al., 2012) personality disorders and substance use disorders are a very common comorbidity among forensic patients and that this comorbidity appears to be significant risk factor for reoffending. This combination occurs in 50-75% of the offenders (Bloem et al., 2011) and cost the society about 600 milliard euros due to cost of crime, work-related problems and especially (mental) healthcare. These problems are an enormous burden to the community in finance and safety and further supports the view that the treatment of patients with this combination should not just focus on behavioral treatment, but should encompass interventions to target the processes of personality and substance abuse problems.

As previously noted, research has found that substance use, especially alcohol and cocaine, is related to (violent) criminal behavior (Bennett et al., 2008; Lammers et al., 2014; Ramaekers et al., 2016), and that money invested in treatment may lead to a large reduction in costs associated with substance abuse (National Institute of Drug Abuse; NIDA).

Nevertheless, current interventions seem insufficient in treatment success of substance use in forensic mental health care and are not effective to reduce violence risk (i.e., 66% of the patients reoffend). This could be due to the earlier proposed reasons indicating that the patients have a complex medical-psychiatric disorder, occurring within a poor motivation, side-effects and in which the prolonged usage of substances has led to functional and structural changes in the brain (Capriles et al., 2003; Garavan, 2007; Janes et al., 2010; Kosten et al., 2006; van Holst & Schilt, 2011; Volkow et al., 2001; Volkow et al., 2007). Because traditional interventions induce brain changes in a non-selective manner using psychological and pharmacological treatment, it remains difficult to improve these treatments with neurobiological knowledge (Luigjes et al., 2009). Therefore, the last decade there is an increased interest in more innovative methods that can target specific brain regions of interest to modulate behavior.

Neuromodulation and transcranial Direct Current Stimulation (tDCS)

Recent insights such as the notion that the external activation of neurons in specific brain networks can improve the functioning of an individual has led to the development of techniques to modulate the brain. Modulating a brain region of interest can increase or decrease neuronal activity (Nitsche & Paulhus, 2000; Woods et al., 2016). One of the most feasible devices for non-invasive brain modulation is tDCS. This neuromodulation device delivers low-intensity, direct electrical currents via scalp electrodes. It typically involves two electrodes placed transcranial on the scalp, the electrodes are connected to a stimulator that delivers a constant, direct current (1-2 mA; Nitsche & Paulhus, 2000). Through this current flow from one activating positive electrode to the negative electrode induces an electrical field that produces the modulating of the brain region of interest (Woods et al., 2016).

It has been previously suggested that through anodal stimulation, certain brain areas could be stimulated to enhance cortical excitability, whereas cathodal stimulation would decrease the activity of the modulated brain region (Nitsche & Paulhus, 2000). More recent studies show that the effect of stimulation is based on the strength of the current flow, the direction and the duration of the modulation (Batsikade et al., 2013; Nissim et al., 2019; Samani et al., 2013). The neuroplasticity of the tDCS electrodes have to be indicated specifically for each study to indicate the effect desired. tDCS alters cortical excitability through inducing neuronal membrane potentials in their sub threshold modulation (Nitsche & Paulhus, 2000; Nitsche et al., 2003) and can change neuronal plasticity



(Brunoni et al. 2012). This change in neuronal synaptic plasticity can either produce Long Term Potentiation (LTP) or Long Term Depression (LTD) –like ‘learning’ in the neurons that are modulated and can be achieved through repeated sessions of brain stimulation (Woods et al., 2016).

Multiple studies confirm tDCS as an effective intervention (Barr et al., 2008; George & Aston-Jones, 2010; Jansen et al., 2013) for many different psychiatric disorders (Kuo et al., 2017), including schizophrenia (Brunelin et al., 2012), major depression (Kalu et al., 2012) and addiction (Trojak et al., 2017). In addition, functional alterations in the brain due to the long abuse of substances are hypothesized to improve by an intervention with tDCS and will reduce craving (Boggio et al., 2008b; Batista et al., 2015; Camprodon et al., 2007; Conti et al., 2014; den Uyl et al., 2015; Falcone et al., 2016; Fregni et al., 2008a, 2008b; Jansen et al., 2013; Klauss et al., 2014; Politi et al., 2008; Shahbabaie et al., 2014; Wietschorke et al., 2016).

A promising, more developed tDCS technique is High Definition-tDCS (HD-tDCS), which uses multiple compact (i.e., < 5 cm²) circular electrodes to deliver the low current (Bikson et al., 2019). In this way HD-tDCS can target brain structures with higher focality (DaSilva et al., 2015) and therefore deeper regions of the brain, such as the vmPFC, can be modulated in an optimal way (Edwards et al., 2013; Faria et al., 2011; Hess et al., 2013; Manuel et al., 2014; Nakamura-Palacios et al., 2016; Tanaka et al., 2013; To et al., 2018).



Fig. 3
Starstim High-Definition transcranial
Direct Current Stimulation (HD-tDCS) device
used in this study.

tDCS Modulating Empathic Abilities

Several studies have demonstrated that through anodal tDCS stimulation targeting the PFC emotional processes can be influenced. tDCS has demonstrated to modulate emotional pain (Coll et al., 2016) and can enhance empathy for pain (Davidov et al., 2013; Conti & Nakamura-Palacios, 2014; Wang et al., 2016). Other studies focused on modulating norm-compliance (Cohen, 2015; Fertoni & Miniussi, 2016; Ruff et al., 2013) and self-perception (Coll et al., 2017; Feeser et al., 2014; Kuehne et al., 2015; Maréchal et al., 2017; Rêgo et al., 2015; Wang et al., 2014). Furthermore, studies (Colzato et al., 2016; Darby & Pascual-Leone, 2017; Sellaro et al., 2016) focusing on morality showed that anodal tDCS targeting the PFC could increase morality, since the reinforcement of morality can increase empathic abilities and reduce criminal behavior (Decety & Cowell, 2014).

Although there seems to be a clear association between empathy and the vmPFC, only a few studies have investigated modulation of the vmPFC. Abend and colleagues (2018) show in their study combining tDCS and fMRI that stimulation with tDCS demonstrated increased emotion-related activation in the vmPFC. Other studies demonstrated that anodal stimulation of the (right) vmPFC increases empathic abilities and morality (Abend et al., 2018; Feeser et al., 2014). Other studies using fMRI demonstrated that increased neuronal activity of the vmPFC predicts increased empathic abilities (Mathur et al., 2010; Waytz et al., 2012). In addition to empathy, antisocial behavior and aggression are associated with dysfunctions in the PFC (Anderson et al., 1999b; Blair, 2004; Calzada-Reyes et al., 2013, 2016; Krämer et al., 2009; Raine, 1993; Raine et al., 2000) and will be described in the next paragraph.

tDCS Modulating Aggression

Studies investigating tDCS and aggression showed that targeting the PFC with anodal tDCS reduces social exclusion and the aggressive behavior that emerges from this exclusion (Riva et al., 2015). In addition, studies found that tDCS can reduce intentions of aggressive behavior (Choy et al., 2018; Dambacher et al., 2018) and unprovoked aggression (Riva et al., 2017). Furthermore, the study of Gillam and colleagues (2018) demonstrated the association between violent behavior and the vmPFC using fMRI and increased vmPFC activity after active tDCS stimulation leading to increased self-reported anger and a decrease in violent behavior after the first session of active tDCS.



The previous mentioned findings support a potential causal link between vmPFC functionality, empathy and violent behavior, therefore supporting the role of the vmPFC as a core region, and providing a promising avenue for reducing aggressive behavior using neuromodulation devices. To date most of the studies that investigate the modulation of the PFC in reducing aggression have relied on conventional tDCS, upregulating one side of the brain and downregulating the contralateral region (i.e., Choy et al., 2017; Gilam 2008; Gilam 2018; Ling et al., 2020; Molero-Chamizo et al., 2019). Therefore, using HD-tDCS could be of significant importance to investigate the effectiveness of tDCS in reducing aggression.

Current Dissertation

Several studies have already investigated the effectiveness of tDCS in targeting different brain regions and in several disorders. To the best of our knowledge, targeting the vmPFC to increase empathic abilities and reduce aggression in a forensic sample has not been investigated. Other studies (i.e., Choy et al., 2018; Gilam et al., 2018) have examined tDCS targeting aggressive behavior but used healthy participants instead of a forensic sample.

Furthermore, this dissertation is innovative in the use of a protocol that focuses on enhancing the effect of tDCS by enhancing the brain state during modulation as other studies did not use this approach (i.e. Molero-Chamizo et al., 2019). We used multiple sessions of HD-tDCS as compared to other studies using conventional tDCS (Choy et al., 2018; Gilam et al., 2008; 2018; Molero-Chamizo et al., 2019). In addition to investigating solely empathy and aggressive behavior, we examined the effect of the tDCS on electrophysiological changes measured in ERP's as well as ongoing brain oscillations during rest.

Furthermore, tDCS has demonstrated to have the ability to enhance synchronicity between electrodes and with that enhance functional connectivity between regions of the brain. In addition to examining the spectral power and asymmetrical activity in the rsEEG, we investigated the synchronicity between frontal regions. These insights can be of substantial importance in enhancing knowledge and the implications for tDCS as a forensic intervention. Lastly, we investigated this in a violent offender sample with substance use dependence as most studies mentioned above focus on just one of these disorders.

Outline Dissertation

With this current dissertation we aimed to gain more insight in the effectiveness of HD-tDCS in the treatment of forensic patients in improving empathic abilities and reducing aggression. The neurological underpinnings were investigated using empathy tasks, aggression tasks, ERPs, brain oscillations in rsEEG and synchronicity between electrodes. In addition, we tried to increase the knowledge in understanding the aggressive brain of forensic patients to give an overview to implement tDCS as a forensic intervention for forensic care.

Chapter II reviewed the literature already existing on using tDCS targeting the vmPFC and DLPFC as a treatment to increase empathic abilities and reduce antisocial behavior . We first investigated the neurobiological insights already present in the literature and the interventions using tDCS to modulate empathic abilities and aggression to determine the most optimal target location. The literature collected supported the application of tDCS as a potential tool to modulate empathic abilities and aggression, especially when targeting the vmPFC.

Chapter III describes in a very detailed manner the study protocol that was designed to investigate multi-session HD-tDCS targeting the vmPFC as an add-on treatment. This study will explore the opportunities to introduce a new intervention to improve empathic abilities and reduce violence in forensic substance use offenders. In addition, this study may help to reduce craving. The study will contribute to knowledge about increasing empathic abilities and the functioning of the vmPFC. Specifically, this study may give insight in how to implement the tDCS intervention in the setting of daily clinical practice for this complex, multiple problem target group and with that contribute to reduction of recidivism.



Chapter IV examined HD-tDCS as an intervention to increase empathic abilities and reduce violent behavior in forensic patients. In addition, using electroencephalography (EEG), the effects on the P₃ and the Late Positive Potential (LPP) of the Event-Related Potentials (ERPs) in reaction to situations that depict victims of aggression were examined. Fifty male forensic patients with a substance dependence were tested in a double-blind sham-controlled randomized trial. The patients received HD-tDCS two times a day for 20 min for five consecutive days. Before and after the intervention, the patients completed self-reports, a resting-state EEG task, an empathy task while attached to the EEG and performed the Point Subtraction Aggression Paradigm (PSAP).

Chapter V examined the effect of HD-tDCS targeting the vmPFC on frequency power, asymmetrical activity and synchronicity in the frontal brain region of a violent offender sample during resting-state EEG (rsEEG). Based on previous literature we hypothesize changes in the theta (4-7 Hz), alpha (8-13 Hz) and beta (14-30 Hz) frequency bands in the violent offender sample. Furthermore, since asymmetrical activation has been linked to aggressive traits, we investigated the effect of the HD-tDCS intervention in our violent offender sample and compared this from pretest to posttest. In addition, we investigated the modulation of the HD-tDCS on the synchronicity (phase-correlation) between the frontal electrodes from pre tot posttest as measured by the Phase Locking Value (PLV).

Chapter VI provides a summary and general discussion of this dissertation with a focus on HD-tDCS as an add-on treatment for forensic patients and the neurological underpinnings of criminal behavior. Practical implementations and future recommendations are also provided.

CHAPTER II

This chapter is published as:

Sergiou, C. S., Santarnecchi, E., Franken, I. H. A., & van Dongen, J. D. M. (2020). The effectiveness of Transcranial Direct Current Stimulation as an intervention to improve empathic abilities and reduce violent behavior: A literature review. *Aggression and Violent Behavior*, 55, 101463.

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The background is a solid dark teal color. In the top-left corner, there is a detailed, light-colored illustration of a human brain. From the bottom-left of the brain, several thin, light-colored lines extend downwards and to the left, ending in small circles, resembling a circuit board or neural pathways.

II

THE EFFECTIVENESS OF TRANSCRANIAL DIRECT CURRENT STIMULATION AS AN INTERVENTION TO IMPROVE EMPATHIC ABILITIES AND REDUCE VIOLENT BEHAVIOR: A LITERATURE REVIEW.

An abstract graphic of a brain with circuitry, rendered in a dark green color, serving as a background for the title.

ABSTRACT

Background:

Empathy can be seen as an individual factor decreasing the probability of violent, criminal behavior, whereas a lack of empathy is seen as an increasing factor to antisocial behavior. Antisocial behavior, especially aggression and impulsive behavior, is associated with dysfunctions in the prefrontal cortex. There has been a growing interest in using Transcranial Direct Current Stimulation (tDCS) as an intervention to modulate brain regions of interest and increasing activity in damaged brain areas. This paper reviews the evidence about using tDCS as a potential intervention to increase empathic abilities and decrease antisocial behavior in violent offenders.

Objective/Methods:

This literature review is conducted to examine what is currently known about how tDCS may modulate empathic abilities and aggressive behavior. Articles in which tDCS was used to increase empathic abilities or modulate violent behavior were selected. Brain areas of interest are the Ventromedial Prefrontal Cortex (vmPFC) and the Dorsolateral Prefrontal Cortex (DLPFC).

**Results/Conclusions:**

Literature collected so far support the application of tDCS as a potential tool to increase empathic abilities and reduce violence in forensic patients, especially when targeting vmPFC. Further research on the vast parameter space of tCS (e.g. stimulation intensity, electrode location, polarity) is needed to consolidate tDCS as a tool in forensic science.

Keywords:

tDCS, neuromodulation, empathy, violent behavior, vmPFC.

I. Introduction

Empathy can be described as the ability to share the mental states of other individuals, facilitating the motivation, action and affections of others (Hein et al., 2008; Singer et al., 2006). According to the models of Blair (Blair, 2001; Blair 2005; Blair et al., 2006) individuals with less empathic abilities may be less susceptible and motivated to inhibited violent behavior, because violence is inhibited by empathy for another person. It is hypothesized that less empathy leads to a higher risk of violence. Empathic abilities are crucial for social enhancement, social interactions and relationships, and for our emotional and social life (Bernhardt & Singer, 2012). It has indeed been found that a deficit in empathic abilities is associated with antisocial and deviant behavior and with that a higher risk of violence (Preller et al., 2014). Blair (1995) suggested that humans might hold a functional mechanism to inhibit violence: the violence inhibition mechanism (VIM). This cognitive mechanism when activated by distress cues, such as sad facial expression or the visual or auditory cues of tears, can mediate in the activation of aggression. Blair proposed that the VIM is a precondition for the development of (1) moral emotions such as remorse, guilt and sympathy, (2) non-violent behavior, and (3) the moral and conventional distinction during childhood. A lack of VIM can lead to the core elements of antisocial behavior.

A meta-analysis of Jolliffe & Farrington (2004) revealed indeed a negative correlation between empathy and offending. The ability to not empathize with someone else, including the suppressing of an emphatic response (McGrady et al., 2008) is regarded as a cause of antisocial behavior (Lovett & Sheffield, 2007). A recent study reported that empathy predicted moral disengagement and juvenile petitions for violent crimes; lower empathy predicted increased violent behavior and more arrests (Galan et al., 2017). In addition, Winter and colleagues (2017) showed that men with a history of aggressive behavior showed diminished empathic responses. Further, in studies of criminal offenders, male offenders demonstrate lower empathy levels than males in control samples. In a study of Heilbrun (1982) male prisoners with histories of violent behavior have been assessed and resulted in low levels of empathy may be associated with physical aggression. Other studies show that sex offenders exhibit low levels of empathy (Geer et al., 200; Scully, 1988; Ward et al., 2000).

Antisocial behavior, especially aggression and impulsive behavior, is associated with dysfunctions in the prefrontal cortex (Anderson et al., 1999b; Blair, 2004; Calzada- Reyes et al., 2013, 2016; Krämer et al., 2009; Raine, 1993; Raine et al., 2000). Of special importance are the impairments in the vmPFC (e.g. emotion regulation, moral decision- making), and the DLPFC (e.g. disinhibition and impulsiveness), which is associated aggression and violent behavior (Blair et al., 2006; Giancola, 2004; Hoaken et al., 2003; Morgan & Lillienfeld, 2000). A meta-analysis (Yang & Raine, 2009) of 43 imaging studies showed that impairments of the DLPFC are related to antisocial behavior. Several studies show evidence that impaired prefrontal cortex areas lead to emerge of aggression in multiple situations, inducing psychopathic traits such as blunted emotions and a lack of empathy (Damasio, 1994), impaired perspective taking with and increased egocentrism and rigidity (Anderson et al., 1999b; Price et al., 1990) and diminished moral judgment (Koenings et al., 2007).

The vmPFC appears to play a crucial role in both aggressive behavior and other antisocial behavior and empathy (Blair, 2001; Anderson et al., 2006; Zheng et al., 2016). One of the most substantial causes on a biological level of aggressive behavior is a lesions of the frontal lobe of the brain (Tateno et al., 2003). The study of Grafman and colleagues (1996) showed that war veterans with lesions of the vmPFC exhibited increased aggressive behavior compared to individuals with no lesion. Bufkin and Luttrell (2005) demonstrate that vmPFC dysfunctions can differentiate between violent and non-violent offenders. In addition, Coccaro and colleagues (2007) found that patients with explosive disorder showed reduced activity of the vmPFC and vmPFC-amygdala connectivity when they had to watch angry faces. The study of Anderson and colleagues (1999b) demonstrated that lesions in the first 16 months of an individual can lead to psychopathic antisocial traits and impaired reasoning on a social and moral level. Furthermore, Chester and colleagues (2017) demonstrated that physical aggression was associated with reduced gray matter volume in the vmPFC. The patients with less gray matter density in the vmPFC cluster were much more likely to engage in real-world violence. These results indicate that deficits in individuals with vmPFC gray matter can increase physical aggression. Studies using functional Magnetic Resonance Imaging (fMRI) in combination with tDCS have demonstrated that active tDCS stimulation led to increased vmPFC activity during the processing of unfair offers, increased acceptance rates of unfair offers, and an increase in self-reported anger. They also found a decrease in subsequent aggressive behavior following active tDCS stimulation in the first session (Gilam et al., 2018). Other studies using function neuroimaging (i.e. fMRI) demonstrated that increased neuronal activity of the vmPFC predicts increased empathic abilities (Mathur et al., 2010; Waytz et al., 2012). The previous mentioned findings support a potential causal link between vmPFC functionality and violent behavior and empathy,



supporting the role of the vmPFC in anger regulation, and providing a promising avenue for reducing aggressive behavior using neuromodulation devices.

Recent insights such as the notion that the external activation of neurons in specific brain networks can improve the functioning of an individual has led to the development of techniques to modulate the brain. One of the most feasible devices for non-invasive brain modulation is Transcranial Direct Current Stimulation (tDCS). tDCS delivers low-intensity, direct current stimulation via scalp electrodes, it involves two or more rubber (or metal) electrodes placed transcranial on the scalp, the electrodes are connected to a stimulator that delivers a constant, direct current (1-2 mA) (Nitsche & Paulus, 2000). In the past scientist thought that through anodal stimulation, certain brain areas could be stimulated to enhance cortical excitability, whereas cathodal stimulation would decrease the activity of the modulated brain region (Nitsche & Paulus, 2000). But recent studies show that the stimulation effect is based on the strength of the current (Batsikadze et al., 2013; Nissim et al., 2019; Samani et al., 2013). The neuroplasticity of the tDCS electrodes have to be indicated specified for each study to indicate the effect. tDCS alters cortical excitability through inducing neuronal membrane potentials in their sub threshold modulation (Nitsche & Paulus, 2000; Nitsche et al., 2003) and can change neuronal synaptic plasticity (Brunoni et al., 2012). This change in neuronal synaptic plasticity is thought to produce Long Term Potentiation (LTP)-like 'learning' in the neurons that are modulated. Multiple studies confirm tDCS as an effective intervention to modify brain activity (Barr et al., 2008; George & Aston-Jones, 2010; Jansen et al., 2013). Therefore, tDCS has been investigated as a new treatment for many different psychiatric disorders (Kuo et al., 2017), including schizophrenia (Brunelin et al., 2021), major depression (Kalu et al., 2012), and addiction (Trojak et al., 2017). However, studies on the effectiveness of tDCS as an intervention in forensic populations, in particular to reduce violent behavior is scarce. Since empathy (i.e. the lack thereof) is found to be a critical factor in violent behavior (Blair, 2001), the aim of the current review is to evaluate the effectiveness of tDCS as a new intervention to increase empathic abilities and decrease violent behavior.

Although, as described above, an increasing number of studies is published on the effectiveness of tDCS as a treatment intervention for different psychiatric disorders, there is few literature on the effectiveness of tDCS in forensic violent offenders. The aim of the present study is therefore is to review the current literature on the effectiveness of tDCS in improving empathic abilities and also on the effectiveness of tDCS in reducing aggressive behavior. By doing this, more insight in the effectiveness of tDCS in forensic patients can be established. Subsequently, this new knowledge could be used for the development of new neuromodulation (e.g. tDCS) protocols for interventions in violent forensic populations.

2. Materials and Methods

2.1. Overview

A comprehensive literature search was conducted using different databases: Google Scholar, Research gate and PubMed. Articles selected in peer-reviewed academic journals were included. A combination of search terms was used to find articles for this current review, including: tDCS, empathy or empathic abilities, aggression or violent behavior, vmPFC, DLPFC. Additional references were identified using the “related citations” function of PubMed and by reviewing the reference lists of obtained articles. Only studies which primary outcome related to tDCS and the modulation of empathic abilities or externalizing behavior were selected. Articles were selected based on the abstract. 115 articles were selected to use in this review based on their relevance for the literature. 20 articles were selected to use to investigate tDCS to modulate empathic abilities (12) and externalizing behavior (8).

2.2. Studies using tDCS to modulate empathic abilities

Twelve studies were selected to describe the findings of tDCS and empathic abilities. Of the studies that used tDCS to increase empathic abilities, five studies used the anodal electrode position on the DLPFC, one study used the right Ventrolateral Prefrontal Cortex (VLPFC) as anodal target location, one study used the Dorsomedial Prefrontal Cortex (DMPFC) as the anodal target location, one study used anodal tDCS on the left DLPFC, another study used anodal tDCS on the right Temporo-Parietal Junction (TPJ) and the last study included used the vmPFC as the anodal target location. The cathode electrode position was contralateral most of the times. The current strength was either 1 mA (1), 1,5 mA (3) or 2 mA (7). The stimulation duration and frequency differed between single sessions with a range between five till thirty min. One study used a twenty-minute stimulation session each day for five consecutive days.

2.3. Studies using tDCS to modulate externalizing behavior

Eight studies were selected to describe the findings of using tDCS to modulate externalizing behavior and can be divided in three studies that investigated tDCS and risk- taking and five studies focused on aggressive behavior. Five studies used the anodal location on the right DLPFC, one study used bilateral placement of the DLPFC and the other two used the right VLPFC as an anodal target location. The cathode position was either placed contralateral or above the left eyebrow. The current strength was either 1,5 mA or 2 mA. The stimulation duration and frequency differed between single sessions ranging between twelve to twenty minutes. One study used a twenty-five-minute stimulation session two times a day consecutive for five days.



3. Results

3.1. *tDCS and Empathic abilities*

Empathic abilities are motivating pro-social behaviors, such as caring for others, inhibiting aggression, and contribute to morality (Batson, 2012). Both empathic concern and moral reasoning are found to be associated with functioning of the vmPFC. This brain area is connected with emotional systems in the brainstem, amygdala, and hypothalamus and links conceptual moral judgment with emotional processes (Decety & Cowell, 2014). Decades of research have indicated that empathic abilities play a principal role in pro-social behavior, but previous research has mainly focused on the understanding of other's internal state and the perception of other's pain (i.e. pain empathy; Davidov et al., 2013; Patil & Silani, 2014; Smith et al., 2014; Svetlova et al., 2010; Williams et al., 2014) instead of generally increasing empathic abilities. An interesting study by Balconi, & Bortolotti (2012) indicated that stimulating the frontal sensory motor cortex caused individuals to empathize less with the other subjects. Another study (Zheng et al., 2016) reported an increase in altruistic preference and trustworthiness after tDCS on the vmPFC.

3.1.1. *Moral judgment*

Morality is defined as principles and values of how individuals should treat one another and involves concepts such as justice, fairness and rights. It contributes to the norm of social interaction and making the right decisions. Reinforcement of morality can decrease criminal behavior and increase empathic abilities [56]. Furthermore, a recent study (Dubljevic & Racine, 2017) demonstrated that damage of the vmPFC was associated with choosing the 'utilitarian' options in the moral dilemmas. This utilitarian decision-making is found to be associated with antisocial and psychopathic traits (Bartels & Pizarro, 2011; Koenings et al., 2007, 2012).

Recent studies (Colzato et al., 2016; Darby & Pascual-Leone, 2017; Sellaro et al., 2016) have shown that stimulating the DLPFC through tDCS can enhance empathy and increase the feeling of morality. Other work (Maréchal et al., 2017) showed that honesty can be increased with tDCS over the DLPFC, and cheating can be significantly decreased. Another study (Kuehne et al., 2015) showed that moral decision-making is not a permanent individual trait but can be manipulated, by showing that tDCS over the left DLPFC changed moral judgments to more non-utilitarian actions.

3.1.2. *Norm Compliance*

Other studies have provided evidence that anodal tDCS over the left PFC leads to more social norm compliance (Cohen, 2015; Fertoni & Miniussi, 2016; Ruff et al., 2013). On the contrary, a recent study (Gross et al., 2018) showed that cathodal tDCS on the left PFC increased individuals rule-following whereas anodal tDCS led individuals to violate more rules. In addition, Ruff and colleagues (2013) reported that stimulating the left PFC with tDCS can enhance voluntary and sanction-induced social norm compliance, which can be of major importance not only on a social level, but also to decrease criminal activity (Raine & Yang, 2006). Furthermore, Knoch and colleagues (2008) reported that cathodal tDCS over the right PFC leads to reduced propensity to punish unfair behavior.

3.1.3. *Self-perception*

Furthermore, another study (Rêgo et al., 2015) showed that anodal tDCS of the DLPFC reduced hostility, sadness and self-pain perception. Badran and colleagues (2017) showed that tDCS in combination with mindfulness meditation increased mood and mindfulness, effects that lasted even after eight weeks. Of particular importance for empathy, is the self-construal concept (i.e. the way individuals perceive the self relative to others). Martin and colleagues (2017) proved that anodal tDCS increased the salience of others, integrating information more towards the 'other' than to the 'self'. Another study (Santesteban et al., 2012) showed that anodal tDCS on the right temporoparietal junction (TPJ) enhanced social ability, by increasing the self-other awareness. Riva and colleagues (Riva et al., 2015) tested anodal tDCS on the vLPFC in social excluded patients and found a decrease in feelings of hurt and social pain after treatment.

3.1.4. *Pain empathy*

Wang and colleagues (2016) replicated the previous studies (Conti & Nakamura-Palacios, 2014; Davidov et al., 2013) and again found that anodal tDCS on the DLPFC enhanced empathy for pain. Furthermore, contrary to increasing empathy, inhibiting the frontal sensory motor cortex makes subjects less empathic with others [84]. In addition, Coll and colleagues (2017) showed that cathodal tDCS over the right temporo-parietal junction led to subjects perceiving others pain as less intense, and reported a decrease in event-related potentials (ERP) when participants were watching facial expressions of pain. Furthermore, a study (Feesser et al., 2014) showed that anodal tDCS over the DLPFC increases cognitive control, leading to the reappraisal of negative emotions in relation to empathy for pain. An overview of studies using tDCS to increase empathy or emotional control can be found in Table 1, all studies mentioned in Table 1 use a healthy population sample.



Table 1. Studies using tDCS to modulate empathic abilities.

Anode=excitatory, Cathode=Inhibitory, ERP=Event-Related Potential, DLPFC=Dorsolateral PFC, vmPFC=Ventromedial PFC, dmPFC=Dorsomedial PFC, VLPFC=Ventrolateral PFC, LPFC=Lateral PFC.

Study	Emotional concept	Anode electrode position	Cathode electrode position	Current strength (mA)	Duration, number & frequency	Sample size & design	Outcome
Marechal et al. (2017)	Social norm	Right DLPFC	Left DLPFC	1,5	30 min, Single session	145	Cheating decreased after tDCS and an increase in honesty
Ruff et al. (2013)	Norm compliance	Right LPFC	Left LPFC	N.A.	N.A.	59	tDCS right LPFC can enhance voluntary and sanction induced social norm compliance
Gross et al. (2018)	Pro-social behavior	Right LPFC/ Left LPFC	Left LPFC/ Right LPFC	2	30 min, Single session	103	Stimulating right LPFC led to increase in less sensitive consequences in light of internal goals and outcomes
Kuehne et al. (2015)	Moral judgment	Left DLPFC	Right Parietal cortex	2	20 min, 5 days of 1 session	24	Shifted participants' preference towards non-utilitarian actions influencing moral decision-making.
Régo et al. (2015)	Pain Empathy	Right DLPFC	Left DLPFC	2	5 min, Single session	24	Decrease of hostility, sadness & self-pain perception
Martin et al. (2017)	Saliency Others	DMPPC	DMPPC	1	20 min, Single session	40	Increased saliency of others, integrating info more to the 'other' than the 'self'
Riva et al. (2015)	Social Pain	Right VLPFC	Left Supra-orbital area	2	20 min, Single session	80	Less social pain and feelings of hurt
Knoch et al. (2008)	Moral judgment	Left Orbit	Right DLPFC	1,5	13 min, Single session	64	Cathodal tDCS on right DLPFC reduces pt propensity to punish unfair behavior
Wang et al. (2014)	Pain Empathy	Left DLPFC	Left DLPFC	2	5 min, Single session	27	Increase of pain empathy after anodal tDCS
Coll et al. (2017)	Pain Empathy	Right temporo-parietal junction	Right temporo-parietal junction	2	20 min, Single session	48	Cathodal tDCS led to subjects feeling less empathized with the other, feeling their pain as less intense. Decreased ERP to facial expressions of pain
Fumagalli (2019)	Norm compliance	vmPFC	vmPFC/Occipital	2	20 min, Single session	78	Anodal tDCS increased the utilitarian responses, but only in females
Zheng et al. (2016)	Trust/ Social norm	vmPFC	vmPFC	2	20 min, Single session	60	Altruistic preference + trustworthiness higher after tDCS
Feesser et al. (2014)	Emotion regulation	Right DLPFC	Right supraorbital	1,5	20 min, Single session	21	Enhanced cognitive control during emotion regulation
Abend et al. (2018)	Emotion regulation	vmPFC	Back of head	1,5	20 min, Single session	19	Reduction of negative emotions

3.2. *tDCS and antisocial behavior*

Different neural substrates have been identified that are related to impulsive, antisocial and aggressive behavior in adolescents (White et al., 2013), healthy adults (Krämer et al., 2007, 2011; Lotze et al., 2007) and psychopathic individuals (Veit et al., 2010). These substrates include brain regions within the prefrontal cortex, cingulate cortex, insular cortex, striatal areas and the amygdala (Krämer et al., 2007, 2011; Veit et al., 2010; White et al., 2013). In the following paragraphs the different concepts of tDCS to modulate antisocial behavior will be described.



3.2.1. *Aggressive behavior*

Several studies confirmed that the PFC inhibits aggressive behavior (Anderson et al., 1999b; Blair & Cipolotti, 2000; Halasz et al., 2006; Lotze et al., 2007; Raine et al., 1997, 1998). Interestingly, the work of Peterson and colleagues (2008) shows that increased activation in the left PFC rather than the right PFC is involved in activating behavioral aggression. Based on the previous studies (Hortensius et al., 2012; Peterson et al., 2008) that proved that tDCS applied on the left DLPFC leads to increased aggressive behavior, Dambacher and colleagues (2015) used anodal tDCS over the right DLPFC to induce the right frontal hemispheric dominance and found significant results in decreasing proactive aggression in male individuals.

A study conducted by Juan and Muggleton (2012) showed that cathodal tDCS on the right DLPFC could elevate impulsivity and reduce inhibition. In addition, the study of Hortensius (2012) used anodal tDCS over the right DLPFC and reported a decrease in proactive aggression in men. These results support earlier studies (Peterson et al., 2008) that anger-related relative left frontal cortical activation correlates with behavioral aggression. A recent study of Choy and colleagues (2018) showed that after anodal tDCS stimulation of the PFC, an increase in activity in the prefrontal cortex can reduce intentions to commit aggression and enhance perceptions of the moral wrongfulness of the aggressive acts.

3.2.2. *Social exclusion*

Furthermore, Riva and colleagues (Riva et al., 2015) showed in their study the important relationship between social exclusion and aggressive behavior, and reported that tDCS anodal on the right vLPFC can reduce the feeling of social exclusion and the aggressive behavior that follows this exclusion, by regulating the functional cortical region that controls aggression (Nitsche & Paulhus, 2000). A more recent study (Riva et al., 2017) indicated that tDCS anodal over the right vLPFC could reduce unprovoked aggressive behavior that occurs from playing violent videogames.

3.2.3. Risk taking

Risk-taking behavior and risky decision making is linked to an increased vulnerability for addictive pathological behavior (Epstein et al., 2007) and previous research shows that inhibiting the DLPFC activity leads to increased risk-taking (Knoch et al., 2006). Other studies (Fecteau et al., 2007a, 2007b) reported that anodal tDCS over the DLPFC effectively decreased risk-taking behavior. In addition, the study of Gilmore (2017) showed that tDCS over the DLPFC reduces risk-taking behavior in veterans with clinically relevant impulsivity. Bechara and colleagues (1994,1996,1997) have collected substantial evidence over the past several years that patients with bilateral lesions of the vmPFC cannot predict positive or negative consequences of their actions, although their behavior gets influenced by immediately available rewards and punishments (Bechara et al., 1994). The patients with the vmPFC lesions fail to display anticipatory electrodermal responses compared to healthy controls when confronted by a risky choice. The healthy controls generate such electrodermal responses even prior to explicitly knowing it was a risky choice (Bechara et al., 1996, 1997). An overview of the studies using tDCS for antisocial behavior can be found in Table 2. Only the study of Gilmore (2017) used a clinical sample (e.g., impulsive veterans), the other studies mentioned in Table 2 use a healthy population sample.



Table 2. Studies using tDCS to modulate externalizing behavior.
Anode= Excitatory electrode, Cathode= Inhibitory electrode, DLPFC= Dorsolateral Prefrontal Cortex, VLPFC= Ventrolateral Prefrontal Cortex

Study	Behavior	Anode electrode position	Cathode electrode position	Current strength (mA)	Duration, number & frequency	Sample size	Outcome
Hortensius et al. (2012)	Behavioral Aggression	Left + Right DLPFC	Left + Right DLPFC	2	15 min, Single Session	80	Anodal tDCS on the left DLPFC had an increase in aggressive behavior when angry
Choy et al. (2018)	Behavioral Aggression	Left + Right DLPFC	Left + Right DLPFC	2	20 min, Single Session	86	Reduced intentions to commit aggression and enhance perceptions of the moral wrongfulness of the aggressive acts
Dambacher et al. (2015)	Proactive Aggression	Right DLPFC	Above left eyebrow	2	12,5 min, Single Session	32	Reduced proactive aggression in men after anodal tDCS on right DLPFC
Riva et al. (2015)	Behavioral Aggression	Right VLPFC	Supraorbital region	1,5 mA	20 min, Single Session	80	Aggression decreased following social exclusion
Riva et al. (2017)	Unprovoked Aggression	Right VLPFC	Supraorbital region	1,5 mA	20 min, Single Session	80	Reduced unprovoked aggression in violent game-players.
Fecteau et al. (2007a)	Risk Taking	Right DLPFC	Left DLPFC	2 mA	13 min, Single Session	36	Bilateral tDCS causes behavioral changes in risk-taking. Decreased risky behavior
Fecteau et al. (2007b)	Risk Taking	Right DLPFC	Left DLPFC	2 mA	13 min, Single Session	36	Bilateral tDCS causes behavioral changes in risk-taking. Decreased risky behavior
Gilmore et al. (2017)	Risk Taking	Right PFC	Left PFC	2 mA	25 min, 2 times a day 10 sessions total.	30	tDCS of the DLPFC reduced risk-taking behavior in veterans with clinical relevant impulsivity

4. Discussion

This literature review was conducted to investigate whether tDCS could be an effective tool for a new intervention to increase empathic abilities and decrease violent behavior. Empathy is crucial for social relationships, meaning that a deficit in empathic ability could lead to antisocial and deviant behavior, and with that, a higher risk of violence (Preller et al., 2014). The reviewed studies show some evidence, that empathic abilities can be increased with anodal tDCS over the PFC. These studies mainly focused the vmPFC as target region, where anodal stimulation was found to increase empathic abilities. However, other regions of the brain were also successfully targeted in some studies, for example the DLPFC and TPJ when mentalizing abilities were concerned (Coll et al., 2017; Feeser et al., 2014; Kuehne et al., 2015; Maréchal et al., 2017; Rêgo et al., 2015; Wang et al., 2014). The previously mentioned studies confirm the importance of the DLPFC and vmPFC in regulating (pain) empathy, moral judgment, self-perception and norm compliance. All in all, these findings provide a starting point for the use of tDCS as an intervention to increase empathic abilities in forensic and patient populations. This review also investigated the effect of tDCS on violent behavior and general externalizing (for an overview see Table 2). Literature shows that tDCS can influence aggressive and violent behavior, with studies mainly targeting DLPFC because of its involvement in cognitive and inhibitory control. Overall, results showed that anodal stimulation over the DLPFC seems able to decrease impulsivity and aggression. The previously mentioned studies confirm the importance of the DLPFC and vmPFC in regulating externalizing behavior. The studies provide evidence for the use of tDCS as an intervention to decrease aggression in individuals, reduce the feeling of social exclusion and reduce risk-taking behavior. Although the majority of studies using tDCS in psychiatric disorders target the DLPFC, the current findings support the importance of the vmPFC as a target region of interest because of its importance in various processes, including valued based learning, decision-making and empathic abilities (Anderson et al., 1999b; Damasio, 1994; Hiser & Koenings, 2018; Janowski et al., 2013; Mathur et al., 2010; Waytz et al., 2012; Zheng et al., 2016;). Enhancing empathic abilities and morality could subsequently decrease violence in aggressive individuals. Using tDCS as a treatment intervention, activity in brain areas such as the vmPFC can be restored so that functionality and connectivity in relevant networks becomes effective again.

Although the previously mentioned studies indicate that tDCS could be a promising intervention to increase empathic abilities and reduce antisocial behavior, results still vary on different parameters. Intensity, duration, electrode position, target location and session repetition timing could explain the inconsistent results provided by tDCS trials in pathological conditions (Lefaucheur et al., 2017). These factors should be taken into consideration when designing stimulation protocols and interpreting the effects of tDCS when applied as an intervention for neuropsychiatric diseases. Furthermore, in the studies that modulate empathic abilities some studies use a small sample size with just a single session of tDCS and no follow up. For instance, many studies (Kuehne et al., 2015; Rêgo et al., 2015; Feeser et al., 2014; Wang et al., 2014) report results on singles sessions with not more then 25 participants in their sample. The duration, frequency and current strength differ over the studies mentioned in this review. For instance, in the study of Gross and colleagues (2018) they found that cathodal tDCS increased the rule-following of the participants but on the contrary, anodal tDCS led participants to specifically violate more often the rules they choose freely. Brain stimulation over the rLPFC thus did not simply increase or decrease selfishness, but showed more evidence in to the role of the LPFC.

Nevertheless, even though researchers recently have an increased interest in the advances of using tDCS as an intervention procedure and the process is gradually adding in the knowledge about neuromodulation, the precise neurobiological mechanisms underlying tDCS remain insufficiently understood (Chrysikou et al., 2017). The replication of the findings reported in the existing studies and the specification of the precise conditions under which positive tDCS effects can be obtained is still a concern in the field (Mancuso et al., 2016). Overall, there is a great deal of variability in the outcomes that may be largely attributed to small and heterogeneous sample sizes, diversity across laboratories and countries, target locations and duration of the sessions. These limitations are intensified by the lack of understanding of the precise effects of a given tDCS protocol on the brain over different durations and cognitive tasks. Future research should focus not only on the effectiveness of tDCS on overt treatment outcome, but also on the long- lasting effects of tDCS on brain functionality (e.g., functional connectivity, brain network properties).



5. Conclusions

Literature collected so far support the application of tDCS as a potential tool to increase empathic abilities and reduce violence in forensic patients, especially when targeting vmPFC. Further research on the vast parameter space of tCS (e.g., stimulation intensity, electrode location, polarity) is needed to consolidate tDCS as a tool in forensic science.

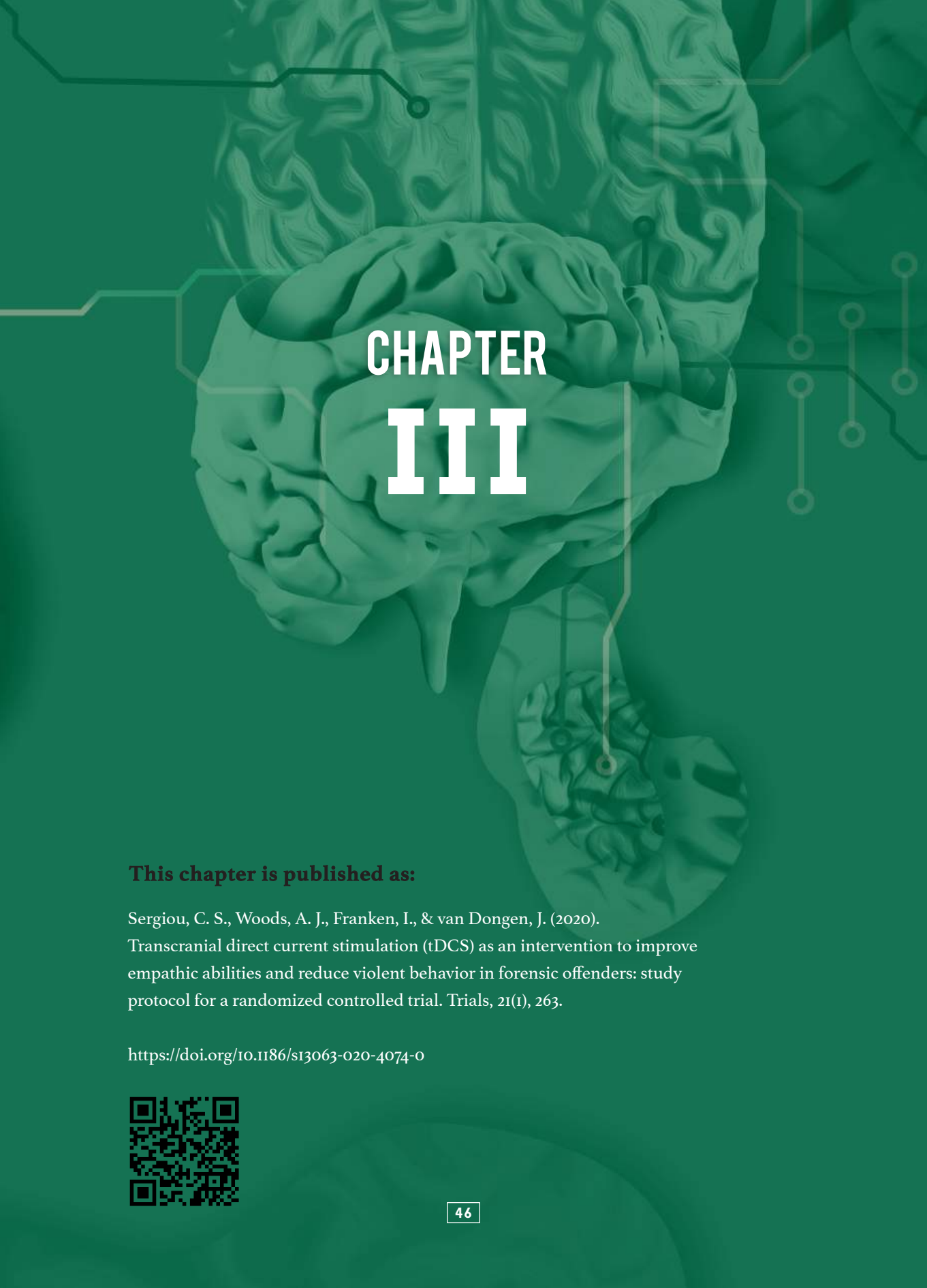
Conflicts of interest

None.



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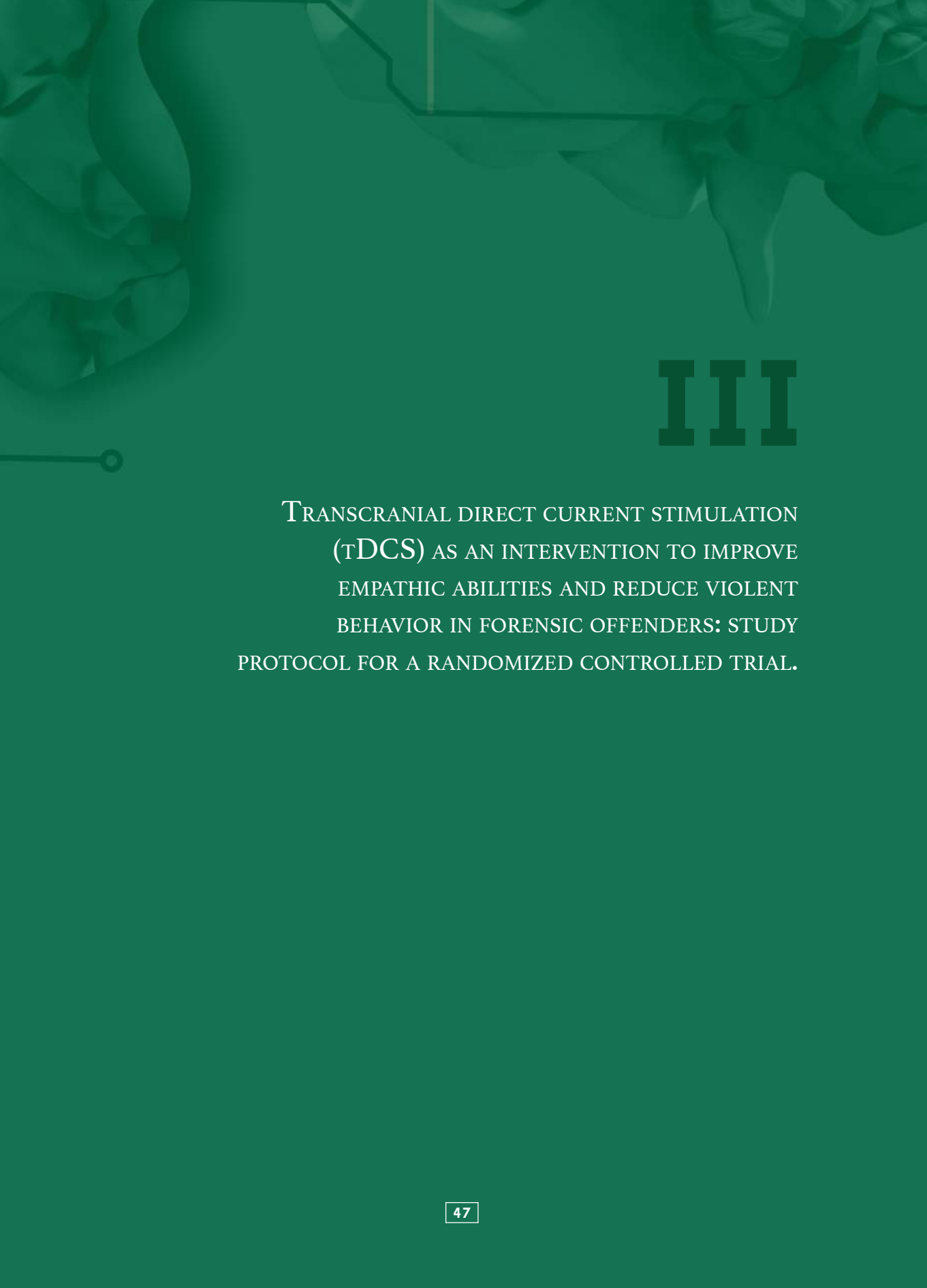
CHAPTER III

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III

TRANSCRANIAL DIRECT CURRENT STIMULATION
(tDCS) AS AN INTERVENTION TO IMPROVE
EMPATHIC ABILITIES AND REDUCE VIOLENT
BEHAVIOR IN FORENSIC OFFENDERS: STUDY
PROTOCOL FOR A RANDOMIZED CONTROLLED TRIAL.

ABSTRACT

Background:

Recent studies show that changes in one of the brain areas related to empathic abilities (i.e. the Ventromedial Prefrontal Cortex (vmPFC)) plays an important role in violent behavior in abusers of alcohol and cocaine. According to the models of James Blair, empathy is a potential inhibitor of violent behavior. Individuals with less empathic abilities may be less susceptible and motivated to inhibit violent behavior, which causes a higher risk of violence. Recent neuroscientific research shows that modulating (stimulation or inhibition) certain brain areas could be a promising new intervention for substance abuse and to reduce violent behavior, such as the neurostimulation technique Transcranial Direct Current Stimulation (tDCS). This study aims to investigate tDCS as an intervention to increase empathic abilities and reduce violent behavior in forensic substance use offenders.

Methods/Design:

A total sample of 50 male forensic substance abuse patients (25 active + 25 sham stimulation) will be tested in a double-blind placebo-controlled study, from which half of the patients will receive an active stimulation + treatment as usual (TAU) and the other half will receive a sham stimulation (placebo) + TAU. The patients in the active condition will receive multichannel tDCS stimulation targeting bilateral vmPFC two times a day for 20 minutes for five consecutive days. Before and after the stimulation period, the patients will complete self-report measurements, perform the Point Subtraction Aggression Paradigm (PSAP) and a passive viewing empathy task. Resting state electroencephalography (rsEEG) will be measured before and after the treatment period. A follow-up will be conducted after six months. Primary outcome is to investigate multichannel tDCS as a new intervention to increase empathic abilities and wit that reduce violent behavior in offenders with substance abuse problems. In addition, it will be studied whether electrophysiological responses in the brain are affected by the tDCS intervention. Lastly the effects of tDCS on reducing craving will be investigated.

Discussion:

This study protocol describes the design of an intervention using multichannel tDCS targeting the vmPFC. This study will explore the opportunities to introduce a new intervention to improve empathic abilities and reduce violence in forensic substance use offenders. In addition, this study may help to reduce craving. The study will contribute to knowledge about increasing empathic abilities and the functioning of the vmPFC. Specifically, this study may give insight in how to implement the tDCS intervention in the setting of daily clinical practice for this complex, multiple problem target group and with that contribute to reduction of recidivism.

Trial registration Dutch Trial Register, identifier: NTR7701. Registered on 12 January 2019; prospectively registered before the recruitment phase. Recruitment started on the 1st of February 2019 and approximately will be finished in the winter of 2019. <https://www.trialregister.nl/trial/7459>.|Protocol version 1. 22th of May 2019.

Keywords:

Transcranial Direct Current Stimulation (tDCS), Empathy, Violent behavior, Substance use, Forensic offenders, Recidivism, Effectiveness, Ventromedial Prefrontal Cortex (vmPFC).



Background

Repeatedly using substances has been found to lead to neuro-adaptations in the ventral striatum and ventral tegmental areas and with that decreased dopamine secretion (Volkow et al., 2009). Decreased dopamine secretion leads to a higher craving for substances and increased saliency for addictive cues (Everitt & Robbins et al., 2005; Hyman, 2005; Perry & Carrol, 2008). Impaired functioning of the Dorsolateral Prefrontal Cortex (DLPFC) in patients with substance abuse disorders, are underlying to a diminished cognitive and inhibitory control and increase the tendency to relapse and maintain in addictive behaviors (Capriles et al., 2003; Garavan et al., 2007; Janes et al., 2010; Paulhus et al., 2005; Kosten et al., 2006; Van Holst & Schilt, 2011; Volkow et al., 1993, 2001, 2007). Recent studies show that changes in the brain areas related to less empathic ability (ea. the Ventromedial Prefrontal Cortex (vmPFC)) in abusers of alcohol and cocaine plays an important role in violent behavior (Gizewski et al., 2012; Peller et al., 2014). Preller and colleagues (2014) found in their study that cocaine users show deficits in emotional empathy and that substance abuse patients are less emotionally responsive to the emotions of other individuals and their mental state.

Empathy is crucial for social enhancement, social interactions and relationships, for our emotional and social life (Bernhardt & Singer, 2012). A deficit in empathic ability could lead to antisocial and deviant behavior and with that a higher risk of aggression (Preller et al., 2014). Aggression is stated here as behavior that is mostly defined by any behavior that is intended to harm someone who is motivated to avoid being harmed (Baron et al., 1994). As said before, antisocial behavior, especially aggression, is associated with dysfunctions in the prefrontal cortex (Anderson et al., 1999b; Blair, 2004; Calzada-Reyers et al., 2013, 2016; Krämer et al., 2009; Raine, 1993; Raine et al., 2000). In addition, research has highlighted the importance of executive functions; the 'higher' cognitive functions that are controlled by the prefrontal cortex in control of aggression (Blair et al., 2006; Giancola, 2004; Hoaken et al., 2003; Morgan & Lillienfeld, 2000). According to the models of James Blair (2001, 2005; Blair et al., 2006) violent behavior is inhibited by empathy. Individuals with less empathic abilities may be less susceptible and motivated to inhibited violent behavior, which causes a higher risk of violence.

The Role of the Prefrontal Cortex

Several studies show evidence that impaired prefrontal cortex areas lead to the emergence of aggressive behavior. Most notably are the impairments in the vmPFC (e.g. emotion regulation, moral decision-making), and the DLPFC (e.g. disinhibition and impulsiveness), which is associated with aggression and violent behavior (Blair et al., 2006; Giancola, 2004; Hoaken et al., 2003; Morgan & Lillienfeld, 2000). These dysfunctional prefrontal cortex areas are inducing psychopathic traits such as blunted emotions and a lack of empathy (Damasio et al., 1994), impaired perspective taking with increased egocentrism and rigidity (Anderson et al., 1999b; Koenings et al., 2007; Price et al., 1990). Research found that children with psychopathic traits showed abnormal activity in the ventromedial prefrontal cortex during a response reversal task in comparison to children without these traits (Finger et al., 2008). In addition, Raine and colleagues (Raine et al., 1993; 1998) demonstrated the relationship between violence and empathy and argued that abnormal brain structures of the vmPFC cause the most extreme form of empathic inability named 'the psychopathic predatory violence'. This psychopathic predatory violence correlates with the lack of empathy, impaired moral judgment (Raine et al., 2003) and hyper metabolism (Raine et al., 1998), which is associated with increased aggressive impulses. Figure 1 shows the hypothesized relationship between the impaired brain areas and the subsequent deficits in empathy and violent behavior.

Although, the DLPFC modulates cognitive control and response inhibition which are associated with aggression and violent behavior (Blair et al., 2006; Giancola, 2004; Hoaken et al., 2003; Morgan & Lillienfeld, 2000), in this study we focus on the vmPFC.



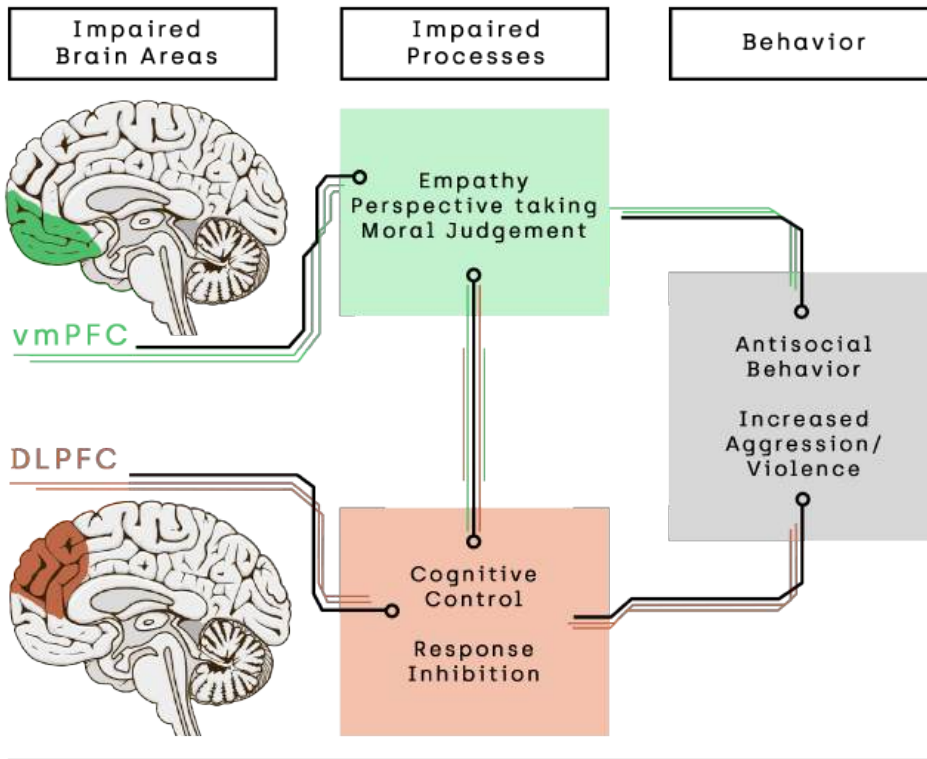


Figure 1.
 Relationship between the impaired brain areas, impaired processes and behavior.
 The highlighted brain area, impaired process and behavior are discussed in this paper.
 *DLPFC= Dorsolateral Prefrontal Cortex, vmPFC= Ventromedial Prefrontal Cortex

Ventromedial Prefrontal Cortex (vmPFC)

The vmPFC is particularly relevant for empathic abilities and antisocial behavior (Anderson et al., 2006; Blair, 2001; Zheng et al., 2016). Studies using functional neuroimaging (i.e. fMRI) have shown that neural activity in the vmPFC predicts empathic abilities and altruistic motivation (Mathur et al., 2010; Waytz et al., 2012). Another longitudinal study (Anderson et al., 1999b) demonstrated that lesions damaging the vmPFC, occurring in the first sixteen months of a human life, result in lifelong psychopathic antisocial traits and will also lead to impaired social and moral reasoning. In addition, damage to the vmPFC is associated with poor decision-making in antisocial behavior (Bechara et al., 1997; Damasio et al., 1990). Furthermore, a recent study (Gilam et al., 2018) combining tDCS with fMRI demonstrated that anodal tDCS placed on the forehead led to increased vmPFC activity and decreased negative emotions. Taken together, these studies suggest a potential link between vmPFC functioning and anger regulation.



Transcranial Direct Current Stimulation (tDCS)

tDCS is a non-invasive neuromodulation technique that modulates the brain region of interest by increasing or decreasing neuronal excitability through constant, low direct current electrodes. tDCS has been proven to be an effective intervention to modify brain activity (Barr et al., 2008; George & Aston-Jones, 2010; DaSilva et al., 2013) and has been investigated in many different disorders (Brunelin et al., 2012; Kalu et al., 2012; Hermann et al., 2017).

In this current study, the effectiveness of tDCS as a new intervention to increase empathy and reduce violent behavior in substance abuse offenders will be investigated. Through modulating (stimulating and inhibiting) certain areas of the brain, tDCS causes a change in the function of the brain, due to an increase in susceptibility to generate and facilitate brain related electrical impulses. This susceptibility is achieved through repeated sessions of brain stimulation and is thought to produce Long Term Potentiation (LTP)-like 'learning' in stimulated neurons. Functional alterations in the brain due to the long abuse of substances are hypothesized to improve by the intervention with tDCS (Boggio et al., 2008b; Batista et al., 2015; Camprodon et al., 2007; Conti et al., 2014; den Uyl et al., 2015; Falcone et al., 2016; Fregni et al., 2008a, 2008b; Jansen et al., 2013; Klauss et al., 2014; Politi et al., 2008; Shahbabaie et al., 2014; Wietschorke et al., 2016) and will reduce craving.

Several studies found that through anodal stimulation of the PFC with tDCS, emotional processes can be influenced. tDCS can modulate emotional pain (Rêgo et al., 2015; Coll et al., 2016) and enhance pain empathy (Coll et al., 2016). Interestingly, studies (Darby et al., 2017; Feeser et al., 2014; Sellaro et al., 2017) suggest that anodal tDCS of the PFC can enhance empathy and increase the feeling of morality. Although there seems to be a clear association between empathy and vmPFC function, only a few studies have investigated modulation of the vmPFC. Abend and colleagues (2018) show in their study combining tDCS and fMRI that stimulating with tDCS demonstrated increased emotion-related activation in the vmPFC. Two studies show that anodal stimulation of the (right) vmPFC increases empathic ability and morality (Abend et al., 2017; Feeser et al., 2014). Furthermore, the study of Gilam and colleagues (2018) used tDCS simultaneously with fMRI and demonstrates that activity of the vmPFC was increased during active compared to sham stimulation.

In addition to empathy, also antisocial behavior, including aggression, are associated with dysfunctions in the prefrontal cortex (Anderson et al., 1999b; Blair, 2004; Calzada-Reyers et al., 2013, 2016; Krämer et al., 2009; Raine, 1993; Raine et al., 2000). Research shows that studies targeting the PFC with anodal tDCS can reduce social exclusion and the aggressive behavior that emerges from this exclusion (Riva et al., 2015) can reduce unprovoked aggressive behavior (Riva et al., 2017) and reduce the intentions of aggressive behavior (Choy et al., 2018; Dambacher et al., 2015). Based on the earlier mentioned theory of Blair, linking impaired empathic abilities to violence, and the model demonstrated (Figure 1), it could be proposed that the vmPFC plays a crucial role in modulating both empathic abilities and thereby also in violent behavior. Previous mentioned study of Gilam and colleagues (2018) also demonstrates the relationship between the vmPFC and violent behavior and showed a decrease in violent behavior in the first session after active tDCS.

In addition to aberrant functioning of specific brain areas, recent research (van Dongen et al., 2018; Tillem et al., 2018) also shows that functional connectivity of brain areas is also affected in individuals with less empathic abilities. Therefore, it is important for the evaluation of the effectiveness of a tDCS intervention, to determine how the resting-state connectivity of the patients, measured with resting state electroencephalography (rsEEG), changes from pre-to post-test to the intervention. Prior research demonstrates that tDCS can alter functional connectivity and that connectivity change is related to the treatment response (i.e., effectiveness; Santarnecchi et al., 2015). To optimize the intervention and increase current focality we use a newer development of the original. This technique, called high-definition tDCS (HD-tDCS or multichannel tDCS) uses gel-based electrodes similar to those used in EEG (Minhas et al., 2010) and is more precise in targeting the brain area of interest.

Therefore, in this study we will investigate the idea that multichannel tDCS applied over the vmPFC will increase empathic abilities and subsequently reduce violence risk.



Aims of the study

The aim of this study is to investigate the effectiveness of tDCS in increasing empathy and reducing violent behavior in offenders with substance abuse problems.

The following research questions will be addressed in this study:

1. Does stimulation with multichannel tDCS targeting the vmPFC increase empathic abilities in forensic substance abuse patients during an empathy task from pre-to post intervention?
2. Does stimulation with multichannel tDCS targeting the vmPFC reduce aggressive responding and violence risk in forensic substance abuse patients from pre-to post intervention?
3. Does multichannel tDCS targeting the vmPFC reduce craving in forensic substance abuse patients?
4. Does multichannel tDCS targeting the vmPFC affect electrophysiological responding in the brain measured through EEG from pre-to post intervention?

Outcomes will contribute to the development of more effective diagnostics and treatment of substance use patients. Furthermore, these data will potentially contribute to improving treatment through increasing understanding of specific targets for treatment interventions. In addition, outcomes may provide better insight in the functioning of the vmPFC and the relationship between empathy and violence risk.

Exclusion

Subjects meeting any of the following criteria will be excluded from participation in this study: major neurological conditions (e.g. traumatic brain injury), or major mental disorders (i.e. major depression, psychotic symptoms). Also, patients taking antipsychotic or other severe medication cannot participate in the study.

Procedure/Design

In this double blind, placebo-controlled study, a total of 50 male participants between the age of 18-60 will be randomly assigned to either the active condition, or the sham condition. Eligible participants will be given written and oral information about the study and will be invited to participate. After informed consent they will participate at the forensic clinic where they are admitted and all the data collected will be anonymous linked to their participant number.

During the first session the following self-reported measures will be assessed. Psychopathic traits will be assessed using the short form of the Self-Report Psychopathy Scale Short-Form (SRP-SF; Gordts et al., 2017) aggression will be measured using the Dutch Reactive and Proactive Aggression Questionnaire (RPQ; Cima et al., 2013) empathy will be measured using the Dutch Interpersonal Reactivity Index (IRI; de Corte et al., 2007). Alexithymia is a supposed cause of disruption in the accurate representation interpretation of affective states, causing empathy deficits. Therefore, alexithymia will be assessed using the Toronto Alexithymia Scale (TAS-20; Bagby et al., 1994). Risky, impulsive and Self-destructive behavior is measured with the Dutch version of the Risky, Impulsive, and Self-destructive behavior Questionnaire (RISQ; van Dongen et al., 2018). To assess impulsive behaviour the Behavioral Impulsivity Scale (BISS-II; Patton et al., 1995) is used. As a screener for Alcohol use the Alcohol Use Disorder Identification test (AUDIT; Babor et al., 2001) is used. As a screener for drugs the Drug Use Disorder Identification Test (DUDIT; Berman et al., 2005) will be used. To measure craving for substance the Obsessive-Compulsive Drug Use Scale (OCDUS-version Cocaine; Franken et al., 2002) will be used. The craving for alcohol will be measured using the Obsessive- Compulsive Drinking Scale (OCDS; Schippers et al., 1997). In addition; EEG will be measured during a resting-state task and during a Passive-viewing Empathy task (Dambacher et al., 2015). Following the participants will perform a rating viewing Empathy task (van Dongen et al., 2018). To measure behavioral indices of the level of aggression the patients will perform in the Point Subtraction Aggression Paradigm (PSAP; Cherek et al., 1992) will be used. After this, a series of ten tDCS sessions (two per day for five consecutive days) will follow. After every tDCS session the participants will fill in a questionnaire that assesses side effects. On Monday, after the last modulation on Friday, the patients will receive a post-intervention in which the rsEEG and the tasks (empathy task and PSAP task) again will be conducted. As a follow up, risk assessment scores of the participant will be obtained after six months to see whether risk reduction is maintained together with one last EEG session with the PSAP task.



Methods/design

Sample

A total sample of 50 male forensic substance abuse patients (25 active + 25 sham stimulation) will be tested in a double-blind placebo-controlled study, from which half of the patients will receive an active stimulation+ treatment as usual (TAU) and the other half will receive a sham stimulation (placebo) + TAU.

Inclusion

In order to be eligible to participate in this study, a subject must meet all of the following criteria: male, age 18-60, good understanding of the Dutch language, diagnosed with an alcohol and/or cocaine SUD according to the DSM-5. The patients have to be abstinent and have an index offense in violence category listed in the HKT-R. These criteria have been selected due to the fact that male subjects will show more aggression on the PSAP task and show more violent behavior in general than females.

Setting

The study will be carried out in two forensic institutions. The sample will be recruited from the Forensische Verslavings Kliniek (FVK), the forensic addiction clinic of Bouman, Antes. The institution is located in Rotterdam, The Netherlands. The sample size is based on other studies, we will operate according to the 'Evidence-based guidelines on the therapeutic use of transcranial direct current stimulation (tDCS)' published in 2017 (LeFaucheur et al., 2017), and seen as achievable, due to the fact that the applicant is currently obtaining research at FVK Bouman Antes, which will make the inclusion of the patients more feasible.

Overview Procedure

- 1). Inclusion of patients after informed consent.
- 2). Collecting information from patients regarding diagnosis, substance abuse, demographic information and HKT-R.
- 3). Pre-intervention in which the patients will receive a resting-state EEG (rsEEG) and perform the aggression task (PSAP) and the empathy task (victims of aggression). In addition, the patients will fill in the self-report questionnaires.
- 4). Intervention multichannel tDCS+TAU or Sham+TAU: twenty minutes of anodal stimulation of the vmPFC and cathodal stimulation of the left supraorbital area with multichannel tDCS, two times daily for a period of 5 consecutive days.
- 5). Post-intervention where the patients receive a rsEEG and will perform the aggression task (PSAP) and the empathy task (victims of aggression), and fill out the self-report questionnaires again.
- 6). Outcome variables: The results on the aggression task and empathy task as well as the score on the HKT-R and the self-report questionnaires will be lower with respect to the results on the pre-test. Furthermore, it will be investigated whether the functional connectivity (EEG) has changed at the post-test in contrast to the pre-test and if this differs within the two conditions (tDCS vs. Sham).
- 7). A follow-up after six months where the patients will perform the aggression task and the empathy task one more time and the HKT-R will be administered.

See Figure 2 and 3 for a detailed overview over the procedure of the trial.

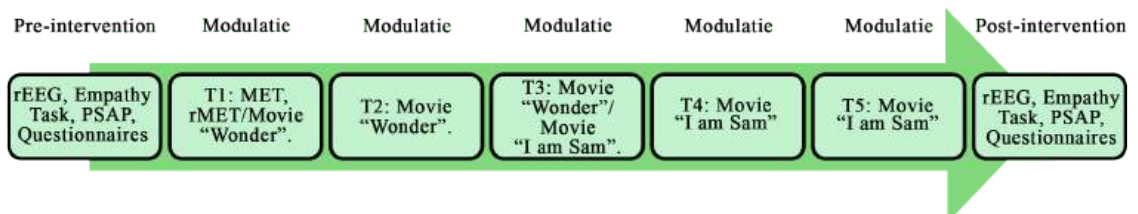


Figure 2.
Flowchart of the trial

Multiple studies (Baxter et al., 2017; Buch et al., 2017; Fisher et al., 2017; Fox et al., 2014;) have concluded that activating the brain state of the area of interest during stimulation, increases the effect of the modulation and contribute to optimize the intervention. An interesting study of Nissim and colleagues (2019) demonstrated with fMRI that the optimal gains from using tDCS can be realized by simultaneously using behavior and modulation for stimulating neural networks. Therefore, to optimize our intervention and increase the activity of the vmPFC and the empathic abilities the participants will be occupied with tasks and movies that trigger these brain states. On the first day, the participants will complete the Multifaceted Empathy Task (MET; Dziobek et al., 2008) and the Reading the Mind in the Eyes test (rMET; Baron-Cohen et al., 2001) to actively enhance their perspective taking and emotion recognition. When finished, they will watch the movie 'Wonder' (2017), for the first part of the stimulation week. For the remaining sessions they will watch the movie 'I am Sam' (2001). The two movies are enhancing the empathic abilities and perspective taking, and therefore will contribute to the optimization of brain state.

Primary Objective

The primary objective is to investigate tDCS as a new intervention to increase empathy and reduce violent behavior in offenders with substance abuse problems. This will be measured with the results of the empathy and aggression task from pre-to posttest. In addition, it will be studied whether electrophysiological responses in the brain are affected by tDCS intervention, this will be demonstrated through comparing the EEG from pre-to posttest. A mixed design will be carried out to test whether ERP amplitudes pre- post intervention differs between the active- and sham group.

Secondary Objective(s)

The secondary objective is to reduce craving in offenders with substance abuse problems. This will be assessed using the four self-report measurements about alcohol- and drug craving, and to compare the results from pre-to posttest. Other study parameters are the results from the self-report questionnaires. These will be compared with the other variables mentioned before and also compared from pre-to posttest.

	STUDY PERIOD								
	Enrolment	Pre-Intervention	Intervention (Modulation)					Post-Intervention	Follow-Up
TIMEPOINT	-T1	T0	T1	T2	T3	T4	T5	T6	T7
ENROLMENT:									
Eligibility screen	X								
Informed consent	X	X							
HKT-R	X								X
Allocation		X							
INTERVENTIONS:									
tDCS Active Stimulation + TAU			←————→						
Sham Stimulation + TAU			←————→						
MET, rMET/ Movies			←————→						
ASSESSMENTS:									
Resting state EEG		X						X	
Passive Viewing Empathy Task (EEG)		X						X	
PSAP		X						X	X
Self-report Questionnaires		X						X	X

Figure 3.

SPIRIT schedule of enrollment, intervention and assessments



Outcomes

To test the effect of the intervention on aggression the PSAP will be used. As a standard for effectiveness of the reduction of violent behavior next to the PSAP, the results of a violence risk assessment instrument (HKT-R; Spren et al., 2014) will be assessed before the intervention starts. The HKT-R is one of the mandatory risk-assessment tools in forensic institutions.

To measure functional changes in brain function during empathic induction caused by the intervention, rsEEG will be used. The expectation is that patients who receive tDCS intervention will show higher event-related potential (ERP) towards the pictures of the victims after the intervention and compared patients who have received the sham- condition.

After six months, as a standard for effectiveness of the reduction of violent behavior, the results of the aggression task (PSAP) and the risk-assessment tool (HKT-R) will be compared to the results previous to the intervention. The HKT-R risk level will be obtained to see whether there is a longitudinal effect of tDCS on aggression and risk reduction. In addition to check whether the effect of the tDCS intervention is long lasting, the participants will perform the self-report questionnaires, empathy task and the aggression task once more.

Recruitment

The recruitment will be active within the forensic institutions. Patients will be actively approached by the PhD student, or can sign up through a form at the department in which they receive their treatment. Research assistants will assist with the assessments. The entire research team will be trained extensively before they have an active role in the re- search. After a screening for exclusion criteria, the patients will receive more detailed information about the study and can decide to participate definitive. An informed consent will be signed prior to the actual test day.

Conditions

Experimental

The patients participating in the active, experimental condition start on Friday with completing the before mentioned self-report questionnaires. Also, EEG will be measured during a resting-state task (rsEEG) and during the passive viewing empathy task. After disconnecting the EEG device, participants will perform the empathy rating task and the PSAP. On Monday, the participants will start with the tDCS intervention; they will receive two times 20-minute tDCS stimulation each day.

There will be a time interval of 3-4 hours between the sessions, depending on the schedule of the patients. This will last for five consecutive days. One anodal electrode is placed on the position of Fpz and the other five cathodal electrodes are placed on AF3, Fz, AF4, F3, F4, according to the international 10-10 EEG system. The electrodes used are the Pistim EEG & tDCS hybrid electrodes with a 12 mm diameter.

Sham

The patients participating in the sham condition will follow exactly the same routine as the participants in the active condition except they will receive 2 mA during 30 seconds instead of the 20 minute stimulation. This has proven to be effective for blinding as participants habituate to the sensation of stimulation within seconds of current initiation (Gandiga et al., 2006).

Blinding

Participants and experimenters are blind to the tDCS condition allocation. An external researcher is the only one who knows which participant number corresponds with each condition. The principal experimenter, neither the patients nor research assistants know which condition is being executed. The trial established procedures to maintain separation in knowledge between the head researcher and the principal experimenter.

tDCS

For the intervention we will operate according to the 'Evidence-based guidelines on the therapeutic use of transcranial direct current stimulation (tDCS)' (Lefaucheur et al., 2017). Multichannel tDCS will be administered with a CE-certified neurostimulator (StarStim-8, NeuroElectrics) following the protocol (Minhas et al., 2010) the device is preprogrammed for stimulation with 2mA during 20 minutes (experimental condition), or with 2 mA during 30 seconds (sham condition). We program the device for each participant, ensuring that the participant and experimenter are blinded for experimental condition. The experimental treatment is a 20 minutes tDCS session, two times a day for five consecutive days (ten sessions per participant). Multichannel tDCS stimulation will be applied over the vmPFC. Sham condition is the same as the experimental condition with the exception that there will only be a ramp up of the electrical stimulation to mimic the sensation of the stimulation.



Instruments

Passive-viewing Empathy task (van Dongen et al., 2018)

To measure empathic abilities and how it changes between pre-and post-intervention, patients will participate in a passive-viewing empathy task. This measurement is based on previous research (van Dongen et al., 2018). The pictures used in the passive-viewing task are selected based on ratings conducted through an anonymous online study (i.e. Amazon's Mechanical Turk). 188 individuals participated in this study and each individual rated 45 pairs of pictures. These were aggression pictures and matched neutral pictures, both were rated on levels of arousal, aggressiveness and valence. The study resulted in 40 pairs of pictures selected for the empathy tasks. The pictures display scenes with either two males, or one male and one female aged between 20 and 25 years old. The males have a white complexion and the female has a black skin tone. For the aggression pictures, the majority of cases (99%) had a male perpetrator and either a male or a female as the victim. The scenes involved physical, sexual, and verbal aggression. To control for stimulus- related confounding factors, all neutral pictures are carefully matched to the aggressive pictures. The neutral photographs were identical to the aggression-related photographs (pair-wise; i.e. the same persons, same location, same colors, same light), only without the aggressive action. Three types of pictures will be used in the experiment: (1) 40 pictures displaying an interaction between two individuals that is of a violent tone; (2) 40 pictures displaying an interaction between two individuals that is neutral and (3) fifteen pictures displaying neutral objects like a bridge or a lamp (fillers). The fillers will not be used for further analyses. 95 pictures in total will be presented, for 6 seconds with intervals of 1.8 seconds between the pictures, randomly. Participants are instructed to look at each picture passively, because then the automatic neural responding in the brain can be determined (van Dongen et al., 2018).

Rating Empathy task (van Dongen et al., 2018).

Following the Passive Viewing Empathy task, the pictures displaying the neutral and aggressive situations will be presented a second time. Participants are now instructed to rate the pictures by answering four questions. The first question is 'Does this picture give you arousal?' assessed to score arousal on a 9-point Likert scale (1 = no arousal to 9 = very high arousal). The following question was to assess the valence of the picture 'Does this give you a negative or a positive feeling?' on a 9-point Likert scale (1 = negative emotions, 5 = no emotions, 9 = positive emotions). The last two questions assess the measure of state empathy. Lastly, there are two questions concerning empathy (i.e. measure of state empathy). One question assessed to what extent the participant could empathize with the perpetrator and the other question assessed to what extent the participants could

empathize with the victim in the aggressive situations. Both questions will be scored on a 9-point Likert scale (1 = no empathy to 9 = high empathy). The total empathy task (passive viewing and rating) on average lasts 40 minutes (van Dongen et al., 2018).

The Point Subtraction Aggression Paradigm (PSAP; Cherek, 1992).

Although multiple paradigms are used to measure aggression on an experimental manner in the labs, the PSAP-task is known to be one of the best-validated instruments (Geniole et al., 2016). During the PSAP-task participants are playing a game against a (fictive) opponent. The goal of the game is to earn more points than your opponent. Participants are offered 3 choices: 1). Participants can earn points by pressing a 100 times on the '1' button on the keyboard. 2). Stealing points from your opponent by pressing ten times on the '2' button or 3). Guarding your own points as an escape so the opponent cannot steal from you by pressing ten times on the '3' button. When a participant picks option number 2 (aggressive response), the points that they steal will not add up at their own score, but only will subtracted from the score of the opponent. If the (fictive) opponent is stealing points from the participants it will be shown in red letters on the screen, in this way the participant will be 'provoked'. If this provoking leads to the participant pressing the '2' (stealing) button more frequently, this can be seen as reactive aggression. When the participant is not provoked by this event, the aggression can be seen as proactive aggression. In this study the e-prime version of the PSAP consists of three sessions of each twelve minutes. The outcome of the PSAP, the aggressive response, is the amount of '2' responses of the participant. When the '2' response is a result of the provocation (the fictive opponent stealing points), then this indicates a reactive aggression response. If the '2' response is not a reaction on the provocation, then the reaction will be seen as proactive aggression response. Research (Geniole et al., 2016) concludes that the PSAP has more ecological validity than for example the Taylor Aggression Paradigm (TAP) (Taylor, 1967). This is due to the following advantages: First, the PSAP task offers the participant an option to actively withdraw (pressing the '3' key), something that is not included in other paradigm tasks. Secondly, in the PSAP task the aggressive response (pressing the '2' button is not receiving points) and therefore more ecological valid than other tasks that do not have this option. An aggressive reaction is a cost-benefit consideration, so choosing an option to steal points that will not add up to your own score of winning could be considered as a costly option.



Questionnaires

Self-Report Psychopathy Scale (SRP-SF; Gordts et al., 2017)

In order to assess psychopathic traits we use the Dutch short version of the SRP-SF. The SRP-SF consists of a subset of 29 of the 64 original items and is a self-report questionnaire in which participants are asked to rate statements with a 5-point Likert scale. The questionnaire consists of four subscales: Interpersonal manipulation (manipulation and deception), Callous affect (lack of empathy or regret), Instable lifestyle (impulsivity and sensation seeking behavior) and Criminal behavior (delinquency and criminal behavior).

Reactive and Proactive Aggression Questionnaire (RPQ; Cima et al., 2013).

In order to assess aggression we use the Dutch version of the RPQ. The RPQ is a 23-item self-report questionnaire in which the participant has to give a rating on how often this behavior has occurred in the past on a 3-point scale (“never”, “sometimes” or “often”). Next to an overall total aggression score, the test provides two separate measures to proactive aggression (deliberate and planned aggression) and reactive aggression (aggression as a reaction on an unplanned circumstance).

Interpersonal Reactivity Index (IRI; de Corte et al., 2007).

In order to assess empathy we use the Dutch version of the IRI, this is a commonly used self-report instrument designed to assess empathic tendencies. The IRI consists of four separate subscales: Perspective Taking (PT), Fantasy (FS), Empathic Concern (EC), and Personal Distress (PD).

Toronto Alexithymia Scale (TAS-20; Bagby et al., 1994).

In order to assess alexithymia we use the Dutch version of the TAS-20. The TAS-20 is a self-report scale comprised of 20 items. Each item is rated on a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). The TAS20 is a reliable and valid measure of emotion processing in adults that includes a total score and three subscales: Difficulty Identifying Feelings (DIF), Difficulty Describing Feelings (DDF), and Externally-Oriented Thinking (EOT).

Risky, Impulsive, and Self-Destructive Behavior Questionnaire (RISQ; van Dongen et al., 2018).

In order to assess risky and impulsive behavior we use the Dutch version of the RISQ. The RISQ is a 38-item self-report questionnaire-based measure, assessing eight domainspecific factors (measuring drug use, aggression, self-harm, gambling, risky sexual behavior, impulsive eating, heavy alcohol use, and reckless behavior).

Behavioral Impulsivity Scale (BIS-II; Patton et al., 1995).

In order to assess the personality and behavioral construct of impulsivity we use the Dutch version of the BIS-II. The BIS-II is a questionnaire which consists of one of the three second-order facets of impulsivity. The 30-item self-report questionnaire consists of six subscales: attention, (i.e. focusing on current tasks), cognitive instability, (i.e., intruding thoughts), motor impulsiveness, (i.e. acting quickly), perseverance (stable lifestyle), cognitive complexity, (i.e. enjoys mental challenges) and self-control, (i.e. plans and thinks deliberately).

Alcohol Use Disorders Identification Test (AUDIT; Babor et al., 2001).

In order to assess alcohol use we use the Dutch version of the AUDIT. The 10-item AUDIT includes questions to assess alcohol intake (questions 1–3), alcohol dependence (questions 4–6) and alcohol-related problems (questions 7–10). Questions 1–8 are scored from 0 to 4, questions 9 and 10 are scored 0, 2 or 4, resulting in a maximum AUDIT score of 40.

Drug Use Disorder Identification Test (DUDIT; Berman et al., 2005).

In order to assess drug use we use the Dutch version of the DUDIT. The DUDIT is an 11-item screening instrument to assess non-alcohol drug use patterns and various drug-related problems. The items range from on a 5-point Likert scale for the first nine items, the last two items are scored on a 3-point scale. The higher scores suggest more severe drug problems.

Obsessive Compulsive Drug Use Scale (OC-DUS-version Cocaine; Franken et al., 2002)

In order to assess drug craving we use the Dutch version of the OC-DUS. The OC-DUS version Cocaine is a 13-item self-report questionnaire that assesses the inability to control or resist cocaine-related thoughts and behaviors, frequency and impact of thoughts and impulses related to cocaine use, and the degree of interference.

Obsessive Compulsive Drinking Scale (OCDS; Schippers et al., 1997).

In order to assess alcohol craving we use the Dutch version of the OCDS. The 22-item Dutch version of the OCDS was developed to reflect obsession and compulsivity related to craving and drinking behavior. The OCDS has been shown to be specific for the obsessive and compulsive characteristics of drinking-related thought, urges to drink, and the ability to resist those thoughts and urges in alcohol-abusing and alcohol-dependent population.



Statistical analysis

Different General Linear Models (GLM) in SPSS will be carried out to analyze the main parameters. For instance, analyzing the empathy and aggression outcome (empathy ratings and b responses on PSAP task), the outcome variables will be handled as the dependent variables. In addition, the group variable (active vs. sham) will be used as the independent between-group variables, whereas the pre-post time will be included as a within subject independent variable. A mixed design will be carried out in which the pre-post intervention of the experimental group will be compared with the pre-post intervention aggression outcomes of the sham-group of patients. The mixed design will be carried out for the PSAP outcomes, empathy ratings, self-report questionnaires and violence risk with the HKT-R instrument.

Power analysis

The sample size is based on the primary outcome and other studies from the previous mentioned guidelines (Lefaucheur et al., 2017) and seen as achievable. The sample size is based on research (Quellet et al., 2015) that reported large effect sizes (i.e. partial eta squared 0.25 and 0.21) for the active vs. sham condition using 15 participants per condition for three conditions. Because we have two experimental conditions, but also include covariates, a sample size of 50 subjects (25 per condition) is considered to have enough power to detect an effect, when power is set to 80% and alpha of 5%, two sided.

Discussion

This study protocol describes the design of an intervention with multichannel tDCS targeting bilateral vmPFC next to TAU, in comparison with sham condition and TAU. This study will explore the opportunities to introduce a new intervention to improve empathy, reduce violence and reduce craving in substance use offenders.

The present study has several strengths. First, to our knowledge only a few studies focus on increasing empathic abilities (Abend et al., 2017; Coll et al., 2016; Darby & Pascual-Leone, 2017; Feeser et al., 2014; Hetú et al., 2012; Sellaro et al., 2017; Rêgo et al., 2015) or modulate externalizing behavior (Gilam et al., 2018; Riva et al., 2015; Riva et al., 2017; Choy et al., 2018), but none of the studies focus on the implication of tDCS in a forensic sample. These individuals are in need of effective care, and by modulating the brain activity this can be a first step towards a new treatment program.

Secondly, tDCS may also influence substance abuse (Boggio et al., 2008b; Batista et al., 2015; Camprodon et al., 2007; Conti et al., 2014; den Uyl et al., 2015; Falcone et al., 2016; Fregni et al., 2008a, 2008b; Jansen et al., 2013; Klauss et al., 2014; Politi et al., 2008; Shahbabaie et al., 2014; Wietschorke et al., 2016) and, in turn, influence the relationship between substance abuse and violent behavior. This could lead to a decrease in recidivism in forensic institutions.

Thirdly, we will use a wide range of instruments and will gather information through multiple sources. We will obtain rsEEG, questionnaires about psychopathic traits (SRP-SF; Gordts et al., 2017), aggression (RPQ; Cima et al., 2013), empathy (IRI; de Corte et al., 2007), alexithymia (TAS-20; Bagby et al., 1994), risky and impulsive behavior (RISQ; van Dongen et al., 2018) and impulsivity (BISSII; Patton et al., 1995). For measuring substance abuse the study includes questionnaires that focus on alcohol use (AUDIT; Babor et al., 2001) alcohol craving (OCDS; Franken et al., 2002), drug use (DUDIT; Berman et al., 2005), and drug craving (OC-DUS-version Cocaine; Schippers et al., 1997). In addition, we have the results on the PSAP and empathy task.

Moreover, we will also include the participants who drop out during intervention in our five-day intervention program. This will give insight into the consequences of the formal procedures preceding intervention and may eventually enable us to describe profiles of the intervention with tDCS in treatment success and failure.

Finally, this study will contribute to development of a cheaper and less invasive treatment for substance abuse. As mentioned before, problems that substance abuse patients cause are an enormous burden to the community (financial and safety).



Previous research has found that substance abuse, especially alcohol and cocaine, are related to (violent) criminal behavior. Money invested in treatment may lead to a large reduction in the costs associated with substance abuse. Nevertheless, current interventions seem insufficient in treatment of substance abuse in forensic mental health care and are not sufficient enough to reduce violence risk, 66% of the patients reoffend.

Despite the strengths of this current study, several limitations may threaten the quality of our study. The greatest challenge will be to have an adequate response rate at baseline and at the end of the five-day treatment. Our target group will be conducted in a forensic institution and its participation is voluntary, so the participants can decide to quit at any time. The population has an extensive history of treatment and criminal justice, so they might have negative experiences with treatment or research studies. Moreover, the participants already have filled in numerous questionnaires in their time spent in institutions, that they may have become tired or suspicious about the purpose of the proposed study. To keep the nonresponse rate as low as possible, the researchers conducting this study are trained to motivate and encourage the participants to complete the program. The interaction will be transparent, attentive and flexible and the researchers will be prepared to give clear information about the purpose, goal and aim of the current study at any time. Furthermore, a treatment that uses neurostimulation and EEG might seem scary and new to the participants as compared to TAU, so extra explanation might be needed to ensure the participants that it is safe. So, to recruit as many participants as we aim to in this study it will require a high degree of motivation.

Finally, an active control group (i.e., sham modulation + TAU) might lead to smaller effect sizes of active tDCS than when the tDCS treatment would be compared to no treatment or a wait-list group. However, considering the seriousness of the problems of the forensic substance use offenders it would not be ethical to let them wait for an intervention and it remains important to get a sense of the effects of placebo versus the effects of active stimulation.

In conclusion, with the present study design we are able to explore what the added effectiveness is of tDCS + TAU in comparison to sham + TAU, which could provide valuable information for institutions, researchers, psychologists and the professionals in the field of criminal justice. In addition, this study may help to reduce craving in forensic substance use offenders and with that reduce not only violent behavior but also recidivism. The study will contribute to knowledge about increasing empathic abilities and the functioning of the vmPFC.

Specifically, this study may give insight into implementing tDCS on the vmPFC in the TAU and the setting of daily clinical practice for this complex, multiple problem target group.



Trail status

The current study started February 2019. The data collection will run until the 50 participants all completed the five-day program. After six-months a follow up will be conducted to test whether the effects of the intervention are still present. Until then, the intervention effects are unknown.

DECLARATIONS

Ethics approval and consent to participate

This study is being conducted in accordance with the ethical guidelines of the Declaration of Helsinki and has been approved by the Medical Ethical Review Committee of the Erasmus Medical Center (registration number: 2018.065 – NL65209.078.18.).

Informed consent will be obtained from all participants prior to the baseline measurement.

Consent for publication

Not applicable.

Availability of data and materials

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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In addition Stichting Volksbond Rotterdam and Stichting Koningsheide gave funding for this study. Stichting Volksbond Rotterdam (SVR) initiates and finances innovative research of socially relevant cases of substance abuse and the background of behavior and genetically expression of the addiction. Stichting Koningsheide supports original and promising projects aimed towards forensic psychiatry.

The financers are not involved in the design of the study nor the drafting of the manuscript. Furthermore, the financers are not and will not be involved in the subsequent process of data collection, analysis and interpretation.



Authors' contributions

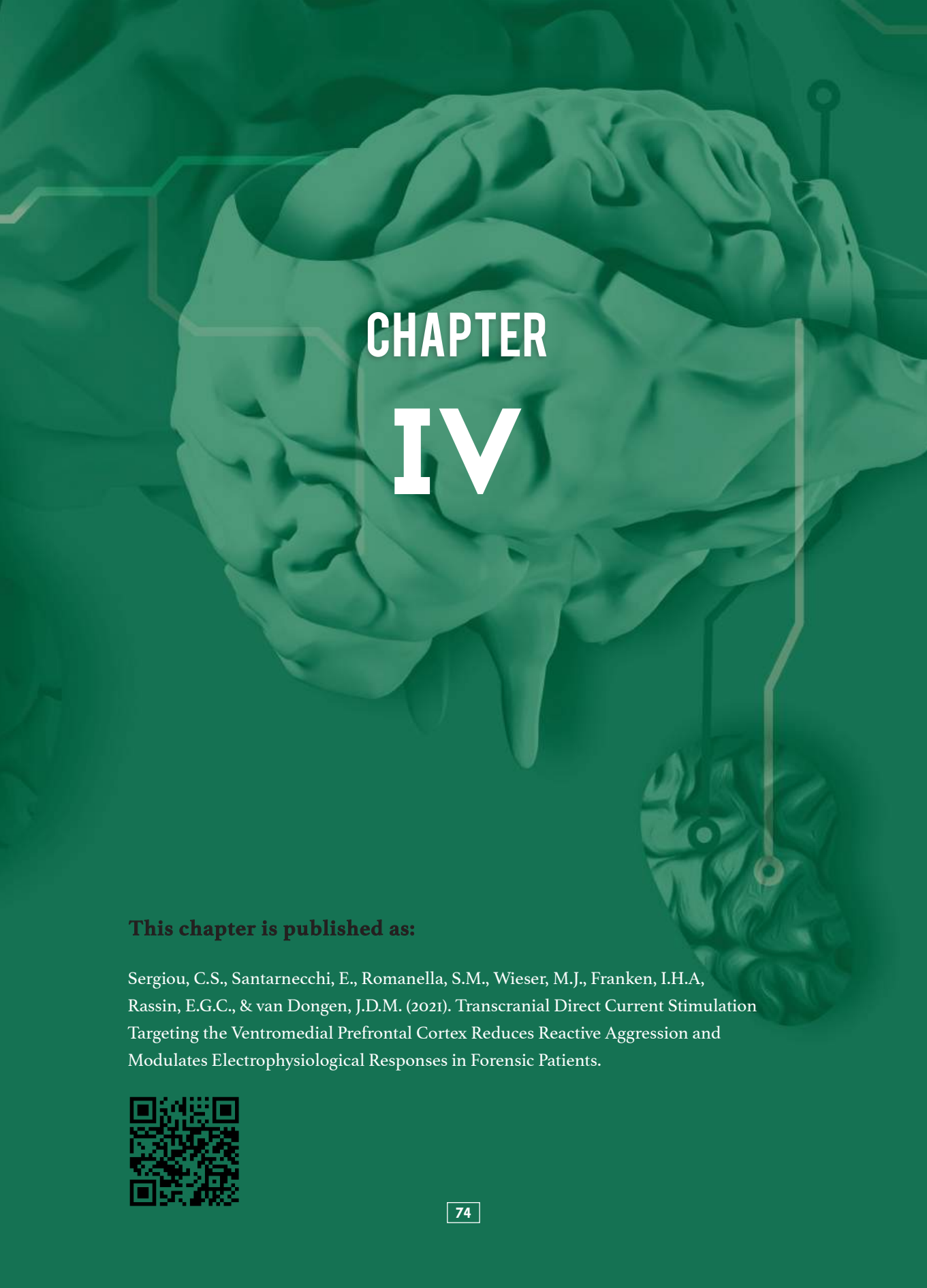
CS and JD are the principal investigators. JD obtained funding for the study. CS designed the trial in collaboration with JD. JD and AW designed the stimulation model for the tDCS.

CS coordinates the recruitment of the participants and the data collection during the study. CS drafted the manuscript with important contributions from JD, IF and AW.

All authors participated in the revision of the manuscript and approved the final version.

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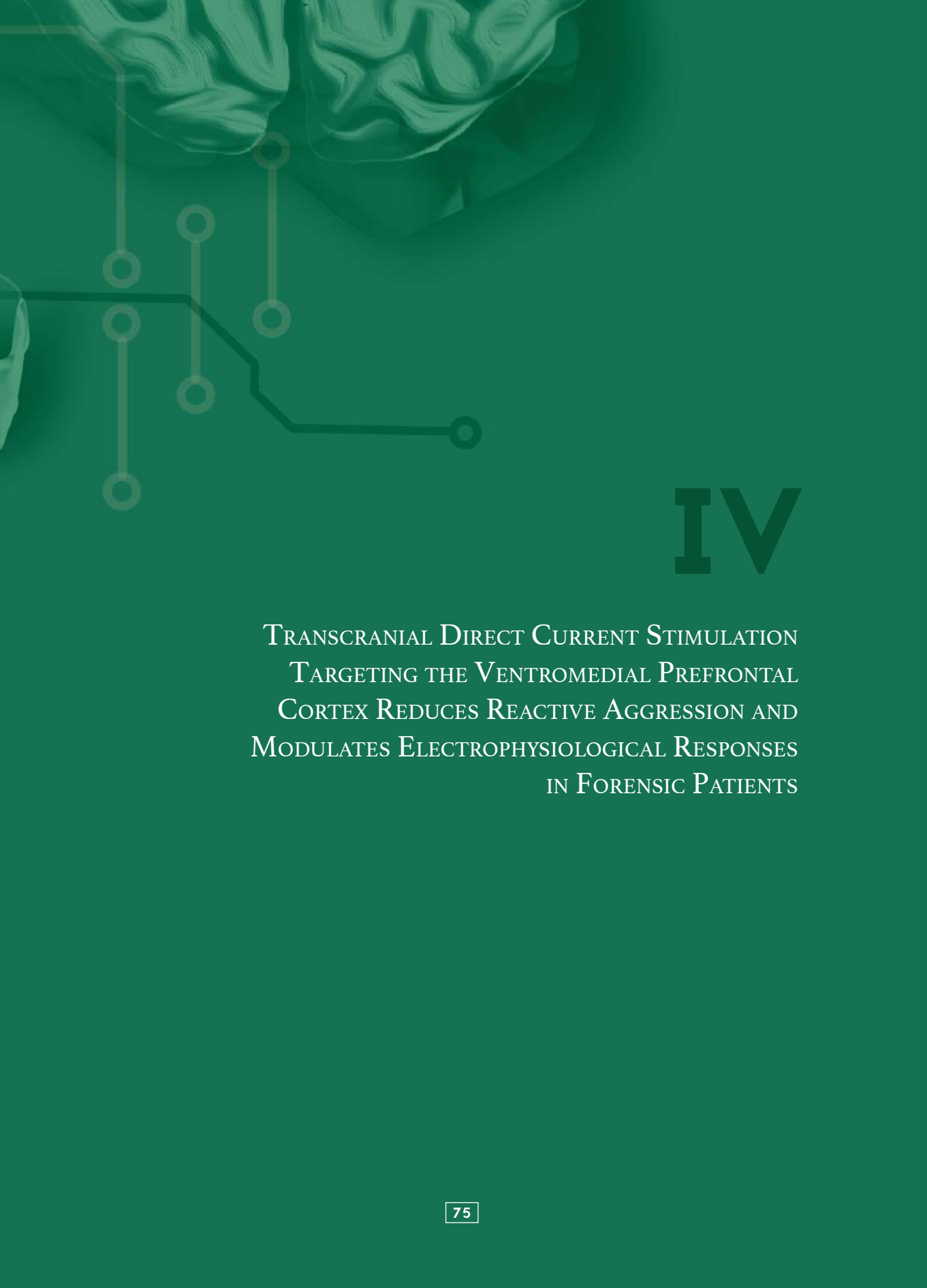


CHAPTER IV

This chapter is published as:

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The background is a solid dark teal color. In the upper left corner, there is a faint, stylized illustration of a human brain. Overlaid on the brain and extending downwards are several thin, light-colored lines that resemble a circuit board or neural pathways. These lines connect various small circles, some of which are also light-colored. The overall aesthetic is scientific and modern.

IV

TRANSCRANIAL DIRECT CURRENT STIMULATION TARGETING THE VENTROMEDIAL PREFRONTAL CORTEX REDUCES REACTIVE AGGRESSION AND MODULATES ELECTROPHYSIOLOGICAL RESPONSES IN FORENSIC PATIENTS

ABSTRACT

Background:

Studies have shown that impairments in the ventromedial prefrontal cortex (vmPFC) in forensic patients who also abuse cocaine and alcohol, play a crucial role in violent behavior. Moreover, interventions that aimed to reduce violence risk in those patients are found not to be optimal. A promising intervention might be to modulate the vmPFC by high-definition (HD) transcranial Direct Current Stimulation (tDCS). The current study aimed to examine (HD)-tDCS as an intervention to increase empathic abilities and reduce violent behavior in forensic substance dependent offenders. In addition, using electroencephalography (EEG), we examined the effects on the P3 and the Late Positive Potential (LPP) of the Event-Related Potentials (ERPs) in reaction to situations that depict victims of aggression.

Methods/Design:

Fifty male forensic patients with a substance dependence were tested in a double-blind sham-controlled randomized study. The patients received HD-tDCS two times a day for 20 min for five consecutive days. Before and after the intervention, the patients completed self-reports, performed the Point Subtraction Aggression Paradigm (PSAP), and EEG was recorded while patients performed an empathy task.

Results:

Results showed a decrease in aggressive responses on the PSAP and in self-reported reactive aggression in the active tDCS group. Additionally, we found a general increase in LPP amplitude after active tDCS. No effects on trait empathy and the P3 were found.

Discussion:

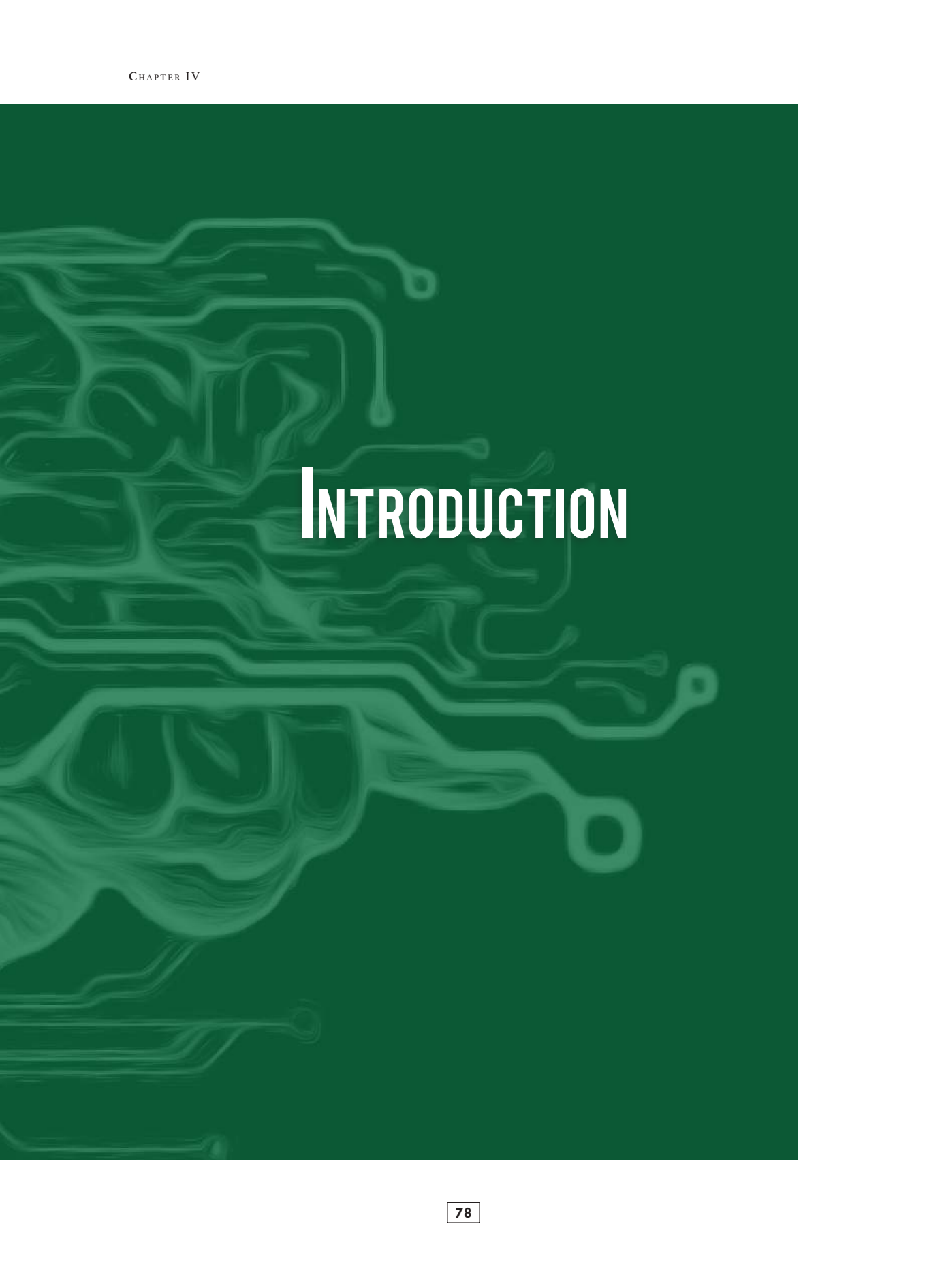
Current findings are the first to find positive effects of HD-tDCS reducing aggression and modulate electrophysiological responses in forensic patients, showing the potential of using tDCS as an intervention to reduce aggression in forensic mental healthcare.

This study was preregistered at the Dutch Trial Register, NTR7701. <https://www.trialregister.nl/trial/7459>

Keywords:

Empathy, aggression, forensic patients, neuromodulation, transcranial direct current stimulation, electroencephalography

INTRODUCTION

An abstract, light green circuit-like pattern flows across the dark green background, primarily concentrated on the left side and extending towards the center.

Substance abuse is a major concern in forensic patients and poses a major risk for violence in these individuals (Pickhard & Fazel, 2013). Substance use offenders are found to have increased odds of violent behavior (Cristello et al., 2020; Friedman, 1998; Gizewski et al., 2012; Gohssoub et al., 2019; Goldstein & Volkow, 2011; Preller et al., 2014; Volkow et al., 2009; Wagner, 1996). Moreover, repeatedly using substances has been found to be associated with functional neuroadaptations in the prefrontal cortex (PFC; Goldstein & Volkow, 2011; Volkow et al., 2009). Especially the Ventromedial Prefrontal Cortex (vmPFC) is found to be associated with substance use disorders (SUD) in regulating reward processing and motivational salience (Goldstein & Volkow, 2011; Engelman et al., 2012; Filbey et al., 2008; Kearney-Ramos et al., 2018; Kilts et al., 2001; Li et al., 2015; Schacht et al., 2013). Recent studies (Gizewski et al., 2012; Preller et al., 2014) show that the vmPFC is also involved in antisocial behavior, especially aggression (Fanning et al., 2017; Yang & Raine, 2009) and impulsivity, in abusers of alcohol and cocaine (Anderson et al., 1999b; Blair, 2004; Calzada-Reyes et al., 2013, 2016; Krämer et al., 2009; Raine, 1993; Raine et al., 2000). The vmPFC is found to be crucial for regulating emotions (Aron et al., 2004; Berkman & Lieberman, 2009; Bernhardt & Singer, 2012; Braver et al., 2009; Decety & Jackson, 2004; Decety & Lamm, 2009; Mathur et al., 2010; Schneider & Koenings, 2007; Shamay-Tsoory et al., 2003; Shamay-Tsoory, 2009; Song et al., 2008; van Dongen, 2020; Wagner et al., 2007) and has a specific role in Theory of Mind (ToM)- processes (Shamay-Tsoory et al., 2003; Bernhardt & Singer, 2012) that are vital in empathic processing due to cognitively adopting the other's perspective (Shamay-Tsoory, 2011).

In addition, neuroimaging studies showed that the structures that are relevant for empathy and the modulation of aggressive behavior, namely the vmPFC and medial frontal regions (Anderson et al., 2006; Blair, 2001; Blair et al., 2006; Covell & Scalora, 2002; Eisenberg et al., 2010; Giancola, 2004; Hoaken & Stewart, 2003; Marshall, 1996; Miller & Eisenberg, 1988; Morgan & Lillienfeld, 2000; Preller et al., 2014; Zheng et al., 2016), are impaired in individuals with antisocial traits (Decety et al., 2013; Grafman et al., 1996; Koenings et al., 2007; Shamay-Tsoory et al., 2010). One study (Skibsted et al., 2017) used the Point Subtraction Aggression Paradigm (PSAP; Cherek, 1992) and also indicated a link between brain activity in the vmPFC and aggressive behavior, proposing the vmPFC to be a neural substrate related to aggressive provocation (Skibsted et al., 2017). Other studies (Blair, 2016, 2018; Gilam et al., 2015, 2017; White et al., 2014, 2016) demonstrated that the vmPFC is related to increasing costs and retaliation of the aggressor and greater punishment choices.

Following the proposed neurocognitive models of Blair (Blair, 2001, 2005; Blair et al., 2006), violence is inhibited by empathy, and therefore individuals with deficits in their empathic abilities are less susceptible and less motivated to inhibit aggression, which increases the risk of violent behavior (Blair, 2013).



Whereas the association between the PFC and aggression has been studied in different neuroimaging, behavioral and clinical studies, to our knowledge little is known about the causal role of the vmPFC in empathic abilities and violent behavior in forensic patients. Therefore, modulating the vmPFC to increase empathic abilities and reduce violent behavior in these patients could be of substantial importance for supporting a causal relation and for formulating new treatment interventions.

A promising tool to modulate empathic abilities and aggression is transcranial Direct Current Stimulation (tDCS). In the review paper of Sergiou et al. (2020b) multiple studies were reviewed that used tDCS to modulate aggression (Chen, 2018; Gilam et al., 2018; Yu et al., 2015) or empathic abilities (Abend et al., 2018; Fumagalli et al., 2010; Riva et al., 2015; Zheng et al., 2016), and concluded that the vmPFC would be the most promising target area for modulating these behaviors. tDCS is a neuromodulation technique that uses electrical currents to increase or decrease activity in the brain region of interest and has been proven effective in many disorders (Barr et al., 2008; George & Aston-Jones, 2010; Janssen et al., 2013). tDCS uses a sub threshold modulation of the membrane potentials of neurons through altering cortical excitability (Nitsche & Paulus, 2000; Nitsche et al., 2003) and can change synaptic neuronal plasticity (Janssen et al., 2013). This change in neuronal plasticity is found to produce Long Term Potentiation (LTP)-like 'learning' in the modulated neurons (Rohan et al., 2015; Woods et al., 2016). In this rapidly evolving field, studies have shown that the stimulation effect is based on the strength, duration of the current and polarity-dependent changes (Batsikadze et al., 2013; Nissim et al. 2019; Samani et al., 2013) and that the results of modulation could be changed due to the nature of previous brain activity (Abraham & Tate, 1997; Hassanzahraee et al., 2018; Karabanov et al., 2015; Krönberg et al., 2018) as described by the Bienenstock- Cooper-Munroe rule (BCM; Bienenstock et al, 1982).

To date, the majority of the studies that investigate the modulation of the PFC have relied on conventional tDCS, upregulating one side of the brain and downregulating the contralateral region (Ling et al., 2020). An even more promising tDCS technique is High-Definition tDCS (HD-tDCS), which uses multiple compact (i.e. < 5 cm²) circular electrodes to deliver the low current (Bikson et al., 2019) and studies have shown that HD-tDCS can target brain structures with higher focality (DaSilva et al., 2015). In this way, deeper regions of the brain, such as the vmPFC, can be modulated in an optimal way (Edwards et al., 2013; Faria et al., 2011; Hess, 2013; Manuel et al., 2014; Nakamura-Palacios et al., 2016; Tanaka et al., 2013; To et al., 2018). Therefore, in the present study we will use a protocol using HD-tDCS targeting the vmPFC, with multiple sessions and with enhancing the brain state during modulation to maximize the effect (Baxter et al., 2017; Buch et al., 2017; Fisher et al., 2017; Fox et al., 2014).

To study the effect of tDCS on brain functioning, the temporal dynamics could be studied using electroencephalography (EEG; Cheng & Decety, 2012; Decety et al., 2015; van Dongen et al., 2018). The event-related potential (ERP) technique has mostly been used to investigate empathy, by measuring brain responses to pictures. Studies have shown that both early and late ERP components are related to perceived pain in others (Cheng & Decety, 2012; Decety et al., 2010; Meng et al., 2012) and in reaction to self-reported trait empathy (van Dongen et al., 2018; Corbera et al., 2014; Fabi & Leuthold, 2017; Vaess et al., 2016). Two ERPs that have been studied extensively are the P₃, and late positive potential (LPP).

The P₃ is an ERP that has a positive voltage in the latency of 300-650 milliseconds (ms; Hajcak et al., 2010). The P₃ is typically associated with controlling sustained attention towards salient stimuli (Hajcak et al., 2010) and has been linked to empathy for pain (Cheng & Decety, 2012; Coll, 2018; Cuthbert et al., 2000; Decety et al., 2010; Fan & Han, 2008; Ibáñez et al., 2011; Li & Han, 2010). The LPP is a positively sustained ERP that occurs around 400-1000 ms and is located at the parietal locations and the central midline (Hajcak et al., 2012) and is associated with empathy and emotional cues. A study of Cuthbert et al. (2000) demonstrated that the LPP indicates an increase in amplitude towards emotional stimuli, arousal, regulation, facial expressions, and affective experience regulation and the study of van Dongen et al., (2018) found reduced P₃ and LPP in individuals with decreased empathic abilities. Therefore, studying these positive potentials contribute to the understanding of emotional regulation (Cheng & Decety, 2012; Cuthbert et al., 2000; Fan et al., 2011; Lang & Bradley, 2010), and serve as an indication that the temporal dynamics of empathic processing in the brain can be studied using late ERPs in the EEG.

The main aim of the current study was to examine whether modulating activity in the vmPFC using HD-tDCS would increase empathic abilities and reduce aggressive behavior in forensic substance dependent patients. In addition, in this study we investigated the effects of the HD-tDCS intervention on electrophysiological responses (P₃ and LPP) to situations that depict victims of aggression (i.e., empathy). Based on prior literature, it was expected that following active tDCS, compared to sham, empathic abilities would increase and aggression would decrease. Additionally, it was expected that the P₃ and LPP amplitude following the viewing of pictures that depict victims of aggression, would be increased after active tDCS compared to sham tDCS.

Knowledge gained from the current study will give insight in the casual relation between activity in the vmPFC, empathy and aggression and can inform in the development of new neuromodulation (e.g., tDCS) protocols for treatment interventions in violent forensic populations.



Materials and Methods

Participants

Fifty male participants (mean age = 37.40 years, SD = 9.19 years, range: 22-62 years) were recruited from two departments of the division for forensic addiction mental healthcare of Antes in Poortugaal, the Netherlands, between February and October 2019. Twenty-one participants were recruited at the Forensic Addiction Clinic (FVK) and 29 from the Department of Forensic Care (AFZ). The patients were randomly assigned to either one of the conditions and participated in a double-blind placebo-controlled study with two conditions. Twenty-five participants received treatment as usual (TAU) + active stimulation, and 25 participants received treatment as usual (TAU) + sham (placebo). For an overview of the demographic characteristics of the sample see Table 1. Inclusion and exclusion criteria are presented in the Supplement S1 in online article version.

The participant flow and recruitment according to the Consolidated Standards of Reporting Trials (CONSORT) can be found in the Supplement S2 in online article version.

TABLE 1

Demographic characteristics

	tDCS group		Sham group	
	<i>n</i>	%	<i>n</i>	%
Caucasian	25	100	23	92
Non- Caucasian	0	-	2	8
Primary education	9	36	8	32
High School	7	28	6	24
Secondary education [VET]	9	36	11	44
DSM-5 Axis I	7	28	10	40
DSM-5 Axis II	8	32	10	40
Mono substance use	8	32	9	36
Poly substance use	17	68	16	64

*Note. Characteristics are displayed in percentage of participants per group, N = 50 (n = 25 for each condition). Participants were on average 36.4 years old (SD = 8.88) in the tDCS group and on average 38.4 years in the sham group (SD = 9.56), and participant age did not differ by condition. VET= Vocational Education and Training, DSM-5= Diagnostic Symptom Manual fifth edition.

Procedure and design

The study consisted of a double-blind, placebo-controlled, randomized trial comparing a group that received active HD-tDCS intervention with a sham control group. Baseline assessments including self-report questionnaires, a resting-state EEG task, passive viewing task (van Dongen et al., 2018), rating empathy task (van Dongen et al., 2018) and the Point Subtraction Aggression Paradigm (PSAP; Cherek, 1992) were conducted during the pre-intervention. During the following week, the participants received 2 times 20 minutes HD-tDCS stimulation or sham intervention targeting the vmPFC for five consecutive days (see the Supplement S3 for a flowchart of the procedure). There was approximately 3-4 hours between the two sessions, depending on the patients schedule and TAU. Outcome measurements were conducted one week later during the post-intervention. Self-report questionnaires and tasks were conducted in a fixed order.

A study by Nissim and colleagues (Nissim et al., 2019) using fMRI demonstrated that the optimal gains from using tDCS could be realized by increasing the activity of the brain area of interest (i.e., the brain state). Therefore, to optimize our intervention, we increased the activity of the vmPFC while stimulating, by showing the patients two empathic movies (Chbosky, 2017; Nelson, 2001) and the Reading the Mind in the Eyes task (RMET; Baron-Cohen, 2001) to trigger these brain states. Patients and investigators were blind to the tDCS allocation. The principal investigator of the project, who was not involved in data collection and initial statistical analysis, preprogrammed the tDCS device in active condition or sham condition matched with a number.

The study was conducted in accordance with the ethical standard of the Declaration of Helsinki (GAWMA, 2014) and is approved by Medical and Ethical Review Board of the Erasmus Medical Centre Rotterdam. The trial protocol was registered at the Dutch Trial Register, (NTR7701). See also Sergiou et al. (2020a) for the complete published study protocol.



Descriptive statistics for self-report questionnaires, rating task, PSAP, P3 and LPP amplitudes

Time I (pretest)						
Variable	tDCS group			Sham group		
	M	SD	Range	M	SD	Range
DIFF_AROUSAL	2.52	2.14	[-] .40-7.28	2.52	2.14	[-] .40-7.28
DIFF_EMOTION	[-] 1.15	2.34	[-] 4.15-4.63	[-] 1.15	2.34	[-] 4.15-4.63
DIFF_VICTIM	1.76	1.89	[-] 3.07-5.05	1.76	1.89	[-] 3.07-5.05
DIFF_PERPETRATOR	[-] 1.22	1.99	[-] 3.75-4.00	[-] 1.22	1.99	[-] 3.75-4.00
PSAP	.12	.09	.0-.47	.12	.09	.0-.47
RPQ Total	21.08	8.14	6-43	21.08	8.14	6-43
RPQ Reactive	12.48	4.01	3-22	12.48	4.01	3-22
RPQ Proactive	8.60	4.69	2-21	8.60	4.69	2-21
IRI Total	55.00	14.73	26-84	55.00	14.73	26-84
SRP-SF Total	78.56	14.28	54-98	78.56	14.28	54-98
AUDIT	12.35	12.61	0-37	12.35	12.61	0-37
DUDIT	22.44	12.51	0-40	22.44	12.51	0-40
P3 AMPLITUDE NEU	2.41	5.23	[-] 7.91-11.59	2.41	5.23	[-] 7.91-11.59
P3 AMPLITUDE AGG	3.34	4.38	[-] 5.56-14.14	3.34	4.38	[-] 5.56-14.14
LPP AMPLITUDE NEU	1.29	6.86	[-] 18.21-15.19	1.29	6.86	[-] 18.21-15.19
LPP AMPLITUDE AGG	2.42	5.48	[-] 13.43-10.23	2.42	5.48	[-] 13.43-10.23

Time II (posttest)						
Variable	tDCS group			Sham group		
	M	SD	Range	M	SD	Range
DIFF_AROUSAL	2.78	2.44	[-] 1.05-7.55	3.29	2.61	[-] .23-7.60
DIFF_EMOTION	[-] 1.38	2.33	[-] 5.72-3.45	[-] 1.73	2.17	[-] 7.25-1.90
DIFF_VICTIM	1.45	2.24	[-] 4.00-3.78	1.27	2.86	[-] 3.75-7.60
DIFF_PERPETRATOR	[-] 1.74	3.77	[-] 3.95-3.77	[-] 1.67	2.34	[-] 7.60-2.60
PSAP	.07	0.7	.0-.33	.15	.19	0-66
RPQ Total	18.58	9.66	8-45	16.36	8.53	2-36
RPQ Reactive	11.20	4.71	3-22	10.32	4.25	2-20
RPQ Proactive	7.37	5.54	0-23	6.04	4.75	0-16
IRI Total	56.30	10.45	39-75	56.45	12.66	37-80
SRP-SF Total	74.96	18.09	37-113	65.32	15.85	28-99
AUDIT	13.76	13.03	0-37	8.36	9.18	0-33
DUDIT	19.95	11.59	0-38	14.47	12.20	0-38
P3 AMPLITUDE NEU	3.13	2.99	[-] 1.87-8.65	0.51	3.96	[-] 8.14-8.89
P3 AMPLITUDE AGG	4.18	3.85	[-] 1.87-13.59	1.79	6.12	[-] 12.57-14.73
LPP AMPLITUDE NEU	2.15	5.87	[-] 19.45-10.44	[-] .11	3.79	[-] 12.88-5.68
LPP AMPLITUDE AGG	5.20	3.65	[-] .89-13.09	2.82	6.04	[-] 17.48-12.18

*Note. PSAP= Point Subtraction Aggression Paradigm, RPQ = Reactive Proactive Aggression Questionnaire, IRI = Interpersonal Reactivity Index, SRP-SF= Self-Report Psychopathy Short Form, AUDIT= Alcohol Use Disorder Identification Test, DUDIT= Drug Use Disorder Identification Test, P3= P300, LPP = Late Positive Potential, NEU = Neutral condition, AGG = aggression condition.

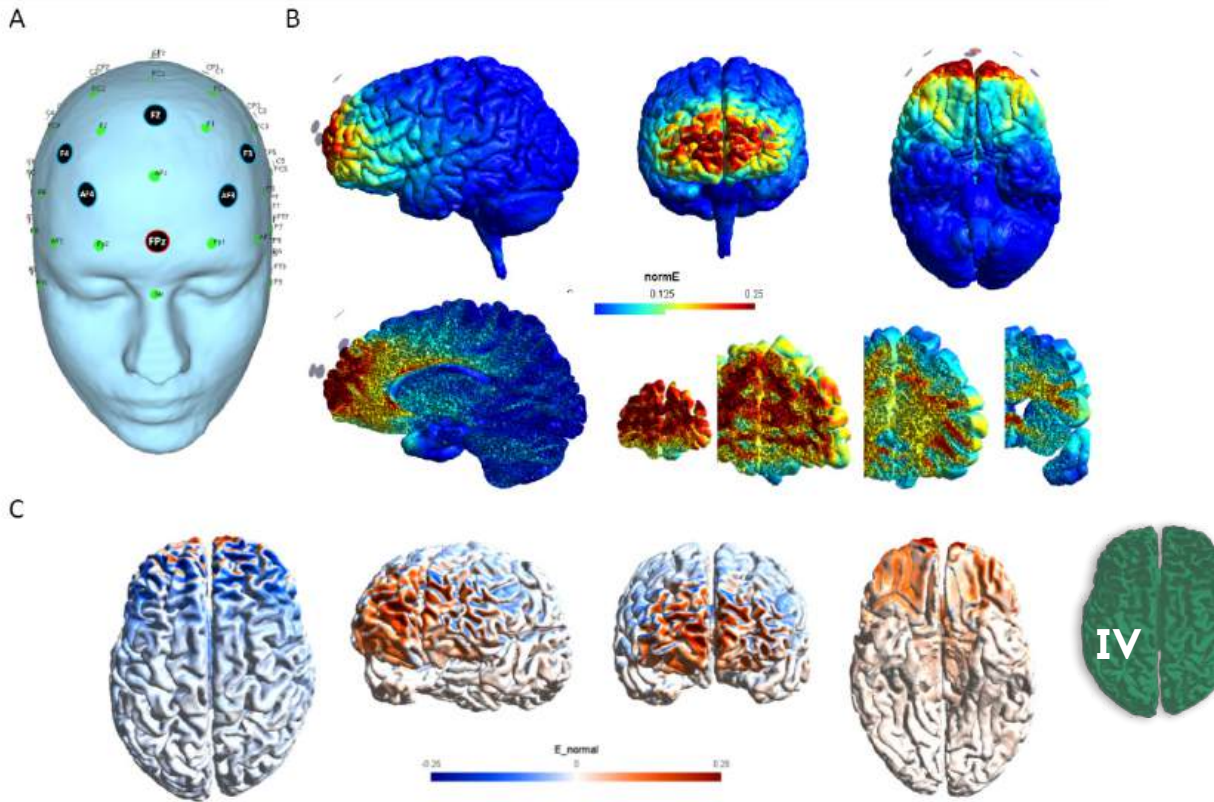


Figure 1.

(A). Placement of the electrodes at 32 standard 10-20 Electroencephalography (EEG) system on the scalp with anodal high-definition transcranial direct current stimulation (HD-tDCS) with 6 x 1cm radius ($\pi \text{ cm}^2$) electrodes over the Fpz (2mA) and cathodal tDCS over AF3, AF4, F3, F4 and Fz (-0.4mA each). (B). Different views and slices of the map of electrical field induced by HD-tDCS montage as expressed in normE (V/m). This measure allows us to see the intensity of the stimulation independently by the polarity. (C). Views of the map of the electrical field expressed in NormalE (V/m) showing us the polarity (anodal/cathodal) of the stimulation.

High-Definition tDCS intervention

HD-tDCS was administered with the CE-certified Neuroelectronics Starstim8® (Neuroelectronics, 2011), operating according to the evidence-based guidelines of LeFaucheur et al. (2017). Before commencement of the study, the HD-tDCS montage optimization was based on the current-flow modeling of the NIC-software of the tDCS Starstim8® system (Neuroelectronics, 2011), (see the protocol (Sergiou et al., 2020a) for detailed description). In addition, the induced E-field of the HD-tDCS montage was computed in SimNibs v3.2 (Thielscher et al., 2015). Detailed description on biophysical modeling can found in the Supplement S5 in online article version and attached on the end of this chapter.

The currents were transmitted through six circular Ag/AgCl PiStim “High Definition” electrodes (1 cm radius, $\pi \text{ cm}^2$) that were applied with conductive gel. The resulting Norm-E field and Normal E-field distribution is created in Gmsh v4.7.1. (Geuzaine & Remacle, 2008) with an output range from 0 to 0.25 V/m (see Figure 1). The HD-tDCS device was programmed for stimulation with 2 mA tDCS during 20 minutes for modulating the vmPFC of the participants in the active condition. The sham condition followed the exact same procedure, with 30 seconds ramping-up and down the tDCS currents at the beginning and end of the protocol, based on earlier research indicating this method being effective for blinding [130]. The anodal electrode was placed on the Fpz location, and the five return or cathodal electrodes were placed on AF3, AF4, F3, Fz and F4 (see Figure 1 for the electrical field model) and the Supplement on end of this chapter for detailed description of the biophysical modeling).

Electrophysiological measurement empathy

To assess EEG measuring empathy, the patients had to perform a *Passive Viewing Empathy* task (van Dongen et al., 2018). This task was designed to measure empathic abilities through EEG. This task consists of 95 pictures displaying scenes for 6000ms with either an aggressive interaction (45), a neutral interaction (45) or a neutral object (i.e. fillers; 45). The aggressive pictures consisted of either a sexual, verbal or physical interaction. Participants were instructed to view all the pictures passively; in this way the automatic neural responding in the brain could be determined in a most optimal way (van Dongen et al., 2018).

State empathy

To assess state empathy, the patients had to perform a *Rating Empathy Task* (van Dongen et al., 2018). Following the *Passive Viewing Empathy* task, the 95 pictures were presented another time to the patients. In this task they were instructed to rate the pictures by answering four questions regarding *arousal*, *emotional valence*, *empathy for victim* and *perpetrator*.

Aggression task

To assess aggression, we used the *Point Subtraction Aggression Paradigm* (PSAP; Cherek, 1992). The PSAP is one of the best-validated paradigms to measure aggression in a lab environment (Geniole et al., 2016). A detailed description of the tasks used in this study included in the Supplement S2 of the online article version.

Self-report questionnaires

For a complete overview of the self-report questionnaires used in the overall study see Sergiou et al., (2020a). The questionnaires used in the analysis of this paper mentioned below and described in more detail in the Supplement S2 of the online article version.

Reactive and Proactive Aggression Questionnaire (RPQ; Raine et al., 2006). In order to index aggression over a weeklong period, we used the Dutch translation of the RPQ (Cima et al., 2013).

Interpersonal Reactivity Index (IRI; Davis, 1980) In order to assess trait empathy, we used the Dutch translation of the IRI (de Corte et al., 2007), this is a commonly used self-report instrument designed to assess empathic tendencies.

The Alcohol Use Disorders Identification Test (AUDIT; Babor et al., 2001). In order to assess alcohol use, we used the Dutch translation of the AUDIT (Allen et al., 1997). The 10-item AUDIT includes questions to assess alcohol, intake, alcohol dependence and alcohol-related problems.

The Drug Use Disorders Identification Test (DUDIT; Berman et al., 2005). In order to assess drug use, we used the Dutch translation of the DUDIT (Berman et al., 2007). The DUDIT is an 11-item screening instrument to assess non-alcohol drug use patterns and various drug-related problems.

EEG recording

EEG was recorded using a mobile version of the BrainProducts (GmbH, 2018) Active-Two System amplifier (Amsterdam, the Netherlands). Thirty-two electrodes were placed on the scalp of each participant following the international 10-20 EEG-system. Two other additional electrodes were placed vertically above and beneath the left eye (electro- oculogram; EOG), the electrodes for the left and right mastoid placement were incorporated in the EEG cap. The EEG and EOG signals were digitized with a sampling rate of 500 Hertz (Hz) and 24-bit analog-to-digital conversion with offline filtering.



EEG data preprocessing

Data were preprocessed offline using Brain Vision Analyzer (2.2.0; 2019). Segmentation was done per condition (neutral vs. aggression) for both P3 and LPP at pretest and posttest, in an interval of 1200 ms (-200ms to 1000ms; see van Dongen et al., 2018) relative to stimulus presentation. Next, the data was filtered using a bandpass filter ranging from .01 Hz to 30 Hz (phase shift-free Butterworth filters; 24 dB/octave slope), the signal was corrected for ocular artifacts using the Gratton and Coles algorithm (Gratton et al., 1983), trials with remaining artifacts (i.e., segments with an EEG signal exceeding an amplitude of 100 μ V) were excluded from further analysis (van Donge et al., 2018), and data was baseline-corrected (200 ms pre-response or pre-stimulus period served as baseline). Nine participants had to be removed from the analyses due to excessive amounts of artifacts in the data (i.e., less than 15 artifact-free segments). Bad channels with too many artifacts were corrected with topographic interpolation. This resulted in a set of EEG data of 41 participants, 21 in the sham condition and 20 in the active tDCS condition.

Because the highest amplitude of P3 and LPP typically is seen at Pz (Ikezawa et al., 2014; Polich, 2007), inspection of the grand average ERPs (see Figure 3) demonstrated a maximal amplitude for aggressive pictures around 400ms after stimulus onset (P3) and followed by the slow wave activity of the LPP, which returned to baseline after 6000ms. For analyzing the ERPs, the P3 amplitude was defined as the mean amplitude of the Pz between 350ms and 450ms after stimulus onset (van Dongen et al., 2018; Decety et al., 2016). The LPP was analyzed at the mean amplitude of the Pz between 500 and 1000ms after stimulus onset (van Dongen et al., 2018; Hajcak et al., 2005). The average signal per condition was then used for determining the characteristics of the ERP.

Statistical analyses

Self-report and the electrophysiological outcomes were analyzed using SPSS 25 (IBM, 2017). To test the differences between the active and the sham group from pre- to posttest on the state empathy following the aggression versus the neutral condition, four 2 (Time; pre vs. post) x 2 (Group; active tDCS vs. sham) ANOVAs were conducted for the scoring on *arousal*, *emotional valence*, *victim* and *perpetrator*. Similarly, to test the differences between the active and the sham group from pre to posttest on state aggression (PSAP), a repeated measures ANOVA was performed. Aggression was indicated by the proportion of aggressive responses. Thus, the number 2 presses divided by the total amount of presses (i.e., [No. option 2]/[No. of total button presses])

To investigate the electrophysiological outcomes of the P3 and LPP, two 2 (Time; pre vs. post) x 2 (Emotion; aggressive vs. neutral) x 2 (Group; active tDCS vs. sham) repeated measures ANOVAs were conducted. For all statistical tests, the level of significance was set at $p < .05$.

Results

The result from the HD-tDCS simulation (Figure 1B) demonstrates that the vmPFC was reached with intensity sufficient to induce effect of the stimulation (0.11-0.2 V/m (see the Supplement S5 of online version article for details).

See the Supplement S6 of online version article for a report on adverse effects. Descriptive statistics for all the measures used in this study, including all the self-report questionnaires, can be found in Table 2. Results showed no significant difference between medication use and the two conditions (tDCS and sham), $X^2(3, N=50) = 3.1, p = .378$ (see the Supplement S8 in the online version of the article).

State empathy outcomes

Concerning state empathy there were no significant main effects for arousal, emotional valence, empathy for victim or empathy for perpetrator between the tDCS group and the sham group from pre-posttest. Furthermore, we did not find any significant inter- action effects.

Aggression outcomes

Regarding the PSAP task there were no significant main effects for Time or Group. We did find a significant Time x Group interaction effect (see Figure 3), with $F(1,48) = 5.87, p < .019, \eta^2 = .11$. Post-hoc tests revealed that the effect is significant ($p = .027$) for the tDCS group and not significant for the sham group ($p = .192$). These results imply that the decrease in aggression from pre to post test was significantly stronger in the tDCS group

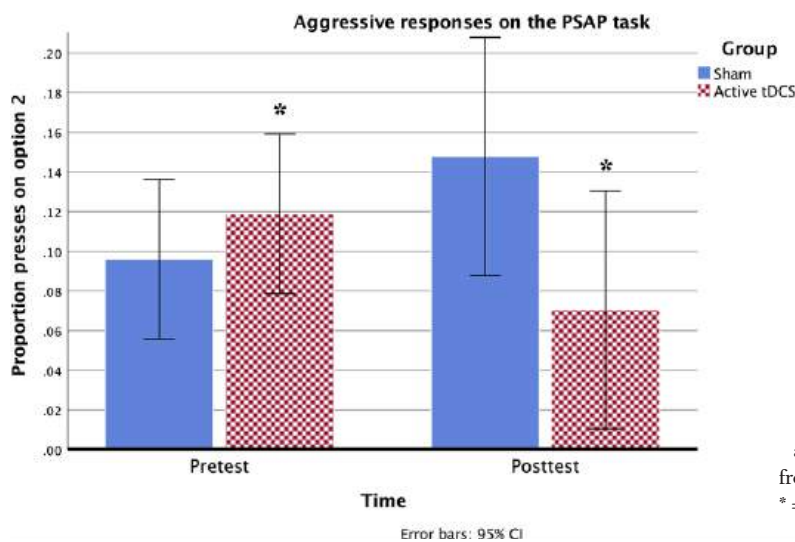


Figure 3. Proportion aggressive responses on the Point Subtraction Aggression Paradigm for the sham group and active tDCS group from pre to posttest. * = Significant effect at $p < .05$.

Self-report questionnaires

Regarding the RPQ we found a significant effect of Time on the RPQ total score $F(1,47) = 9.51, p = .003, \eta^2 = .25$ and RPQ reactive aggression subscale ($F(1,48) = 10.68, p = .002, \eta^2 = .19$) over the two groups. There were no main effects of Group. These results indicate a significant reduction in self-reported reactive aggression after the intervention as compared to the pretest across both groups. There were no significant findings for the SRP-SF, IRI, AUDIT or DUDIT.

P₃ and LPP

See Figure 3 for an overview of the grand average of P₃ and LPP in the active tDCS and sham group from pre- to posttest. Analyses of the P₃ amplitude showed no significant main effect for Time, Group or Emotion. The results indicated a significant interaction effect between Time and Group ($F(1,39) = 4.52, p = .040, \eta^2 = .10$), but not for Emotion. Follow-up t-tests revealed that the sham group showed a significant [$t(20) = 3.07, p < .01$] decrease in P₃ amplitude from pre- ($M = 3.07, SD = 3.82$) to posttest ($M = 1.47, SD = 4.76$). Graphs are displayed in Figure 4.

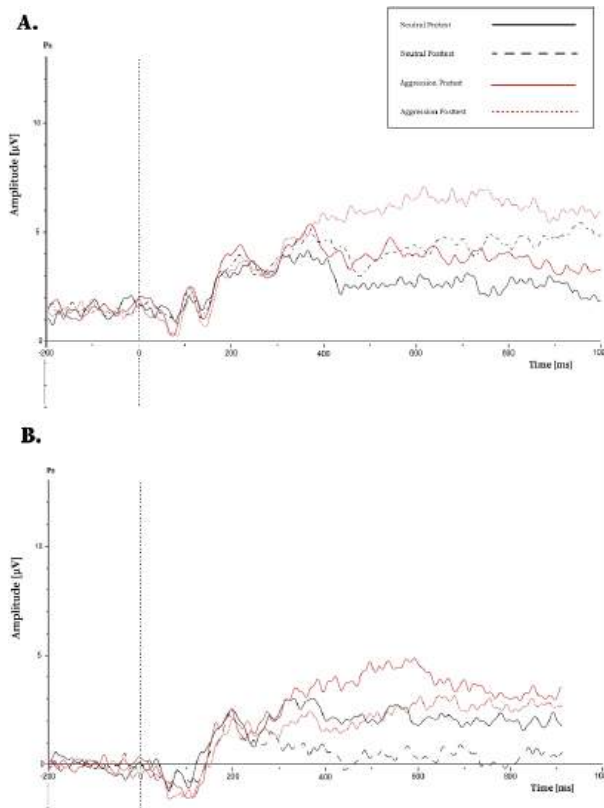


Figure 3.

(A). Event related potentials (ERPs) grand mean recorded at Pz during the passive empathy task for the active HD-tDCS condition.

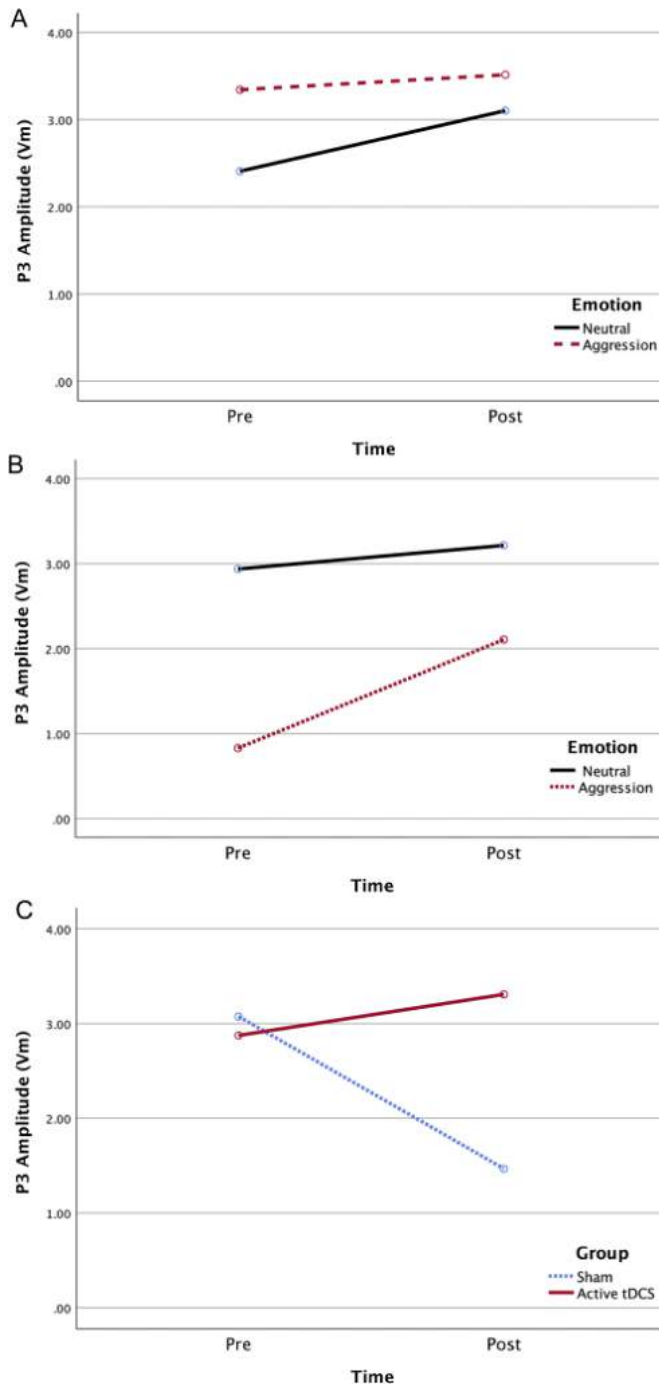
(B). Event related potentials (ERPs) grand mean recorded at Pz during the passive empathy task for sham condition.

Figure 4.

(A). Intervention effects on the P3 amplitude (μV) for the active tDCS group on emotion from pre-to posttest.

(B). Intervention effects on the P3 for the Sham group on emotion from pre-to posttest.

(C). Intervention effects on the P3 post-hoc between the active tDCS group and the Sham group from pre-to posttest.



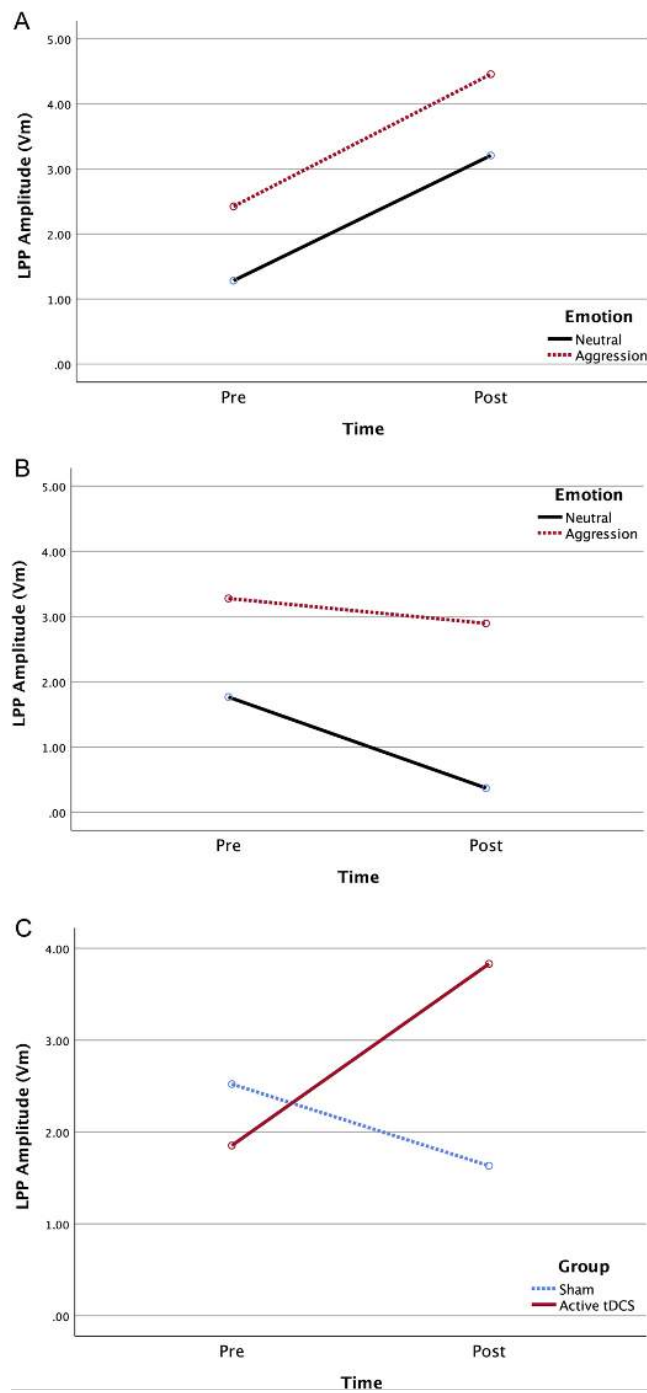


Figure 5.

(A). Intervention effects on the Late Positive Potential (LPP) amplitude (μV) for the active tDCS group on emotion from pre-to posttest.

(B). Intervention effects on the LPP for the Sham group on emotion from pre-to posttest.

(C). Intervention effects on the LPP post-hoc between the active tDCS group and the Sham group from pre-to posttest.

Analyses of the LPP amplitude resulted in a significant main effect for Emotion ($F(1,39) = 5.62, p = .023, \eta p^2 = .13$), meaning that aggression pictures led to higher LPP amplitudes compared to the neutral pictures in both groups. There were no main effects for Time or Group. In addition, we found a significant interaction effect between Time x Group ($F(1,39) = 5.66, p = .022, \eta p^2 = .009$). Follow-up t-tests revealed that the tDCS group showed a significant [$t(19) = -2.29, p = .03$] increase in LPP amplitude compared to the sham group, from pre to posttest, independent of emotion. Meaning an overall increase in amplitude after the intervention for the tDCS group. Graphs are displayed in Figure 5.

To check if the increase in LPP amplitude from pre- to post-test for both aggression and neutral pictures was also present in the filler pictures an additional analysis was performed (see the Supplement S4 in online version). Results showed no significant effects on Time, Group or Emotion.



Discussion

The aim of this study was twofold. First, we examined HD-tDCS as an intervention to increase empathic abilities and reduce violent behavior in forensic patients with a substance dependence. Second, we examined the HD-tDCS effects on the electrophysiological responses (P3 and LPP) to situations depicting victims of aggression. Since the intervention was the tDCS treatment in addition to TAU, all the found effects are a product of the interaction between the intervention (or placebo) and the TAU.

Results showed no effects of tDCS on state empathy. Although this is not what we had expected, it can be explained by the fact that the vmPFC may not be directly related to empathic abilities. As shown in the model of Blair (2005), decreased activity in the vmPFC is linked to impaired social and affective decision making, and the vmPFC is found to be involved in perspective-taking and regulating emotions (Chen et al., 2012; Decety et al., 2015, van Dongen et al., 2018), that are vital for empathic processing (Shamay-Tsoory et al., 2003), but not related to empathy directly.

Consistent with our hypothesis, we found a reduction in reactive aggression on the aggression task from pre-to posttest in the active tDCS group compared to the sham group. Additionally, we found the same effect for self-reported reactive aggression and the total aggression score as measured with the RPQ. The reduction of reactive aggression is consistent with previous findings on the associating between the vmPFC and aggressive behavior (Anderson et al., 1999; Blair, 2004; Calzada-Reyes et al., 2013, 2017; Krämer et al., 2009; Raine, 2003; Raine et al., 2000; Yang & Raine, 2009). Previous research also indicated a link between the brain activity in the vmPFC and the aggressive behavior measured with the PSAP, proposing the vmPFC to be a neural substrate related to aggressive provocation (Blair, 2016; Gilam et al., 2015, 2017; Skibsted et al., 2017; White et al., 2014, 2016). Consistent with these results is the proposed framework of Blair (2013), which demonstrates that diminished activity in the vmPFC leads to frustration-based reactive aggression. Consequently, the present study provides insight into the functional role of the vmPFC in reactive aggression and how modulating activity in the vmPFC can reduce aggressive responses in the laboratory setting. Future research has to show the generalizability of these results to violent behavior outside the laboratory.

With respect to the electrophysiological measures (P3 and LPP), we expected that after a week of tDCS intervention, aggressive pictures would result in larger ERP amplitudes (more arousal, more empathy) compared to neutral pictures. This hypothesis was partly confirmed. Regarding the P3, only an interaction effect was found between Time and Group, showing that the P3 amplitude decreased from pre- to post-test in the sham group. One of the reasons could be that the inhibitory effect was caused by a learning effect. Studies (Hajcak et al., 2005; Sailer et al., 2010; Yeung et al., 2009) have shown that the P3 is very sensitive for learning effects in repeated measure design, and that the amplitude becomes smaller when target probability increases (Donchin & Coles, 1988; Duncan-Johnson et al., 1977; Groen et al., 2007; Jongsma et al., 2006; Kompatsiari et al., 2016; Lindín et al., 2004; Radlo et al., 2001).

Results for the LPP showed an increase in amplitude for the active tDCS group in both emotion conditions, although active tDCS did increase LPP amplitudes as expected, we did not expect that the LPP amplitude would also increase for neutral pictures. An explanation for this finding would be that modulating activity in the vmPFC resulted in a modulation effect in social interactions generally, regardless of the emotions displayed in the pictures. To test this hypothesis, we performed an additional analysis on the filler pictures (i.e., neutral objects) and indeed found no significant effects of tDCS on either the P3 or LPP following viewing filler pictures. Thus, even though the results did not differentiate between neutral or aggressive pictures as proposed, the current results are in line with previous research (Cohen, 1988; Hajcak et al., 2005, 2009; Sailer et al., 2010; Samani et al., 2013; To et al., 2018) indicating that the positive slow potentials indicate an increased emotional regulation and affective experience (see the supplementary materials (S7) for detailed description of the additional analysis).

Our model of current flow indicated that the HD-tDCS stimulation reached the vmPFC with intensity sufficient to induce an effect of the stimulation as also described in earlier studies (Francis et al., 2003; Geuzaine & Remacle, 2008; Liu et al., 2018). We do acknowledge that a factor of inter-individual variability should also be taking in consideration. Future studies could implement specific algorithms or potential biophysical modeling to individualize montage and take into account the potentially influencing factors contributing to the direction of current flow (Bikson et al., 2004; Dmochowski et al., 2017; Huang et al., 2017; Metwally et al., 2015; Rahman et al., 2013).



This paper has several strengths. First, this study is the first to find a decrease in aggression in forensic patients using an EEG-tDCS design. Other studies (Choy et al., 2018; Molero-Chamizo et al., 2019; Nitsche & Paulhus, 2000) that have shown reduced aggression with tDCS did not include EEG in their protocol, although of significant importance for monitoring the activity of the brain region of interest. Second, we used a design that was double blind and with multiple sessions of tDCS. It has been demonstrated in several studies (Afifi et al., 2013; Dell’Osso et al., 2012; Mancuso et al., 2016; Monte-Reis et al., 2009; Silva et al., 2013; Song et al., 2019; Valero-Cabré et al., 2008; Zanardi et al., 2020), that modulation is most effective after multiple sessions, because the induction of synaptic plasticity in the cortex requires a multiple time stimulation to be effective and to mediate the durability of neocortical circuits (Rushmore et al., 2013). Third, in the current study the current flow montage sufficiently reached the vmPFC. We used HD-tDCS which is found to have a better focality (Bikson et al., 2012; Rushmore et al., 2013) than conventional tDCS, and as a result reaches deeper regions of the brain (Edwards et al., 2013; Hess, 2013; Faria et al., 2011; Manuel et al., 2014; Nakamura-Palacios et al., 2016; Tanaka et al., 2013; To et al., 2018). Fourth, compared to previous studies (Faria et al., 2011; Tanaka et al., 2013) we optimized the intervention by enhancing the brain state of interest during the tDCS sessions (Nissim et al., 2019).

Albeit promising, our findings should be interpreted in the context of limitations that are important to consider in future studies investigating this matter. First, we found a significant decrease in the P3 amplitude for the sham condition. Future studies should try to correct for learning effects on the P3 by implementing slightly different pictures in the viewing task or have a longer period from pre to posttest. Second, it might not just be the impairments of the PFC that influence aggression and antisocial behavior, but neural network disruptions associated with that behavior (i.e., Raine, 2019). Therefore, it would be relevant to investigate neural networks and functional connectivity in understanding aggressive behavior in future studies. Third, future studies should highlight the role of tDCS as a modulator, the effects of tDCS on synaptic activity as according to the BCM rule (Abraham & Tate, 1997; Bienenstock et al., 1982; Hassanzahraee et al., 2018; Karabanov et al., 2015; Krönberg et al., 2018) and using potential biophysical modeling to individualize montage. Last, in this study, no long-term effects of the tDCS intervention were examined, nor effects on violence in real life. Future studies should address these issues to further support the effectiveness of this tDCS intervention.

Conclusion

The development of successful evidence-based therapeutic interventions in forensic mental health is crucial for reducing violence risk. To our knowledge this is the first study investigating HD-tDCS as an intervention to increase empathic abilities and reduce violent behavior in a forensic sample, and that also examined the effects on electrophysiology. Our results showed that multiple sessions of HD-tDCS targeting the vmPFC resulted in reduced aggression. In addition, this modulation also resulted in increased LPP amplitudes following aggressive pictures, indicating an increase in attention to, and emotional evaluation of scenes depicting victims of aggression. Hence, our results significantly improved our understanding of the neural correlates of aggression posed by violent individuals and although it has to be interpreted with caution when implicating in a real-life setting, give support for the effectiveness of tDCS as treatment intervention in forensic patients.



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Supplementary Material

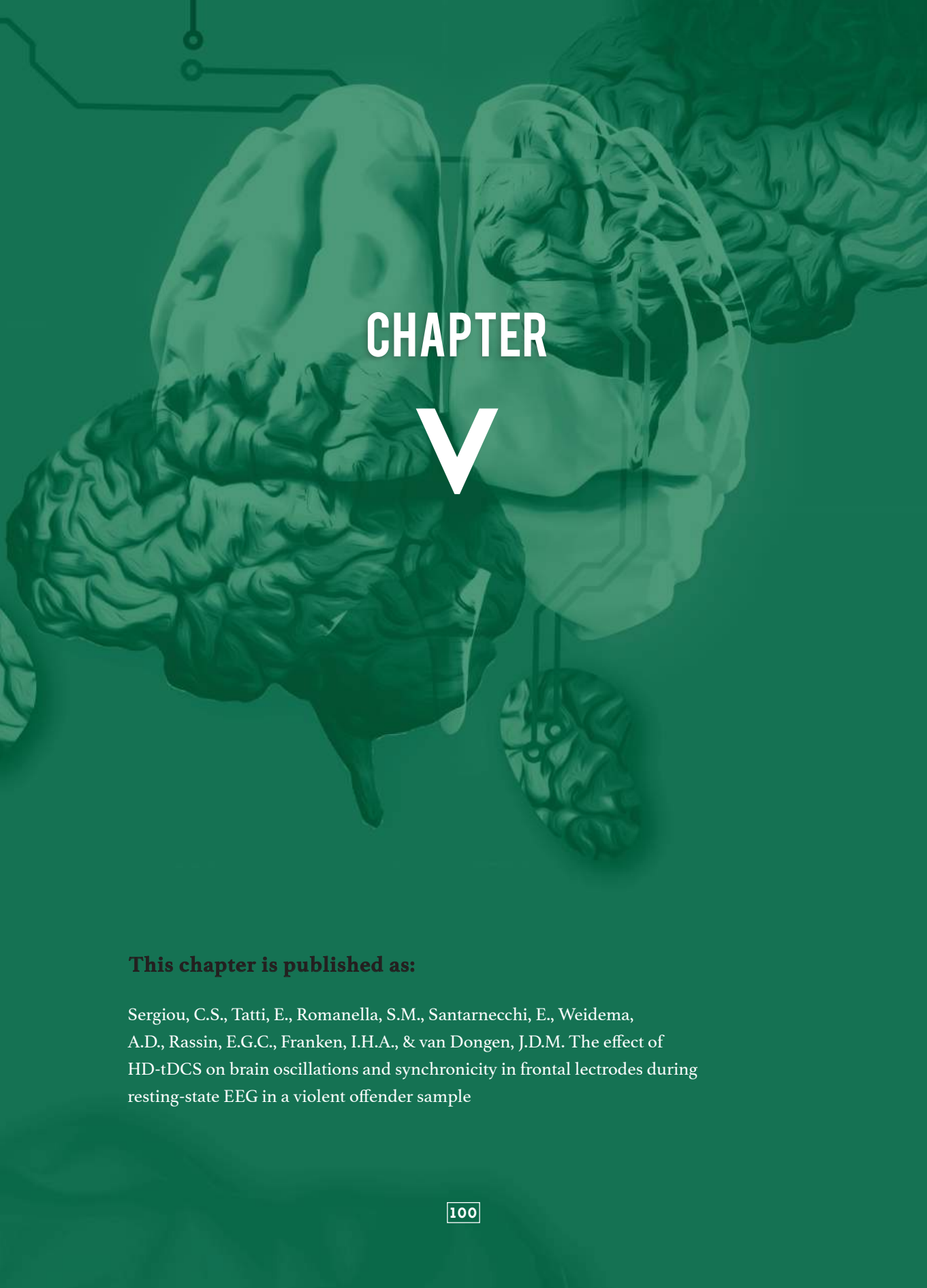
Biophysical modeling and HD-tDCS simulation

The induced E-field of the HD-tDCS montage was computed in SimNIBS v3.2, an open-source simulation package that integrates segmentation of MRI scans, mesh generation, and FEM E-field computations (Thielscher et al., 2015). The software provides a realistic volume conductor head model, created as default in FEM model generated using the T1-and T2-weighted images and segmentation from the SimNIBS example data set. The data sample was acquired from a healthy subject within the approval of the Ethics Committee of the Medical Faculty of the University of Tübingen [15]. The data corresponds to a healthy subject (Ernie), which includes white matter, gray matter, cerebrospinal fluid, bone, and scalp tissue volumes. In our simulation we kept the isotropic conductivities given by default (Thielscher et al., 2015). The final mesh, comprehensive of gray and white matter, scalp, bone, and cerebrospinal fluid, comprises approximately 200,000 nodes and 3.6 million tetrahedral elements (see Windhoff et al., 2013 for further modeling details).

E-field distribution was computed within the SimNIBS model. We modeled the electrodes starting from the set used in the experiment, as small cylinders, 6 x 1cm radius ($\pi \text{ cm}^2$), with homogeneous conductivity and added to a 0.5mm in height corresponding approximately to electrode gel. The placement of the electrodes followed the 32 standard 10-20 Electroencephalography (EEG) system on the scalp with anodal high definition transcranial direct current stimulation (HD-tDCS) with electrodes over the Fpz (2mA) and cathodal tDCS over AF3, AF4, F3, F4 and Fz (-0.4mA each). The resulting Norm E-field and Normal E-field distribution is shown in Gmsh v4.7.1 (Geuzaine & Remacle, 2008) with an output range from 0 to 0.25 V/m.

We chose the measure of NormE in order to test the absolute intensity of the E-field induced, but this only give us an information of the strength of the electric field as vector, and therefore irrespective of the direction. Because we are interested in activating the vmPFC thanks to the anodal current in FPz, we also computed the NormalE of the E-field (see Figure 1, panel C). The resulting map represents the excitatory stimulation in ventral prefrontal areas, opposed to the cathodal stimulation spreading over the dorsal frontal and parietal regions.





CHAPTER V

This chapter is published as:

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THE EFFECT OF HD-tDCS ON BRAIN
OSCILLATIONS AND SYNCHRONICITY
IN FRONTAL ELECTRODES DURING
RESTING-STATE EEG IN A VIOLENT
OFFENDER SAMPLE

ABSTRACT

Violence is a major problem in our society and therefore research into the neural underpinnings of aggression has grown exponentially. Although in the past decade the biological underpinnings of aggressive behavior have been examined, research on neural oscillations in violent offenders during resting-state electroencephalography (rsEEG) remains scarce. In this study we aimed to investigate the effect of HD-tDCS on frequency power, asymmetrical activity and synchronicity in the frontal brain region of a violent offender sample.

Fifty male violent forensic patients were included in a double-blind sham- controlled randomized study. The patients received 20 minutes of HD-tDCS two times a day on five consecutive days. Before and after the intervention the patients underwent a rsEEG task.

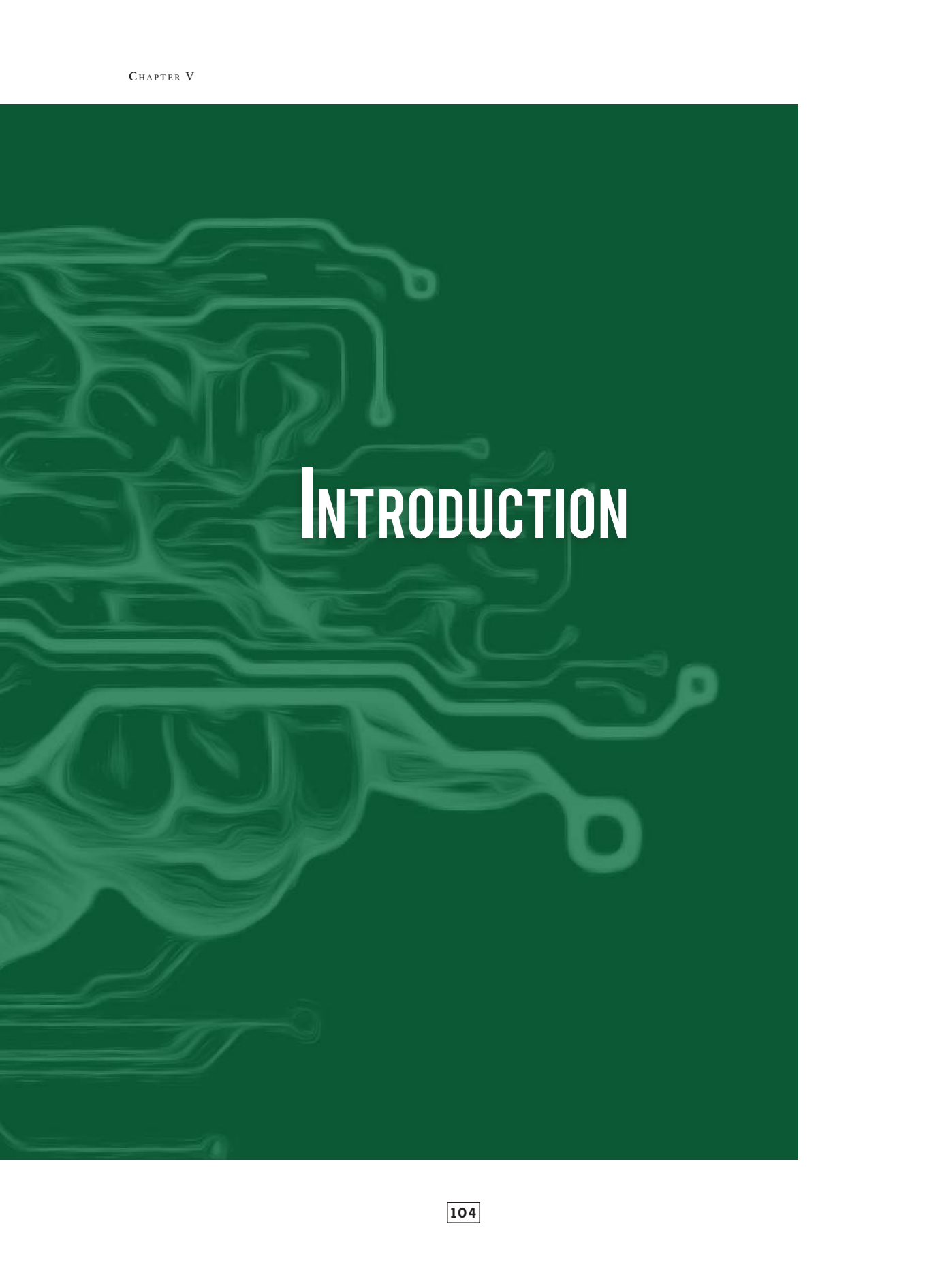
Results showed no effect of the tDCS modulation on the power in the frequency bands. Also, no increase in asymmetrical activity was found. However, we found increased synchronicity in frontal regions in the alpha and beta frequency band indicating enhanced connectivity between the electrodes as a result of the tDCS-intervention.

Despite support for the effect of HD-tDCS on enhancing synchronicity in violent offenders, further research is needed. In order to maximize the targeted effect of tDCS, the interaction of modulation with the dynamics of endogenous activity needs to be examined comprehensively to be better understood. We suggest with caution that HD-tDCS could be an appropriate method to restore affected brain connectivity within violent offenders.

Keywords:

High Definition-transcranial Direct Current Stimulation; resting-state electroencephalography; violent offenders; brain oscillations; frontal brain region;

INTRODUCTION

An abstract, light green circuit-like pattern on a dark green background. The pattern consists of various lines, loops, and nodes, resembling a complex network or a stylized map. It is positioned on the left side of the page, with the word 'INTRODUCTION' centered over it.

Violence is a major problem in our society (Cunha-Bang et al., 2017; Grochowska and Kossowska 2012; Mooney and Daffern 2011, 2015), and consequently, research into the neural underpinnings of aggression has grown exponentially. Studies on aggressive behavior indicate that dysfunction within the temporal and frontal lobe is associated with violence (Convit, 1991; Gatzke-Kopp et al., 2001; Krynicki, 1978; Tancredi & Volkow, 1988; Volkow & Tancredi, 1987; Volavka, 1990; Wong et al., 1994), and predicts later criminal behavior (Lewis et al., 1985; Raine et al., 1990). Other studies have suggested that the prefrontal cortex (PFC) is the core brain structure in regulating aggression, particularly the ventromedial prefrontal cortex (vmPFC; Kose et al., 2015; Marxen et al., 2016; Sergiou et al., 2020a, Sergiou et al., 2020b) and deficits in the vmPFC could lead to increased violent behavior (Blair, 2001, 2005, 2006). For instance, a review study by Bufkin and Lutrell (2005) described at least 17 neuroimaging studies of aggressive, violent or antisocial individuals demonstrating that diminished functioning of the prefrontal cortex is associated to their aggressive behavior.

One way for studying dysfunctions in these brain structures is using resting-state electroencephalography (rsEEG). In the past decades, rsEEG has received increased attention due to its unique advantages in investigating brain connectivity and characterizing changes in brain functions among different psychiatric disorders. Resting-state is the state in which an individual is awake, but not engaged in a specific task allowing to reflect the intrinsic activity of the brain (Duan et al., 2020; Deco et al., 2011; Greicus et al., 2003). Temporal dynamic patterns of the brain activity in resting state measured by the electroencephalography (EEG) are suggested to change more rapidly than can be detected using techniques such as functional Magnetic Resonance Imaging (fMRI; Baker et al., 2014; Deco & Corbetta, 2011; Siegel et al., 2012). In addition, EEG is a favorable technique to measure temporal dynamics because it is more suited to investigate the mechanisms underlying neural communication as EEG can directly measure the electrophysiological activity (Ganzetti & Mantini, 2013; Pfurtscheller & Lopes da Silva, 1999). In that matter it can demonstrate the organization of spontaneous brain oscillations and interactions between different networks with high temporal resolution (de Pasqualle et al., 2016; Lui et al., 2017; Siegel et al., 2012).

rsEEG enables the investigation of electrical fluctuations in neural signals (Howells et al., 2018; Jorge et al., 2014), allowing brainwave patterns during rest and thought processing to be detected (Nasser et al., 2019). These neural oscillations can be divided into different frequency bands, i.e., the delta wave (δ ; 0.3–3.5 Hz), theta wave (θ ; 4–7.5 Hz), alpha wave (α ; 8–13 Hz), beta wave (β ; 14–30 Hz) and gamma wave (γ ; >30 Hz), which are linked to different emotional states and cognition (Di et al., 2018). Various formations of neural synchrony in oscillations over the different frequency bands are proposed to be the major structure of neural integration (Munia & Aviyente, 2019). Studies investigating



the electrophysiological estimation of neural activity suggest that these different frequency bands can be linked to distinct central roles (da Silva, 2013) and that the oscillations create a synchronization over various brain regions to endorse cognitive processing (Engel et al., 2013; Hall et al., 2014; Schölvinck et al., 2013; Stam, 2014; Tewarie et al., 2015).

With regard to criminal behavior and violence, previous research has shown decreased arousal in the frontal brain regions (Decety et al., 2015); demonstrating enhanced slow-wave activity of the delta (0.5-3.5 Hz) and theta band (3.5-7.5 Hz) in the brain of antisocial individuals, and decreased alpha activity in criminals (Ellingson, 1954; Mednick et al., 1981; Raine et al., 1990). Common findings in these studies investigating EEG of antisocial and violent individuals show that excessive slow-wave activity is related to aggressive tendencies within these individuals (Hill & Waterson, 1987; Scarpa et al., 1997; Brower & Price, 2001; Moya-Albiol, 2004). These aggressive tendencies such as to aggressiveness, elevated anger and violence (Convit et al., 1991) that are linked to this slow-wave activity are all core elements of antisocial personality disorder.

Furthermore, besides the excessive increase of activity in slow waves, the reduced power in the alpha frequency band (8-12 Hz) also seems one of the more robust findings in individuals with antisocial and behavioral disorders (Calzada-Reyes et al., 2013; Convit et al., 1991; Fishbein et al., 1989; Forssmann & Frey, 1953; Lindberg et al., 2005; Keune et al., 2012). Alpha frequency has been linked to inhibition (Klimesch et al., 2007; Weisz et al., 2011) and is proposed to represent communication between different brain regions on a long range (von Stein et al., 2000). Given the impaired neural activity of violent offenders in the alpha frequency band, it is plausible that impaired brain communication reflects one of the neuroanatomical key factors to antisocial and violent individuals (Motzkin et al., 2011; Yang & Raine, 2009). In sum, these studies indicate that violent offenders show frontal region abnormalities in enhanced power in the theta frequency band and decreased alpha power and have an impaired brain communication or connectivity.

The origins of aggressive behavior remain highly complex, several patterns of functional connectivity (Hofman & Schutter, 2009; Hoptman et al., 2010; Nelson & Trainor, 2007) and frontal cortical activity (Harmon-Jones, 2003; Harmon-Jones et al., 2010) have been proposed in relation to aggressive behavior. Increasing evidence suggests asymmetrical involvement of the frontal hemispheres in aggressive behavior (for review see Harmon-Jones, 2003). Particularly, the left hemisphere has been found dominant in the involvement of aggressive behavior (Carvor & Harmon-Jones, 2009; Peterson et al., 2008; Schutter et al., 2008). Further support for the asymmetrical cortical activity demonstrates that the left-sided compared to right-sided excitability underlies aggressive tendencies, and link increasing activity with higher levels of experienced anger and aggressive behavior (Peterson et al., 2008). Furthermore, the studies of Hofman and Schutter (2009, 2012) demonstrated that the asymmetrical activity in the beta frequency band has been

found to be related to the development and expression of aggressive behavior (Hofman & Schutter, 2012) and in trait aggression in healthy individuals. In further support of this assumed link between asymmetry and aggression, the theory behind this indicates that relatively higher levels of left-sided as compared to right-sided cortical excitability is associated with increased approach-related motivational tendencies and decreased levels of avoidance-related motivational tendencies ((Harmon- Jones and Allen, 1998; Harmon-Jones, 2007; Schutter et al., 2008).

In addition to investigating the power and asymmetrical activity, an interesting technique to examine the synchronicity and communication between frontal regions is to measure the intercorrelation in phase between electrodes. Synchronization can be described as a type of connectivity measure and refers to a process in which linear or nonlinear oscillatory components reflect a property in activity over time, demonstrating a collective behavior (Boccaletti et al., 2018). In this way the interaction between brain areas can be detected (Sengupta et al., 2014). Several methods can be used to determine the intercorrelation in phase between electrodes, one of these methods is the Phase Locking Value (PLV; Lachaux et al., 1999), which determines the coherence in phase between electrodes. The PLV is found to have the capability to determine phase coherence independent of the spectral power of the recorded signals and to be a robust measure in case of noisy signals (Cohen, 2015). Finally, the PLV is a more sensitive measure of synchronicity when an investigation is driven by a hypothesis (Cohen, 2015) and best fitted to use when conducted rsEEG.

A compelling way of investigating brain oscillations is to combine EEG with non- invasive brain stimulation (NIBS), measuring changes in brain activity and capturing individual variability or specific disorder alterations could be possible (Benussi et al., 2020; Massimini et al., 2012; Voineskos et al., 2010). A promising NIBS device is High- Definition transcranial Direct Current Stimulation (HD-tDCS), which uses a high- definition technique targeting brain regions with higher focality to localize brain regions (DaSilva et al., 2015) and in this way, modulates deeper regions in the most optimal way (Edwards et al., 2013; Hess, 2013; Nakamura-Palacios et al., 2016; Manuel et al., 2014; Tanaka et al., 2013).

Although brain stimulation was traditionally used to target single brain regions of interest to modulate brain activity and neurological deficit, recently multiple studies demonstrate that the effect of the neuromodulation device is not limited to the stimulated region underlying the electrodes, but it can resonate into cortical networks and changes in functional connectivity (Di Lazzaro, 2004; Massimini et al., 2005; Siebner et al., 2001). As the neural signal reflects fluctuation of potentials resulting from postsynaptic potentials of cortical neurons which can be measured with EEG and therefore the brain areas directly or indirectly affected by the tDCS can be investigated (Mangia et al., 2014). Multiple studies have demonstrated that NIBS can modulate brain synchronicity, both during task



performance as in resting state (Alon et al., 2011; Chib et al., 2013; Ding et al., 2013; Grefkens et al., 2010; Keeser et al., 2011a; Meinzer et al. 2012, 2013; Peña-Gómez et al., 2012; Park et al., 2013; Polanía et al., 2011a, 2012; Stagg et al., 2013; Vanneste & de Ridder, 2011; Weber et al., 2014). Investigating the synchronicity may thus be very important to picture deficits in the brain in different neurological or psychiatric disorders and dysfunctions such as in aggressive behavior (Drysdale et al., 2017; Fox, 2018; Fox et al., 2012; Sheffield & Barch, 2016).

Several studies have investigated the change in neural activity in different frequency bands using neuromodulation during task-related EEG in healthy individuals (Antal et al., 2004a, 2004b, Adrolino et al., 2005; Notturmo et al., 2013; Polanía et al., 2012) and found increased functional connectivity patterns after anodal stimulation. Other studies found enhanced power in alpha and theta after anodal stimulation of the DLPFC, and demonstrate the feasibility of NIBS in modulating oscillatory brain activation (Zaehle et al., 2011a, 2011b).

However, fewer studies have investigated changes in oscillatory brain activity through tDCS during resting state (Adrolino et al., 2005; Spitoni et al., 2013; Zaehle et al., 2011b). Spitoni et al., (2013) found that anodal tDCS alters ongoing brain activity in the alpha frequency band in the postero-parietal cortex during rest. Another study of Mangia et al., (2014) found an increase in alpha and beta power during and after stimulation of anodal tDCS over the posterior-parietal cortex. Furthermore, other studies (Keeser et al., 2011a; Hampstead et al., 2013) demonstrated that tDCS increases the interaction between inter- and intra-cerebral cortexes and an enhancement of the coherence of the cerebral rhythm and functional organization at rest (Polanía et al., 2012). Since all the above-mentioned studies used healthy individuals, it would be interesting to expand these studies by investigate the deficits in the frontal region of violent offenders in resting-state to determine the effect of HD-tDCS in altering activity.

Although in the past decade the biological underpinnings of antisocial personality, violent behavior and crime have been received increased attention (i.e., Convit, 1991; Gatzke-Kopp et al., 2001; Krynicki, 1978; Tancredi & Volkow, 1988; Volkow & Tancredi, 1987; Volavka, 1990; Wong et al., 1994) in healthy subjects, investigating associated deficits in neural oscillations during rsEEG in a violent offenders remains scarce (i.e. Calzada-Reyes et al, 2013; Convit et al., 1991; Fishbein et al., 1989). Furthermore, using neuromodulation as a tool to modulate the brain oscillations in a violent offender sample measured in power, asymmetrical activity and phase-synchronization is not yet investigated.

In the current study we aimed to expand the scarce tDCS-rsEEG research in violent offenders by investigating whether HD-tDCS can modulate brain oscillations in a violent offender sample. The aim of this paper was threefold: (i) first we examined whether HD- tDCS targeting the vmPFC could modulate a change in power within the proposed

frequency bands (theta, alpha, beta) over our electrodes of interest (i.e., the prefrontal cortex). Based on previous literature we hypothesized an increase in activity the alpha frequency band (8-12 Hz), and a decrease in the activity of the slow frequencies in theta frequency bands (4-7 Hz) in the violent offender sample after the HD-tDCS intervention.

Second, (ii) asymmetrical activity in the frontal cortex has been linked to aggressive traits (i.e., Hortensius & Schutter, 2012; Schutter & Hortensius, 2009) in the alpha and beta frequency band. Therefore, we wanted to investigate whether the asymmetrical alpha and beta change from pre to poststimulation. We expected an increase for the activity in both frequency bands. Last, (ii) this study examined the phase- correlation or phase-locking value (PLV) between each possible pair of the eight frontal electrodes and compare this from pre to posttest to see whether the HD-tDCS intervention affected the strengthening of synchronization between the electrodes in the specific frequency bands. We hypothesized that there would be in an increase in the PLV in the alpha and beta frequency bands.



Materials and Methods

Participants

Data were collected as part of a study examining the effects of HD-tDCS in increasing empathy and reducing violent behavior in forensic substance use offenders (reported elsewhere see Sergiou et al., 2020a; Sergiou et al., 2021). A total of 50 male participants (mean age = 37.40 years, SD = 9.19 years, range: 22-62 years) were recruited from two departments of the division for forensic addiction mental healthcare of Antes in Poortugaal, the Netherlands. Twenty-one participants were recruited at the Forensic Addiction Clinic (FVK) and 29 from the Department of Forensic Care (AFZ). Inclusion criteria were: age 18-60, a good understanding of the Dutch language, diagnosed with an alcohol and/or cocaine substance use disorder (SUD) according to The Diagnostic and Statistical Manual of Mental Disorders (DSM-5; American Psychiatric Association, 2013), patients had to be abstinent and had to be sentenced for a violent offense as determined by one of the most used mandatory risk assessment tools in the Netherlands (HKT-R; Spreen et al., 2014). Exclusion criteria included: major neurological conditions, or major mental disorders, taking antipsychotics or other confounding medication. All participants gave written consent to participate in the study. The study was conducted in accordance with the ethical standard of the Declaration of Helsinki (General Assembly of the World Medical Association, 2014) and was approved by the Medical and Ethical Review Board of the Erasmus Medical Center Rotterdam.

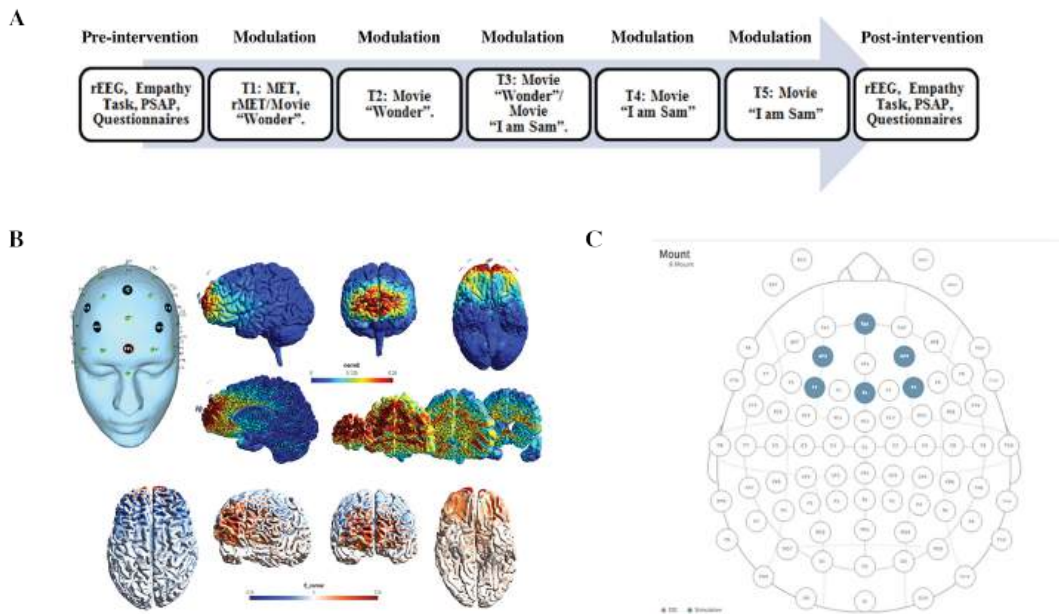


Figure 1. Procedure of the intervention week with the High-Definition transcranial Direct Current Stimulation (HD-tDCS). A). Intervention week, B). Biophysical modelling montage targeting the ventromedial Prefrontal Cortex (vmPFC), C). Electrode placement of the HD-tDCS montage.

High-Definition transcranial Direct Current Stimulation

HD-tDCS was administered with the CE-certified Neuroelectronics Starstim8 (Neuroelectronics Barcelona, SLU; 2011), and the protocol was based on the evidence-based guidelines of LeFaucheur et al. (2017). Before the commencement of the study, the HD-tDCS montage optimization was based on the current flow modeling of the NIC-software of the tDCS Starstim8 system (Neuroelectronics Barcelona, SLU; 2011), see the protocol Sergiou et al. (2020a) for a detailed description. In addition, the induced E-field of the HD-tDCS montage was computed in SimNIBS v3.2 (Thielscher et al., 2015).

The currents were transmitted through six circular Ag/AgCl PiStim “High Definition” electrodes (1 cm radius) that were applied with conductive gel. For the active condition the HD-tDCS device was programmed for stimulation targeting the vmPFC, with 2 mA tDCS during 20 minutes. The sham condition followed the exact same procedure, only with 30 seconds ramping up and down of the tDCS currents at the beginning and end of the protocol, based on earlier research indicating this method being effective for blinding (Gandiga et al., 2006). The anodal electrode was placed on the Fpz location, and the five return or cathodal electrodes were placed on AF3, AF4, F3, Fz and F4 (see Sergiou et al., 2021 for electrical field modeling and biophysical modeling). In Figure 1 an overview of the HD-tDCS montage is presented.

Resting-state EEG (rsEEG) task

To assess the brain state in resting condition using rsEEG, the patients underwent a rsEEG task with eyes open condition (EO) and eyes closed (EC) condition oscillated every minute, for four consecutive minutes.

Data were collected when participants were sat comfortably with their eyes fixated on the screen. The recording lasted for about four minutes and was followed by a passive viewing empathy task (see Sergiou et al., 2020a; Sergiou et al., 2021). Since the cue for switching between the conditions of the EO and EC conditions was based on the sound of a bird (2130ms) per trial, this was removed before further analysis, since this can evoke an auditory response.

EEG recording and preprocessing

Activity of the EEG was recorded using a mobile version of the BrainProducts (Live Amp, 2018) Active-Two System amplifier (Amsterdam, the Netherlands). Resting-state was recorded using a 4min eyes open-eyes closed-eyes open-eyes closed recording session. Thirty-two electrodes have been placed on the scalp of each participant following the international 10-20 EEG system. Two other additional electrodes were placed vertically above and beneath the left eye (electro-oculogram; EOG), the electrodes for the left and right mastoid placement were incorporated in the EEG cap.



The recording electrodes were positioned according to the extended 10–20 system and digitized at a sampling rate of 500 Hz. The impedance for all electrodes was maintained below 10 K Ω . The ground electrode was placed at the Pz position in both systems. Preprocessing was conducted in the open-source EEGLAB toolbox (Delorme and Makeig 2004) and custom MATLAB 2018b scripts (the Mathworks, Inc.).

The preprocessing pipeline included 1–70 Hz bandpass filtering, a notch filter at 50 Hz. Epochs of two seconds were created, resulting in four segments per participant (pre and post, EO and EC). Data were excluded based on data segments containing artifacts (EEGLAB, personalized pipeline) and the use of independent component analysis (ICA) removal of eye movement and muscle artifacts. If any of the two data sets (pre and post) of EO and EC, recorded from a given participant needed to be removed due to excessive artifacts or missing signal (i.e., if the number of remaining trials was less than 50%), that data was excluded from the analysis. Similarly, data from participants were removed if the signal from their resting-state recording that remained after cleaning had a total length of less than 240s. The signals were re-referenced to average reference at the end of the script. Ultimately, we included only the EC condition for further analysis, since the EO condition was not of good quality. In addition, studies (i.e., Barry et al., 2007) recommend using eyes-closed resting EEG as a baseline for all studies which do not use visual stimuli in their task, indicating that EC is a more valid way of measuring resting-state in our study.

Spectral Power Analysis

The power of three EEG bands (theta, 4–7 Hz; alpha, 8–13 Hz; and beta (14–30), was calculated for each electrode individually (averaged over trials) for each window of spectral power. The frequency-domain analysis was performed using the Darbeliai EEGLAB extension (Baranauskas, 2008), which makes use of a Fast Fourier Transform (FFT) algorithm to extract the absolute ($\mu\text{V}^2/\text{Hz}$) power density, the relative (%) power density and the mean frequency (Hz) within each of the three frequency bands. The absolute power (AP) can be described as the integral of all of the power values in the frequency range of that band. Relative power (RP) is the absolute power calculated over the four frequency bands together (1–32 Hz). Mean (total) frequency (Hz) was also derived from the entire analyzed spectrum (1–32 Hz). Mean frequency was calculated over the following electrodes for the theta, alpha and beta frequency band: F3, F4, F7, F8, Fz, Fp1, FC1 and FC2. Asymmetrical Frontal Activity

Measures of inter-hemispheric (absolute) power asymmetry for each band were computed for three homologous sites (F3–F4, F7–F8, FC1–FC2) and mean spectral power for left vs right frontal activity was measured by averaging the four corresponding electrodes (see Schutter et al., 2012). Accordingly, the asymmetrical power distribution was calculated by $\text{powerasymmetry} = (\text{power}^{\text{right}} - \text{power}^{\text{left}}) / (\text{power}^{\text{right}} + \text{power}^{\text{left}})$ (Schutter et al., 2008)

meaning that a greater value or increase in asymmetrical activity is right over left cortical activity. In this way there can be controlled for individual differences in skull thickness, skull-to-cortex distance and orientation of the underlying cortical tissue (Herbsman et al., 2009).

EEG Connectivity Analysis

To examine the connectivity between the frontal electrodes from pre to posttest, we used the Phase Locking Value (PLV), to calculate the phase symmetry between couples of electrodes (Fz, Fp1, F3, F4, F7, F8, FC1, FC2). We calculated this separately for the theta (4-7 Hz), alpha (8-13 Hz) and beta (14-30 Hz) frequency band and the two measurement moments (pretest vs. posttest). The PLV was calculated using a customized script in Matlab. Phase locking between selected electrode pairs of recording sites was calculated based on the PLV-method described in (Lachaux et al., 1999).

We extracted EEG whole-brain connectivity over 26 numbers of couples of electrodes in three EEG frequency bands (theta, alpha, beta) using a sliding windows approach. We chose to look at data for 1.998 sec continuously. For each frequency band, we then computed the Phase-Locking Value (PLV), a phase synchronization measure that corresponds to the mean phase difference between signals from two electrodes. This allowed us to investigate the phase synchronization between two narrow-band signals.

To do so we use the Hilbert transform: $z_i(t) = s_i(t) + jHT(s_i(t))$

We then computed the relative phase between two signals:

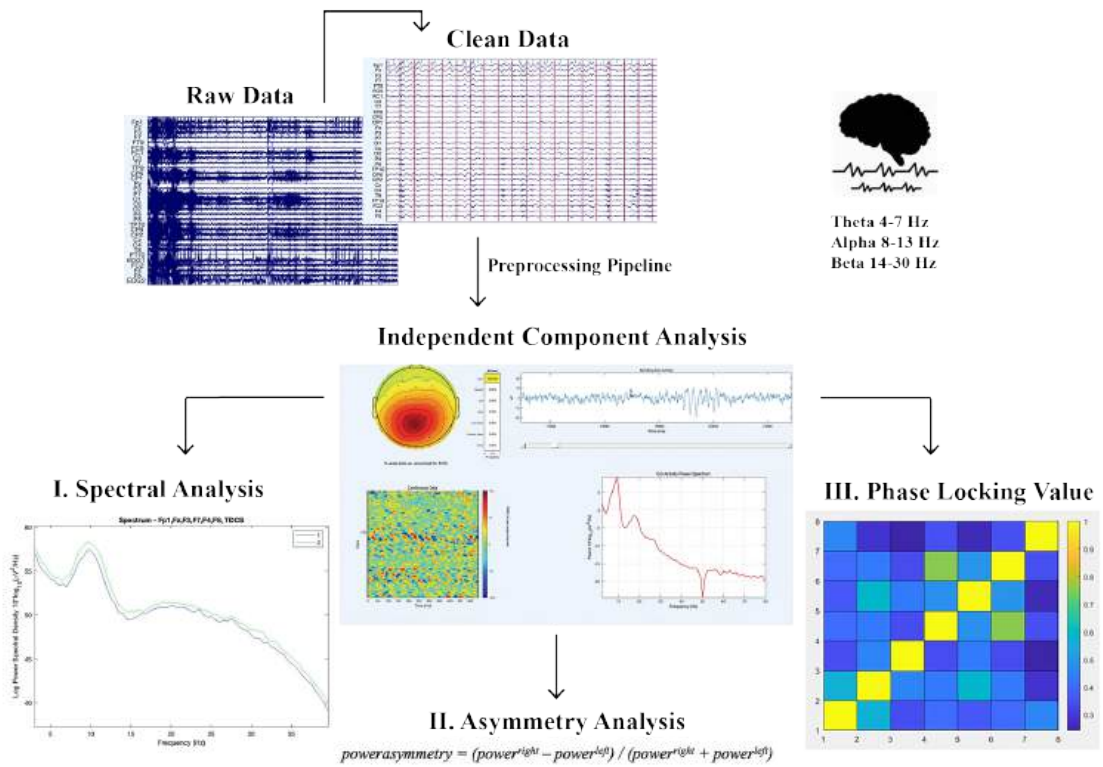
$$\Delta\phi(t) = \arg \left(\frac{z_1(t)z_2^*(t)}{|z_1(t)||z_2(t)|} \right)$$

The instantaneous PLV is computed with the formula: $PLV(t) = \left| E \left[e^{j\Delta\phi(t)} \right] \right|$

We created a matrix of 26 PLV correlations of chosen electrodes in the prefrontal area (Fz-F3, Fz-F4, Fz-F7, Fz-F8, Fz-FP1, Fz-FC1, Fz-FC2, FP1-F3, FP1-F4, FP1-F7, FP1-F8, FP1-FC1, FP1-FC2, F3-F4, F3-F7, F3-F8, F3-FC1, F3-FC2, F4-F7, F4-F8, F4-FC1, F4-FC2, F7-FC1, F7-FC2, F8-FC1, F8-FC2). The PLV indicates the inter-trial variability of the phase association between pairs of brain regions as a function of time. Values closer to 1 indicate strong synchronicity (i.e., phase-locking) between two regions within the indicated time window, whereas values closer to 0 indicate phase variation between two electrodes, and thus represent a weak synchronicity between regions (Spooner et al., 2020).

For an overview of the described analysis see Figure 2.



**Figure 2.**

Flowchart of study design for the pretest and posttest. I) Spectral analysis (power), II) Asymmetrical activity analysis, III) Phase correlation analysis (PLV).

Experimental procedure

For the experimental procedure of the study see the protocol paper (Sergiou et al., 2020a) and our other recently published paper (Sergiou et al., 2021).

Statistical Analyses

We used analysis of variance for repeated measures (ANOVA) to examine if there was a difference between active tDCS and the sham condition for our main region of interest (i.e., the prefrontal cortex). Data are reported as means and standard deviations. Mauchly's test was used to test for sphericity, and the Greenhouse–Geisser correction was applied if necessary. A *p*-value of < 0.05 was considered significant. Statistics were performed using the SPSS 25.0 software (Statistical Package for Social Sciences, SPSS Inc, Chicago)

A three-way repeated-measures ANOVA was used to test for the effect of condition (active tDCS vs. sham) $\times$ lead (Fp1, Fz, F3, F4, F7, F8, FC1, FC2) $\times$ Frequency band (theta, alpha, beta) calculated in rsEEG absolute power (μV^2).

For the calculation of the asymmetrical activity three repeated measures ANOVA for each frequency band (theta, alpha, beta) were conducted to test whether this activity changed from pre to posttest for the two groups (tDCS vs. Sham).

To compute phase synchronization, we extracted the PLV using the method described by Lachaux et al., (1999) and calculated the difference in symmetry between the electrode pairs that are part of the frontal region (Fz, Fp1, F3, F4, F7, F8, FC1, FC2) using a customized MATLAB script. We measured the connectivity between the electrodes using the Phase Locking Value (PLV). We conducted separate repeated-measures ANOVAs for the PLV of each chosen frontal electrode pair over Time (pretest vs posttest) with group (tDCS vs Sham) as a between subjects factor.



Results

Overall, neither side effects nor any relevant discomfort was observed during or after the HD-tDCS intervention (see Sergiou et al., 2021).

Effect of tDCS on the Power in frequency bands

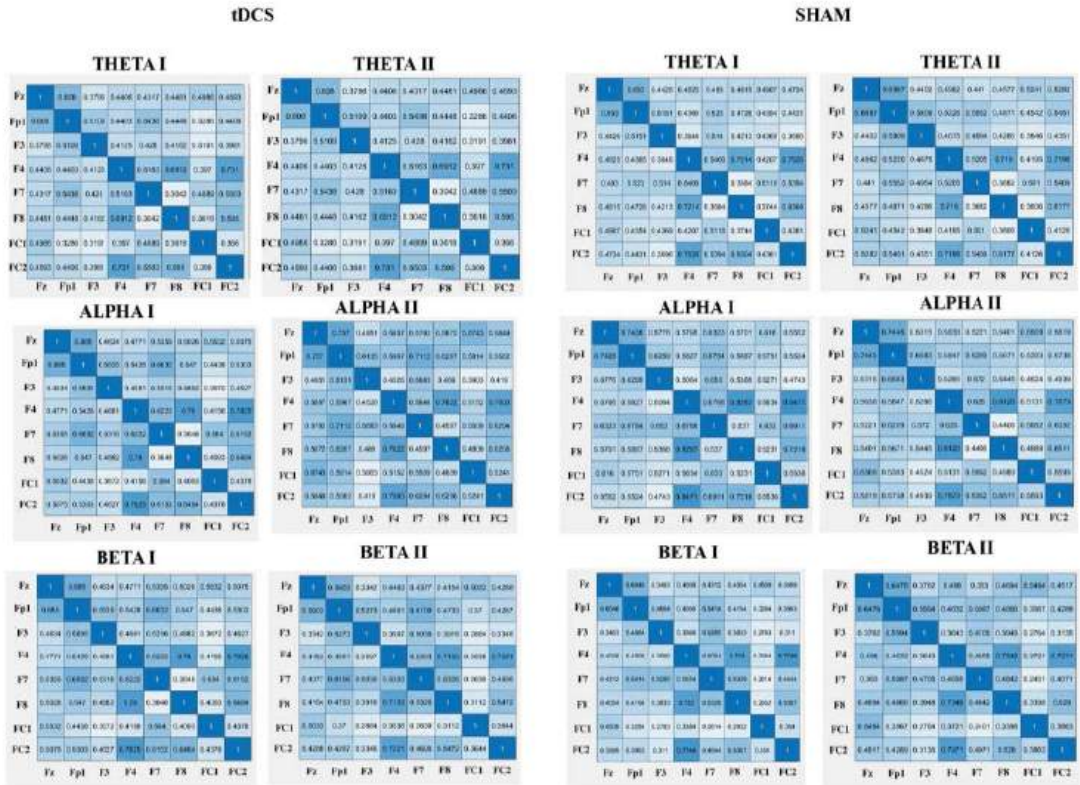
The results of the analysis on the absolute EEG power revealed significant main effects of lead ($F(1.79,73.7) = 1.09, p < .01, \eta^2 = .34$), and frequency band ($F(1.31,53.7) = 22.7, p < .01, \eta^2 = .36$). In addition, we found a significant interaction effect for lead x frequency band ($F(1.64,67.8) = 20.9, p < .01, \eta^2 = .34$). Because we did not find any significant results for time or condition, we did not further investigate these effects.

Effect of tDCS on the Asymmetrical Frontal Activity

Regarding the asymmetrical activity we did not find a significant effect for the theta or alpha frequency bands. For the asymmetrical activity in the beta frequency, we did not find a significant main effect for Time, but we did find a significant interaction of Time vs. Group $F(1,41) = 4.39, p = .042, \eta^2 = .09$). An exploratory post-hoc analysis revealed that the effect was significant between groups at the pretest. When we examined this in more detail, we found that for a large subgroup in the tDCS condition ($n=11$) the baseline values were starting at negative (indicating a greater left-sided activity) and that for this subgroup the increase was significant ($p < .001$) towards a left to right sided cortical activity from pre- to postintervention.

Effect of tDCS on the Connectivity (PLV)

Regarding the effects of tDCS on the PLV's, we did not find any significant results for the theta frequency band. However, for the alpha frequency band, we found a significant interaction for FPr-FCI alpha ($F(1,41) = 5.67, p = .022, \eta^2 = .12$) in tDCS vs. sham in pre- vs. poststimulation. The tDCS groups showed an increase of connectivity as reported by higher PLV values from pretest ($M = .44, SD = .26$) to posttest ($M = .58, SD = .19$). In addition, we found a significant interaction for F7-FCI alpha ($F(1,41) = 6.15, p = .017, \eta^2 = .13$). The tDCS group showed an increase of connectivity as reported by higher PLV values from pretest ($M = .36, SD = .22$) to posttest ($M = .46, SD = .21$). Regarding the beta frequency band, we found a significant interaction for F7-F8 beta ($F(1,41) = 4.294, p = .045, \eta^2 = .10$) with an increase for the tDCS group from pretest ($M = .49, SD = .13$) to posttest ($M = .53, SD = .12$) and a decrease for the sham group from pretest ($M = .53, SD = .12$) to posttest ($M = .48, SD = .12$).



Discussion

Using HD-tDCS targeting the vmPFC, we investigated the effect of brain modulation on neural oscillations during rsEEG in a sample of violent forensic patients. The aim of the current study was threefold.

First, we investigated the effects of the modulation on the power of frontal electrodes in the theta, alpha, and beta frequency band. We did not find an effect of the tDCS modulation on the brain activity by the frontal electrodes. This finding may be explained in several ways: first, it could be possible that our specific focus on frontal electrodes contributed to our findings. Although the importance of the frontal region in violent offenders has emerged from multiple studies, it could be that the modulation in spectral power was more prominent in other regions of the brain. Future studies should investigate a whole brain analysis to examine if the effect of tDCS on power might be detected in other brain areas. For example, in the study of Konicar et al. (2021) they found the excessive slow-wave activity limited to temporal and parietal regions instead of frontal, this could explain we did not find an effect of the intervention in the theta frequency band from pre- to posttest. Future studies will have to determine to what extent these characteristically aberrant brain processes in power in the frequency bands of antisocial individuals are specifically linked to certain regions, which could provide important implications for treatment.

Our second aim was to investigate the effects of the modulation on the asymmetrical activity of frontal electrodes. For the theta frequency band, we did not find an effect of the tDCS modulation on the asymmetrical activity. Our failure to find an effect of the modulation on asymmetrical activity in the alpha frequency band contradicts our hypothesis, but closer inspection of the literature shows that this is not uncommon. For example, our null finding is in line with the findings reported in (Hofman & Schutter, 2012). In their study, they did not find any evidence for an association between tDCS and asymmetrical alpha activity and explain this by the assumption that alpha oscillations tend to be dominant over the posterior scalp locations during resting-state EEG recording. Therefore, a posterior-to-anterior direction of the configuration of the posterior dipole could explain the absent association between the alpha oscillations and frontal excitability (Hofman & Schutter, 2012) and this could also explain or lack of evidence for this in the current study. Another explanation could be that the association between asymmetrical activity in forensic patients may also be dependent on individual characteristics in personality which might not have been studied in the present study (Knyazev et al., 2018). For example, in the study of Savostyanov et al. (2009) the synchronization of electrophysiological cortical activity posttest was modulated by trait anxiety.

Last, it could be the timing of the sessions on which day since other studies found that these variables are correlated with asymmetrical frontal cortical activity (i.e., Peterson & Harmon-Jones, 2009). Therefore, more research is needed to better understand state variance in resting baseline EEG measures.

For the beta frequency band, we observed an interaction between asymmetrical activity and group at the pretest, but we did not find increased asymmetrical activity using the tDCS intervention over time for the complete tDCS group. The results partly confirm our hypothesis and are therefore partly consistent with previous studies (Hoffman & Schutter, 2012; Schutter et al., 2008; Schutter & Hofman, 2009) that also reported an increase in beta asymmetrical activity after neuromodulation. Furthermore, the results provide additional support for the finding that beta activity and faster brain oscillations are related to the underlying cortical activity (Lubar, 1997). The results of the current study on asymmetrical activity in the beta frequency band suggest the possibility to alter cortical activity after a tDCS intervention and should be investigated more in-depth in future studies due to our exploratory findings for the high left-cortical activity subgroup. This is also in line with previous research suggesting that more left than right-sided activity may interact with interventions in terms of positive treatment outcomes (Smith et al., 2017).

Finally, our third aim was to investigate the effect of the modulation on the resting-state connectivity (expressed as PLV) between frontal electrodes. For the theta frequency band, we did not have a hypothesis and we did not find an effect. Our finding of an increase in resting-state connectivity from pre- to posttest in the alpha and beta frequency band is consistent with our hypothesis. We found a significant increase for the tDCS group in the alpha frequency band for the electrode pairs FC1-Fp1 and FC1-F7. Both these electrode pairs are in the left hemisphere, this means that the tDCS intervention enhanced the connectivity between these electrodes in the alpha frequency range. Higher connectivity between the electrodes can be seen as increased functionality in controlling cognitive abilities, and with increased cognitive abilities it could be promising to enhance the firing within this hemisphere to decrease deficits in activity in violent offenders and their aggressive behavior.

For the beta frequency band, we found a significant increase for the F7-F8 electrode pair, this means there is an increase in connectivity indicating that the two hemispheres in the prefrontal areas are more functionally connected and with that more synchronous in their activity. Since the brain relies on the interhemispheric transfer of information to integrate behavior and cognition (Doron & Gazzaniga, 2008), for the larger part in the axonal transmissions between the two hemispheres sub-served by the corpus callosum (Banich, 1998). Thus, decreased connectivity is associate with cognitive impairment (Babiloni et al., 2006; Keary et al., 2009; Kumar et al., 2009; Thiruvady et al., 2007), meaning that



increased connectivity affects the transfer between the hemispheres and with that interhemispheric connectivity is reinforced to influence cognitive performance and improve the functioning of the frontal brain region. Furthermore, to conclude results also indicated trends towards significance in the alpha and beta frequency bands in comparison with no results on the theta synchronicity, indicating that the modulation had an effect on the two frequency bands that are the most linked to frontal activity and our violent offender sample.

The outcomes of this study could provide important knowledge about dysfunctions in neural oscillations associated with aggression in offenders, and the possible modulation of this neural activity through HD-tDCS. Furthermore, it could contribute to the effectiveness of HD-tDCS as a supplementary treatment in forensic institutions.

Limitations

Some limitations on the generalizability and specificity of the present results must be noted. First, it must be acknowledged that the sample consisted of male forensic patients, and therefore we were not able to distinguish between a control non-violent sample and a violent offender sample. To further examine the brain activity of aggressive individuals, future studies should include a non-violent matched sample. Second, as shown in earlier studies (Hofman & Schutter, 2012) the characteristics of the EEG recording in the localization of cortical activity sources and the claims made in this study regarding the activity over the brain regions should be interpreted with caution to ensure the effects originates from the cortical tissue beneath the electrodepositions. Further studies are warranted to further investigate the causal relationship between tDCS-related behavioral performance enhancement and EEG activity by simultaneously record the EEG while modulating the brain region of interest. Despite modest support for the effect of asymmetrical beta frontal cortical activity on aggression and significantly enhanced connectivity, further research is needed.

Third, we decided to examine the EC condition while it could be that the EO condition would have interesting results that have not been analyzed in this study. Other studies used the EO condition to prevent participants from falling asleep (Boonstra et al., 2016) and found changes in mean frequency in rsEEG after tDCS. In contrast other studies (Jacobson et al., 2012a; Keeser et al., 2011b) examined resting state during the EC condition. The methodological differences may have affected the outcomes of this study as studies (i.e., Antal et al., 2007) have indicated that the aftereffects of tDCS depend on the physiological state during the administration of tDCS. Additionally, differences in the effects of tDCS intervention may also arise from variability in anatomy (Seibt et al., 2015).

Conclusion

The present study was the first to investigate modulation of neural dynamics using HD-tDCS in rsEEG of a forensic sample using HD-tDCS and analyzing three different aspects of the electrophysiological signals. We found no evidence for the effect of tDCS in the power of the theta, alpha and beta frequency bands. However, we find interesting effects in increased asymmetrical activity in the beta frequency band after an exploratory analysis in the tDCS condition, supportive for increased right frontal activity after the intervention. Furthermore, we found increased connectivity in frontal regions in the alpha and beta frequency band as a result of the intervention, indicating enhanced synchronicity between frontal regions. In order to maximize the targeted effect of tDCS, the interaction of modulation with the dynamics of endogenous activity needs to be examined comprehensively in the future to be better understood. Although future studies should further investigate the complex neural underpinnings of aggression, it can be suggested that tDCS is an appropriate method to restore affected connectivity within a violent offender sample and with that enhance cognitive functioning in the frontal region.

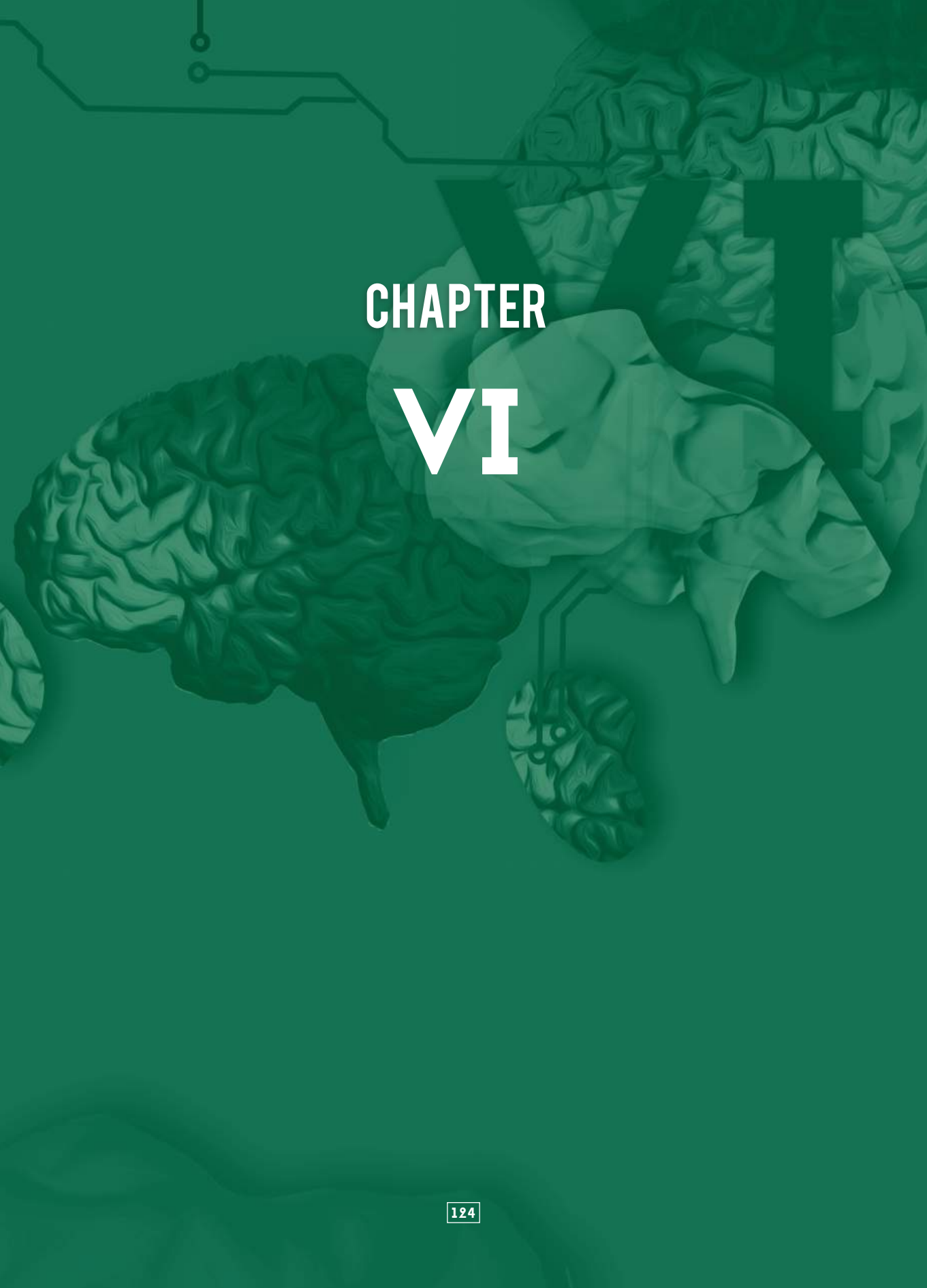


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A stylized illustration of a human brain in shades of green, with white circuit lines overlaid on it. The brain is shown from a side profile, with the cerebellum visible at the bottom. The circuit lines are composed of straight segments and small circles, resembling a printed circuit board. The background is a solid dark green.

CHAPTER

VI



VI

GENERAL DISCUSSION

Individuals with a substance use dependence have lower empathic abilities leading towards violence. This finding has been explained by the underlying brain abnormalities in the ventromedial Prefrontal Cortex (vmPFC). Since the treatment of this forensic patient group seems insufficient there is growing interest in a new add-on treatment option using tDCS. This dissertation explored the effects of HD-tDCS in a forensic sample with substance use dependence to see whether empathic abilities could be enhanced and subsequent aggression could be reduced. Furthermore, this dissertation investigated the behavioral and neurophysiological measures of aggression and empathy to get a better insight into the effects of tDCS in the treatment of forensic patients. To examine these aims we studied the effect of using multiple session HD-tDCS in a forensic sample and investigated the neural mechanisms underlying the aggressive brain. The ultimate goal of this dissertation was to investigate whether non-invasive neuromodulation, in specific the use of High-Definition transcranial Direct Current Stimulation (HD-tDCS), could be an effective intervention to reduce aggression in a forensic care setting.

Summary of Results

The second chapter investigated the existing literature in the use of tDCS in modulating empathy and aggression. The review describes several studies using the DLPFC as a target region to increase empathic abilities and reduce antisocial behavior. A few studies used the vmPFC as a target region, this could be because the studies used conventional tDCS and therefore could not reach deeper brain regions such as the vmPFC. Furthermore, studies did not investigate the association between empathy and aggression but solely focused on a subcomponent of either empathy or externalizing behavior. Several studies investigating tDCS modulating empathic abilities show some evidence that empathic abilities can be increased with anodal tDCS over the DLPFC. Regarding studies using tDCS in modulating externalizing behavior, most research has indeed been conducted using stimulation of the PFC. Studies differ substantially on parameter such as intensity, duration, electrode position, target location, session repetition, and timing. This could explain the inconsistent results provided by tDCS trials in the treatment of psychiatric disorders.

The third chapter describes the background, study protocol and planning in detail based on the investigated literature and to propose our aims to increase empathic abilities and reduce aggression in forensic patients.

In the fourth chapter the results of the HD-tDCS study were discussed. Although we didn't find significant results for trait or state empathy, we did find a significant decrease on the aggression task (PSAP) and the self-report aggression questionnaire (RPQ) in the active tDCS group. In addition, the HD-tDCS intervention was associated with a change in electrophysiological brain responses. We found a general increase in the LPP amplitude for social interactions and a higher LPP amplitude for the aggressive pictures vs. neutral pictures. This partly confirmed our hypothesis, showing that the positive slow potentials indicate an increased emotional regulation and affective experience. We did not find any results for the P₃ amplitude. The use and craving of substances seemed unaffected by the tDCS intervention, or at least with the questionnaires that we used to measure this.

As an additional measure of electrophysiological responses next to the ERPs, we explored the temporal dynamics in the brain of our forensic sample. Therefore, in the fifth chapter, we investigated if the HD-tDCS intervention also had an effect on the brain oscillations of the resting state EEG (rsEEG). We analyzed three frequency bands (theta, alpha, and beta) in the frontal electrodes. We examined the frequency power and the change from pretest to posttest. In addition, we investigated the change in asymmetrical activity in the alpha and beta bands. Furthermore, to determine the synchronization between the frontal electrodes and if this synchronicity was enhanced after the tDCS intervention we used a Phase Locking Value (PLV). In contrast to our expectations, we didn't find an effect of the HD-tDCS intervention on the power in the frequency bands. Also, the predicted increase in asymmetrical activity couldn't be observed. However, we found an increased synchronicity in frontal regions for the alpha and beta frequency bands indicating an enhanced connectivity between the electrodes as a result of the tDCS-intervention. Although future studies should further investigate the complex neural underpinnings of aggression, the present study provides a first indication that tDCS could be an effective method to restore affected connectivity within a violent offender sample and with that enhance cognitive functioning in the frontal region.



Discussion

In this chapter the results will be discussed per variable that was investigated in this dissertation. Furthermore, the results we found to improve treatment and future directions will be described.

Empathy

Empathy is a complex, psychological phenomenon that is crucial for understanding other people's cognitive and emotional processes in social contexts (Keuken et al., 2011; Krall et al., 2016; The, 2003; Yang et al., 2018; Young et al., 2010). Due to its complexity, universal consensus on its definition and conceptualization is not yet established. Although there is a large body of research dedicated to an enhanced understanding of empathy, studies differ enormously in their conceptualization. For example, Hetú and colleagues (2015) define empathy as a set of mental abilities to allow individuals to understand and detect affective states of other individuals. In contrast, other studies (The, 2003; Völlm et al., 2006) describe empathy as the ability to share and interfere emotional experiences of other individuals. These are just a few examples to indicate that the complexity of conceptualization also leads to a complexity in investigating empathy. Therefore, it is helpful to state that in our studies we conceptualized empathy as 'the multi-dimensional concept that can be described as the ability to understand the emotions of others and to share the mental states of other individuals in social contexts, to facilitate their motivation, action and affections' (Bahji et al., 2021; Hein et al., 2008; Hétu et al., 2012; Singer et al., 2006; van Dongen, 2020).

In addition, not only the conceptualisation is complex; several brain areas are involved in the neural regulation of empathy, such as the medial prefrontal cortex, dorsolateral prefrontal cortex, temporal junction (TJP), the inferior frontal gyrus (IFG), the inferior parietal lobe (IPL), the amygdala, the supplementary motor area, the anterior insula, and the prefrontal cortex (PFC) and the vmPFC (Carrington & Bailey, 2009; Fan et al., 2011; Gonzalez-Liencre et al., 2013; Lockwood, 2016; Miller et al., 2020; Morrison et al., 2004; Reniers et al., 2014; Saxe & Wexler, 2005; Santiesteban et al., 2012; Shamay-Tsoory & Aharon-Peretz, 2007; Shamay-Tsoory et al., 2009). In our studies, we focused on the vmPFC as the brain region of interest to modulate empathic abilities, due to extensive literature on this region being the core element of emotional processing. Neuromodulation seems like a potential promise to modulate empathic abilities, but also the indicated brain regions that are involved in empathic abilities differ per study.

Therefore, it seems difficult to modulate empathy with the use of tDCS. A recent meta-analysis (Bahji et al., 2021) found that clinical studies using neuromodulating to

enhance empathy are still limited, and that only a few studies have managed to enhance empathy (Yang et al., 2018a;2018b). In line with this scarce evidence, our study did not manage to modulate the empathic abilities of our forensic sample on the behavioral empathy task (rating empathy task; van Dongen et al., 2018) or the self-report empathy questionnaire (IRI; Corte et al., 2007). This could be due to a few factors; first, the empathy task that we used could be too complex for our sample. On an anecdotal level, the patients indicated after the session that they found the task confusing due to the answering options. In addition, they indicated that looking at pictures with the same people in different scenarios was not enough to actually empathize with the victims, although when watching the two empathic movies during the stimulation they indicated how they felt for the characters and could empathize with the main characters in the movies and their experiences. That said, it could also be inherent to the antisocial personality disorder that they found this especially challenging. Therefore, investigating empathic abilities in future studies could focus on using another empathy task, make it more real through watching empathic movies or scenes so the patients can relate more with the main character.

Second, it might be possible that the vmPFC may not be directly related to empathic abilities but more to the regulating of emotions. As suggested by Blair (2005), decreased activity in the vmPFC is linked to impaired social and affective decision making, and the vmPFC is found to be involved in perspective-taking and regulating emotions (Chen et al., 2012; Decety et al., 2015, van Dongen et al., 2018), that are vital for empathic processing (Shamay-Tsoory et al., 2003), but not related to empathy directly. In addition, the previously mentioned studies presented in the literature review (Chapter II) all specified their outcome variable to a certain characteristic of empathic abilities, whereas in this dissertation we took empathy as a broader construct. It may be that in order to investigate and modulate the association between the vmPFC and empathic abilities, the task should focus more on empathic concepts such as emotion regulation or perspective taking.

Another future goal may be using tDCS in combination with fMRI, so the activated brain regions can be more precisely determined. In this way when the participants are focusing on an empathy task, it can be easier detected which brain regions are involved and that knowledge can then be used to develop a neuromodulation protocol specified to that brain region of interest to modulate empathic abilities.

Hence, it is not clear if moderate improvements in speed and accuracy on lab-based computerized measures of empathy would lead to meaningful clinical improvements. Additionally, combining tDCS with other treatment modalities, such as cognitive training, is worthy of consideration, as is the use of imaging techniques to help elucidate the mechanism of action of tDCS. Altogether, there remains a need for high-quality evidence on the effects of tDCS on empathic abilities across multiple, large, pre-registered clinical trials before tDCS can be applied in clinical populations.



Aggression

To our knowledge, this was the first research investigating multiple sessions of HD-tDCS targeting the vmPFC in a forensic sample that was successful in reducing laboratory aggression. We found a significant decrease in the aggressive responses on the PSAP task in the active tDCS session from pretest to posttest. This is in line with previous findings using the PSAP as a laboratory aggression tool and proposing the vmPFC to be a neural substrate related to aggressive provocation (Blair et al., 2016; Gilam et al., 2015, 2017; Skibsted et al., 2017; White et al., 2014, 2016). These results are consistent with the previously described models of Blair (2013), which demonstrated that diminished activation of the vmPFC can lead to frustration-based reactive aggression. Furthermore, we also found a significant reduction in self-reported aggression on the total subscale of the RPQ and the reactive subscale of the RPQ. Interestingly, this decrease in self-reported reactive aggression can be linked to the behavioral PSAP measure, both measuring reactive aggression. The reduction of reactive aggression is consistent with previous findings on the associating between the vmPFC and aggressive behavior (Anderson et al., 1999; Blair, 2004; Calzada-Reyes et al., 2013, 2017; Krämer et al., 2009; Raine, 2003; Raine et al., 2000; Yang & Raine, 2009).

Although one other study (Molero-Chamizo et al., 2018) demonstrated a reduction in aggression in a forensic sample using tDCS, that study did not also measure electrophysiological responses. The current study is the first study combining HD-tDCS with EEG, which is crucial for monitoring electrophysiological effects of tDCS.

Consequently, the present dissertation provides insight into the functional role of the vmPFC in reactive aggression and how modulating activity in the vmPFC can reduce aggressive responses in the laboratory setting. Future research should measure the long-term effects and investigate if the decrease of aggression could also be detected on the ward, towards staff and other patients, and ultimately if this decrease transfers to the “integration into society” phase. Hence, our results significantly improved our understanding of the neural correlates of aggression posed by violent individuals. Despite the fact that the results have to be interpreted with caution before they can be implemented in clinical practice, the findings give support for the effectiveness of tDCS as a treatment intervention in forensic patients.

Regarding aggression and the treatment of violent offenders, it would be very interesting to further develop this intervention and measure over different kinds of violent offenders. From a meta-analysis, we know that the deficits in brain regions differ between psychopathic, antisocial, violent and non-violent offenders (Yang & Raine, 2009). Accordingly, the use of neuroimaging tools such as fMRI would be of great importance to further develop the knowledge in the neurological underpinnings of aggression. Based on the gained knowledge throughout this dissertation and previous literature the critical

need for longitudinal studies to investigate the effectiveness on aggression for a longer period of time. In addition, as well the need for studies that include other types of violent offenders or female antisocial individuals and which assess potential mediating variables (e.g., empathy, emotional regulation) are needed to optimize neuromodulation as a tool in forensic practice and to further develop the understanding of the criminal brain.

Furthermore, it is important to combine the knowledge of different fields to indicate the individual spectrum of aggression per patient. Some of the patients may have their aggressive tendencies from biological underpinnings such as heritage. Some of the patients may have their aggressive tendencies through their environment (social), and others have their predisposition to a deficit in their brain through a disorder or substance dependence. Therefore, combining these fields also from a younger age could contribute significantly to the understanding of aggressive behavior. Research that combines the social, biological, and neurological fields in adolescents with externalizing problems is increasing, and recognizing the importance of follow-up every several years to indicate the effects of certain treatment on a longitudinal level. In this way, it would be easier to predict and prevent certain problematic adolescents and decrease the level of aggressive and delinquent behavior in the future.

Substance use

Few studies have investigated the effects of tDCS on craving and the use of substances targeting the vmPFC. Unfortunately, in this dissertation, we were unable to find any significant effects on the self-reported substance use and craving. This could be due to a few reasons. First, the studies that were successful in reducing the use of substances targeted the DLPFC and were measured over a longer period of time. A meta-analysis of Song and colleagues (2019) demonstrated that the DLPFC has potential to reduce drug use in multiple substance use disorders. Second, the questionnaires that we used were aimed on the use of substances at that moment and the testing took place in a forensic addiction clinic, meaning that the patients in the clinic were abstinent and not using substances at all. In the future a better fitting questionnaire should be implemented, that investigates the craving and substance dependence in line with the interventions already ongoing in the forensic addiction clinic.

In addition, it could be that the type of intervention was not successful in capturing the changes in the craving towards substance use. A recent study demonstrates that brain modulation provides better clinical outcomes when it is combined with alcohol specific inhibitory control training (ICT; Dubuson et al., 2021). Therefore, by modifying the suboptimal interaction between the strong substance-dependent response and the weakened control over this response, combining tDCS and an alcohol version, or other substance, of the ICT has the potential to reduce substance relapse and craving (Bechara,



2005; Lindgren et al., 2019; Noël et al., 2013a, 2013b; Stacy et al., 2010).

Furthermore, the timing of the intervention in the stage of the treatment as usual could also be of importance. Each patient that participated in our study had a different time schedule in the forensic clinic; some of them were already there for a long time, some just arrived, or were just there for a period in between clinics. Hence, it could be that those individual differences in the stage of their treatment contributed to the effectiveness of tDCS as an additional treatment in reducing craving and substance use. For example, Volkow et al. (1991) demonstrated that during the first week of abstinence, an elevated cerebral metabolism in the PFC has been found that after this first week decreased the activity of the PFC, due to lower brain dopamine. Meaning that the metabolic changes seen in substance users during this detoxification stages are related to the brain dopamine activity. Therefore, the impact of tDCS on altering synaptic plasticity could be of importance in the different stages of neurocognitive recovery (Verveer et al., 2021). Lastly, the motivation of the patients to engage in the modules and their treatment is very important to strengthen the effectiveness of the tDCS treatment.

Electrophysiological Outcome Measures

Regarding the use of electrophysiological measures to examine changes in brain fluctuations we used two methods of analyzing the EEG (ERPs and rsEEG). Previous studies (i.e., Cuthbert et al., 2000) indicated that the LPP indicates an increase in amplitude towards emotional stimuli and affective experiences. In addition, the study of van Dongen et al., (2018) demonstrated that individuals with decreased empathic abilities show a reduced P3 and LPP. Therefore, studying these positive potentials contribute to the understanding of emotional regulation (Cheng et al., 2012; Cuthbert et al., 2000; Fan et al., 2011; Lang & Bradley, 2010), and serve as an indication that the temporal dynamics of empathic processing in the brain can be studied using late ERPs in the EEG. Increased empathic abilities would therefore result in more empathic arousal towards the pictures that depict aggressive situations and would result in enhanced P3 and LPP amplitudes in the EEG. Accordingly, with respect to the electrophysiological measures (P3 and LPP), we expected that after a week of tDCS intervention, aggressive pictures would result in larger ERP amplitudes (more arousal, more empathy) compared to neutral pictures. This hypothesis was partly confirmed.

P₃

For the P₃ we did not find the increase in amplitude that we hypothesized but instead a significant decrease in P₃ amplitude for the sham condition. One of the reasons could be that the reduced P₃ was caused by a learning effect. Multiple studies (Hajcak et al., 2015; Sailer et al., 2010; Yeung et al., 2004) have shown that the P₃ is sensitive for learning effects in a repeated measures design and that the P₃ becomes smaller when target probability increases (Duncan-Johnson et al., 1977; Donchin & Coles, 1988; Kompatsiari et al., 2016). Possibly due to an enhanced conformity to expectancy (Hajcak et al., 2015; Yeung et al., 2004) and have been reported to decrease across various tasks (Jongsma et al., 2006; Groen et al., 2007; Lindín et al., 2004; Radlo et al., 2001). Therefore, it could be that the P₃ was significantly reduced in the sham group, and that the tDCS neutralized the learning effect, which resulted in only a slight lesser increase in amplitude than for the LPP amplitude.

Furthermore, studies have shown that subjects with a substance use dependence (Biggins et al., 1997; Branchey et al., 1993), or with an antisocial personality disorder (ASPD; Bauer et al., 1994; Bauer et al., 1997; Costa et al., 2000) have demonstrated reduced P₃ amplitudes. In the same vein, reduced P₃ amplitudes tend to be associated with the risk in using substances and might predict substance use dependence (Brigham et al., 1995; Herning et al., 1989) and alcoholism in adolescence (Berman et al., 1993; Hill et al., 1995). Therefore, since our sample consisted of 21 patients with an alcohol disorder or alcohol comorbidity classification and 8 patients were diagnosed with ASPD, this might have influenced the amplitude reduction. However, there was no significant difference between the tDCS or sham group.



Late Positive Potential (LPP)

For the LPP we found a general increase for the (affective) pictures regarding social interactions in comparison to the filler pictures. This partly confirms our hypothesis that active tDCS targeting the vmPFC increased LPP amplitudes following the social-emotional event as in line with previous research (Balconi & Pozzoli, 2003; Coll, 2018; Eimer et al., 2003; Herrmann et al., 2002; Smith et al., 2013; van Dongen et al., 2018; Winker et al., 2019), showing that the positive slow potentials indicate an increased emotional regulation and affective experience. However, we did not find this increase in amplitude for the aggressive pictures compared to the neutral interactions. It could be due to the fact that previous findings of an increase in LPP amplitude for a specific emotion-focused more on pain empathy (for an overview see Coll, 2018). In addition, the ‘affective sharing response’ to pain in others might not be as fast and as automatic as previously thought and might rely more on the initial identification of the stimulus (Coll et al., 2017b). Nevertheless, the processing of the emotional content of a situation is obviously crucial for feeling empathy, which could lead to increased prosocial behavior and the inhibition of antisocial-aggressive responding (Blair, 1995).

Since another way of measuring the effectiveness of HD-tDCS on electrophysiological responses is to investigate the ongoing brain oscillations, we investigated this exploring the rsEEG data of our sample. We examined three measurements of electrophysiological activity to investigate whether the intervention would induce a change in oscillatory dynamics.

Spectral Power in rsEEG

First, we wanted to review the already existing research on abnormalities of the spectral power in the brain of violent offenders. Multiple studies have indicated that violent offenders have increased slow-wave activity in the theta frequency band. In addition, these studies demonstrated decreased alpha activity in these samples. Therefore, we wanted to investigate if the same characteristics were found in our sample and most important if our HD-tDCS intervention could influence the ongoing brain oscillations. Unfortunately, in the paper presented in Chapter V, we couldn't demonstrate an effect of HD-tDCS on the electrophysiological signals in the theta, alpha or beta bands. This finding could be explained in several ways: first, the specific focus on the frontal electrodes. Because the literature indicated several dysfunctions in power in the frontal region of the brain in violent offenders, and since the frontal region is our main target region in reducing aggression and enhancing empathy in our sample, we decided to focus solely on the frontal electrodes in the analysis of the rsEEG. It could be that our focus on the frontal regions contributed to the null finding. A more recent study has also included the parietal and temporal regions and did find an effect when using tDCS (Konikar et al., 2021) and even found that these regions are more dominant for the power in slow-wave activity. If the activity in our sample was also limited to these regions, this could explain why we did not find any effect regarding our exploration.

Although no studies are reporting the same effect for alpha or beta in antisocial individuals, it might be that these frequency bands also have a more pronounced effect located in other brain regions. Future studies should investigate whether these aberrant brain processes, characteristically for antisocial or violent individuals, are linked to specific regions because this could provide important treatment implications. In addition, investigating a whole-brain analysis should be a logical follow-up to this study to see whether the effects were present in other brain regions in our sample.



Asymmetrical Activity in rsEEG

Since increasing evidence suggested asymmetrical involvement of the frontal hemispheres in aggressive behavior, we investigated this in our violent offender sample. We examined the effect of HD-tDCS on the asymmetrical activity in the alpha and beta frequency band. We did not find any significant effects of the tDCS intervention in asymmetrical activity in the alpha frequency band, which is in line with previous studies (Hofman & Schutter, 2009, 2012). In those studies, no association between tDCS and alpha asymmetry was reported and this is explained this by the assumption that alpha oscillations tend to be dominant over the posterior scalp locations during resting-state EEG recording. Therefore, a posterior-to-anterior direction of the configuration of the posterior dipole could explain the absent association between the alpha oscillations and frontal excitability (Hofman & Schutter, 2012) and this could also explain or lack of evidence for this in the current study. In line with the null-finding of the power, this could be interesting to examine in future studies.

Although we did find a significant interaction in the asymmetrical activity in the beta frequency band, there was no an effect over time. When we examined this effect in more detail, it did emerge that a larger part of the active tDCS group showed a significant increase, and more specifically, all patients that had a significantly higher score on proactive aggression. This is definitely worth investigating in future studies.

The results on asymmetrical activity in the beta frequency band suggest the possibility to alter cortical activity after a tDCS intervention and should be investigated more in-depth in future studies. This is also in line with previous research suggesting that more left than right-sided activity may interact with interventions in terms of positive treatment outcomes (Smith et al., 2017). In addition, for future studies it would be very interesting to link asymmetrical activity to aggression outcomes, since multiple studies demonstrate the association between asymmetrical frontal activity and aggressive tendencies (Hofman & Schutter 2009, 2012; Schutter et al., 2008). In that matter, it could enhance our understanding of the aggressive behavior of forensic patients and how underlying cortical activity plays a role in this.

Synchronicity in rsEEG

An interesting method to examine the synchronicity and communication between frontal regions is to measure the intercorrelation in the phase between electrodes. We investigated the effect of HD-tDCS on the synchronicity between the frontal electrodes using the technique of Phase Locking Value (PLV). The HD-tDCS intervention resulted in a significant enhancement of the synchronicity in the alpha and beta frequency band in the frontal region. For the alpha frequency the enhanced PLV's were in the same hemisphere. This enhanced connectivity can be described as increased functioning communication within this region and with that increased functionally in the use of cognitive abilities.

For the beta frequency band, the enhanced synchronicity occurred between the two hemispheres. This means that the two hemispheres in the prefrontal areas are more functionally connected and with that more synchronous in their activity. Since the brain relies on the interhemispheric transfer of information to integrate behavior and cognition (Doron & Gazzaniga, 2008), for the larger part executed in the axonal transmissions between the two hemispheres sub-served by the corpus callosum (Banich, 1998). Thus, decreased connectivity is associated with cognitive impairment (Babiloni et al., 2006; Keary et al., 2009; Kumar et al., 2009; Thiruvady et al., 2007), meaning that increased connectivity affects the transfer between the hemispheres and with that interhemispheric connectivity is reinforced to influence cognitive performance and improve the functioning of the frontal brain region. This implies that tDCS might be able to enhance connectivity between electrodes in the frontal regions and with that increase functional communication within this region and strengthen the use of cognitive abilities.

One of the main takeaway notes is to investigate the brain regions that are characteristic of the forensic sample tested. Results can vary based on electrode locations (frontal, parietal, temporal, etc.) and the methods used to calculate connectivity. For example, we used the PLV, but recent studies also indicate the use of the Phase Lag Index (PLI) as a value measurement in calculation synchronicity. The choice of all these parameters contributes to a better understanding of modulating electrophysiological measures and the possible modulation of neural activity with HD-tDCS leading towards an implementation in treatment.

In conclusion, we have demonstrated that HD-tDCS can modulate electrophysiological response in task-related stimuli (i.e., LPP). In addition, the present study provides the first indication that HD-tDCS could be an effective method to restore affected connectivity within a violent offender sample and with that enhance cognitive functioning in the frontal region within one hemisphere as well as interhemispheric.



HD-tDCS

To our knowledge, this is the first time that multiple sessions of HD-tDCS targeting the vmPFC has been investigated in a forensic sample. This innovative version of tDCS has been used because of its capability to target brain regions with higher focality (DaSilva et al., 2015) and reach deeper regions of the brain (Edwards et al., 2013; Faria et al., 2011; Hess et al., 2013; Manuel et al., 2014; Nakamura-Palacios et al., 2016; Tanaka et al., 2013; To et al., 2018). In line with these studies, in this dissertation we demonstrated (Chapter IV) that the vmPFC was reached with an intensity sufficient to generate an effect and that the tDCS intervention resulted in reducing aggression and modulating electrophysiological responses. In addition, we demonstrated that we effectively modulated activity in a brain network.

Regarding the rsEEG of our forensic sample, we did not find any results that the tDCS intervention had an effect on the power in the frequency bands or results on modulating asymmetrical activity. We did find results on modulating the phase-correlation in the synchronicity between frontal electrodes. Using biophysical modeling we demonstrated that we modulated the vmPFC with activity sufficient to induce an effect, but with the use of modeling beforehand using a program such as the StimWeaver algorithm of Neuroelectronics, the individual characteristics of a sample and procedure can be decided to optimize the effect of the modulation of the brain region of interest, selecting electrodes positions and current that best reach the target (either making the stimulation the most focal or the most intense). Since it is of great importance to individualize montage and modeling, for future studies this optimizing current-flow programming (i.e., Stimweaver) should be investigated to increase the stimulation effect. Furthermore, the protocols should be matched on the characteristics of the sample studied taking into account several parameters, like current characteristics, electrode sizing, stimulation site, montage selection, tDCS devices, single versus multi-site stimulation, frequency of stimulation to optimize the use for interventions in psychiatric disorders.

However, we did find that the tDCS intervention significantly enhanced synchronicity between the frontal electrodes in the alpha and beta frequency band as measured by the Phase-Locking Value (PLV). Although future studies should investigate the neural underpinnings of aggression, it can be suggested that tDCS is an appropriate method to restore affected connectivity within a violent offender sample and with that enhance cognitive functioning in the frontal region.

Limitations and Future Directions

Although we found the first promising effects in reducing aggression in a forensic substance use sample in a protocol with multiple sessions and HD-tDCS, there is still a long way to go before we can implement this treatment into forensic practice.

Unfortunately, we did not find any results on increasing empathy or reducing craving, so before HD-tDCS can be implemented in this forensic sample it is necessary to further investigate the effects in different forensic patients, different brain regions, the strength and duration for long-lasting effects, and the laboratory tools that are most useable to gain more knowledge into reducing aggression. Furthermore, regarding modulating of the electrode position using HD-tDCS device there is still no gold standard in which electrodes to include or exclude and the number of electrodes to use. In addition, the literature still differs on the use of current flow and montage. Per different disorders or brain regions of interest, these characteristics should be carefully evaluated.

Using neuroimaging methods could contribute to determine the effect of the tDCS. Not only for modulating effects but also to determine the brain state during the modulation in the most optimal way. Although we have implemented the use of two empathic movies during modulation to enhance the brain state in order to increase the effectiveness of the modulation, the combination with neuroimaging methods can help to better study brain state in parallel to the behavioral and clinical effects of tDCS (See, e.g., Nord et al., 2019).

As increasing evidence suggests, brain regions function in a continuous network instead of single regions predicting certain behavior or disorders. In this dissertation we focused on targeting the vmPFC but in the future, it would be an even better approach to focus on enhancing brain networks and using network tDCS (Mencarelli et al., 2020).

The next steps for implementing HD-tDCS as an add-on for treatment as usual (TAU) is to do multicenter research and to study the effects over different types of violent offenders and forensic patients. Although there is a lot of research on the use of NIBS in several disorders, its use in forensic practice is still scarce. Forensic patients are a difficult sample to work with in terms of comorbidity, medication, scheduling and personal motivation. Therefore, every step of the intervention protocol should be very well planned from the beginning. Hence, the timing of tDCS treatment and the motivation of patients to actively engage in treatment programs should be considered in future research.



Implications for Forensic Practice

The ultimate aim of the current study was to investigate the effectiveness of HD-tDCS as an add-on treatment in forensic care. Before this step can be accomplished a few considerations for future research should be mentioned.

First of all, more research is needed. The specific neurobiological correlates of aggression and empathy should be further investigated, especially in a forensic sample since research still remains scarce. This could be done by using neuroimaging methods such as fMRI or a combined tDCS-fMRI study in addition to solely compare effects from pre- to post-intervention, also visualize the activity to enhance the understanding of the intervention. Since a broader range of brain regions is involved in violent behavior and the impaired processes leading to increased aggression, and not the brain region in isolation, investigating brain networks is a propitious next step. A promising research line is investigating structural and functional connectivity exploring brain networks in relation to different psychological disorders. In the same vein, it would thus be very interesting to focus on network-based tDCS to modulate the connectivity between regions associated with aggression.

Regarding the effectiveness of reducing aggression, it is important to first investigate this in several other clinics in the Netherlands and internationally as a multicenter research design, before tDCS as a clinical treatment can be implemented. As mentioned before, there are very different findings in brain deficits regarding ASPD, violent offenders, and non-violent offenders (Yang & Raine, 2009). Therefore, before the treatment can be generalized not only different clinics but also different forensic patient samples should be investigated and in addition, different types of violent offenders. It would be helpful to connect with a broader network of researchers internationally to increase sample sizes, combine neuromodulation techniques and knowledge. Hence, to implement HD-tDCS as an add-on therapy, the therapists and practitioners working in forensic care should receive both relevant knowledge and training to understand the neurological underpinnings of the forensic treatment group and to be able to design and adjust current-flow modulating regarding the specific needs of their client group.

Furthermore, according to my perspective, future treatment should include different research fields into achieving the goal of reducing aggression. Biological and psychosocial would be an ideal combination to individually determine what is the best for each patient to increase treatment success. Since each individual may have symptoms that are a combination of an individual's cognitive and biological functioning (Brazil et al., 2018; Montague et al., 2012). Hence, to gain knowledge in the bio-cognitive markers of antisocial disorders and the underlying aggressive behavior and how they combine in an individual is unique and should be approached as each case individually (Brazil et al., 2018). Investigating the link between these different factors to further strengthen these crucial fields in obtaining research on the same path is of substantial importance to increase the understanding of the aggressive brain.

Promising Forensic Intervention

Altogether, tDCS is a promising forensic intervention to implement in practice. It is very practical in use for research and treatment due to mobility, tolerability, affordability and the option to have a reliable procedure to use a control (sham) condition (Priori et al., 2009). In addition, as also used in this dissertation, the tDCS device is programmed in such a way that a reliable double-blind procedure can be used. The device uses a preprogrammed code and with that, blinds the conditions for researchers, practitioners and patients. Furthermore, the modulation has no side effects, although asking the patients whether they felt any sensations is preferable. The same can be done for the blinding conditions, asking the patients afterward if they thought they had a tDCS intervention or the placebo. Last, the device is easy to use when training is completed and the options to use for multiple brain regions and disorders are endless.



General Conclusion

In general, the results described in this dissertation support the use of HD-tDCS as a forensic intervention. In our proof-of-concept study, we demonstrated the effectiveness in reducing aggression. We have demonstrated that reactive aggression both using a laboratory task and a self-report questionnaire can be reduced after a week of intervention. In addition, we demonstrated the effectiveness of tDCS in modulating electrophysiological responses. Unfortunately, we did not find any results for empathic abilities or substance craving, we suggest using a different empathy task based more specific on the construct related to the vmPFC, if the vmPFC as a solely brain region is the aim of a future study. Even more promising would be to investigate network-based tDCS instead of focusing solely on one region of the brain.

Regarding substance use, it is very important to time the intervention due to motivation, the plasticity of the brain, and time after the last use of the substances. Regarding the modulation of connectivity and the frequency bands, it is of great importance to select the frequency bands and regions of the brain with great caution and take into account all the parameters that fit the sample you want to investigate.

Taken together, the overall conclusion of this dissertation is that HD-tDCS seems to be an effective intervention to reduce aggressive behavior and modulate electrophysiological responses. Although effective, more research is needed to implement HD-tDCS in forensic clinical practice. The most important next step is to conduct multicenter research investigating different forensic patients and violent offenders. In this way, the underlying processes and application of HD-tDCS can be further determined and ultimately be able to improve treatment success in forensic patients. Nevertheless, this thesis supports HD-tDCS as a promising tool to be implemented as an add-on treatment in forensic care.



The background of the page is a solid green color. Overlaid on this are two crumpled paper-like shapes, one in the upper left and one in the lower right, both in a slightly darker shade of green. A thin, light-colored line, resembling a circuit or a path, starts from the top right and winds its way down towards the lower right, ending near the crumpled paper shape. The word "REFERENCES" is centered in the upper half of the page in a white, bold, sans-serif font.

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Samenvatting

Gewelddadig gedrag is nog steeds een groot probleem in de maatschappij. Ook al komt geweld voor in meerdere omstandigheden, individuele uitingen van agressief gedrag bepalen het grootste gedeelte van geweld in de samenleving. Het aantal onderzoeken naar de neurologische processen onderliggend aan agressie is de laatste jaren exponentieel toegenomen, want het voorspellen van agressief gedrag in forensische patiënten is cruciaal voor de behandelingen in de forensische zorg. Huidige interventies, zoals cognitieve gedragstherapie, blijken vaak niet voldoende te werken binnen de forensische sector en ook niet voldoende te leiden tot een vermindering in gewelddadige recidive (Greeven & Bosma, 2012; Koeter et al., 2009). Een onderliggend probleem die de effectiviteit van de huidige behandelingen belemmert is dat een groot gedeelte van deze patiënten ook een verslavingsachtergrond heeft. Deze doelgroep is vaak slecht gemotiveerd om deel te nemen aan behandeling omdat er vaak sprake is van een complex medisch-psychiatrische aandoening betreft, waarbij het langdurig gebruik van middelen tot functionele en structurele veranderingen in de hersenen heeft geleid (Volkow et al., 2009). Het gebruik van middelen kan gewelddadig gedrag vergroten door de psychofarmaca (stoffen die invloed hebben op het brein) in drugs en alcohol, waardoor agressie toeneemt en daardoor ook een verhoging in gewelddadige delicten (Collins, 1981; De Looft et al., 2017; Virkkunen & Linnoila, 1993; White & Gorman, 2000). Een effectieve behandeling voor forensische patiënten met een afhankelijkheid van middelen is dus van groot belang voor de veiligheid van de maatschappij.

Recentelijk is de interesse voor het onderzoeken van innovatieve methoden op het gebied van neuropsychologie duidelijk toegenomen. In de neurowetenschap wordt er gekeken naar bepaalde afwijkende activiteit of tekorten in het brein, die mogelijk in relatie kunnen staan met gedragingen passend bij een stoornis of neurologische tekorten. Een veelbelovende interventie die de afwijkingen in het brein kan beïnvloeden is transcraniële Direct Current Stimulation (tDCS). Op deze manier kan het gedeelte van het brein dat onderliggend is aan agressief gedrag worden gemoduleerd. Voordat ik meer uitleg geef over deze nieuwe techniek, eerst wat achterliggende informatie over welk deel van het brein betrokken is bij agressie.

Het brein kan worden ingedeeld in verschillende hersendelen, en met name het voorste gedeelte, de prefrontale cortex (PFC) staat in relatie met agressie en antisociaal gedrag. Studies hebben aangetoond dat met name de dysfuncties in de ventromediale prefrontale cortex (vmPFC) een cruciale rol spelen in het agressieve gedrag van justitiabelen met een verslavingsachtergrond (met name alcohol en cocaïne). De vmPFC is een gebied in het frontale (voorste) deel van het brein die een belangrijke rol speelt in emotieregulatie, sociale interacties, het maken van beslissingen en het kunnen onderdrukken van bepaalde impulsen. Omdat de vmPFC zo belangrijk is voor het reguleren van emoties, heeft het ook een specifieke rol in het adapteren aan een andermans perspectief en empathie. Uit onderzoeken die gebruik maken van neuro imaging, een techniek die het brein visueel in kaart kan brengen, is gebleken in dat de vmPFC van antisociale individuen beschadigd is, met als gevolg een stoornis in het modeleren van agressief gedrag. Bovendien, is er uit neurocognitieve modellen (Blair et al., 2001, 2005, 2006) gebleken dat empathie een grote rol speelt in de remming van gewelddadig gedrag geremd. Een individu die een tekort heeft aan empathisch vermogen zou hierdoor minder gevoelig zijn voor de effecten van agressie en hierdoor ook minder motivatie hebben om agressie te onderdrukken, waardoor het risico op het uiten van gewelddadig gedrag toeneemt. Ook is er in onderzoeken aangetoond dat de hersenstructuren die relevant zijn voor empathie, zoals de vmPFC, aangetast zijn bij personen met antisociale trekken. Deze hersenstructuren zijn betrokken bij het geven van een moreel oordeel, impulsiviteit en staan in relatie met agressie en gewelddadig gedrag.

Terwijl de associatie van de prefrontale cortex en agressief gedrag al eerder is onderzocht in neuroimaging-, gedrags- en klinische studies, is er zover wij weten nog weinig bekend over de causale rol van de vmPFC in empathische vaardigheden en agressief gedrag in verslaafde justitiabelen. Gezien de relatie tussen de vmPFC met empathisch vermogen en agressief gedrag, kan het moduleren van de vmPFC om empathisch vermogens te vergroten en agressief gedrag te verlagen binnen deze doelgroep van groot belang zijn. Zoals eerder al kort genoemd, is tDCS een interventie die kan inspelen op het moduleren van hersengebieden. tDCS is een neuromodulatie techniek die op een non-invasieve (niet-schadelijke) manier de vmPFC kan moduleren. Deze techniek maakt gebruik van zwakke elektrische stroompjes om activiteit in het betreffende hersengebied te verhogen of te verlagen. Hierdoor wordt de prikkelbaarheid van hersencellen veranderd en op die manier kunnen membraampotentialen worden gemoduleerd wat weer leidt door de veranderbaarheid van de synapsen (punt waart twee hersencellen met elkaar communiceren). Het zorgt er dus voor dat de hersencellen gevoeliger worden en sneller kunnen 'vuren' waardoor de neurale plasticiteit kan worden aangepast en de leerbaarheid



van de hersencellen kan worden verhoogd.

Een goede manier om de effecten van tDCS te meten op de activiteit in het brein is elektro-encefalografie (EEG). EEG is een methode waarbij elektroden op de hoofdhuid worden geplakt en hierdoor kan de elektrische activiteit van het brein worden onderzocht. Doormiddel van EEG kan gekeken worden naar event-gerelateerde potentialen (ERPs) en naar het frequentiespectrum. ERPs beschrijven de elektrofysiologische reacties van de hersenen op gebeurtenissen (events). Hierdoor kan er worden gekeken naar hoe een individu reageert op stimuli; in dit proefschrift hebben we ons gericht op emotionele stimuli in lijn met agressie en empathie (sociale interacties). Het frequentiespectrum is een range van hersensignalen in het brein dat wordt uitgedrukt in hertz (Hz). De golflengtes binnen deze range kunnen worden onderverdeeld in delta (1-4 Hz), theta- (4-8 Hz), alpha- (8-12 Hz), beta- (12-25 Hz) en gamma-activiteit (>25 Hz). Elk van deze frequentiebanden duidt een verschillend moment aan in de brein oscillaties en kan worden gelinkt aan verschillend gedrag. Een verandering hierin of een afwijkend patroon kan dus worden gekoppeld aan bijvoorbeeld agressie of een antisociale stoornis. Hierdoor wordt naast het meten van de effectiviteit van High-Definition transcranial Direct Current Stimulation (HD-tDCS) als behandeling, de verandering in agressie en empathie, ook het effect van tDCS op het EEG meegenomen in dit proefschrift.

Allereerst is er een onderzoek gedaan in de bestaande literatuur naar wat er al bekend is over het gebruik van tDCS in het verhogen van empathische vermogens en het verlagen van agressief gedrag doormiddel van een review onderzoek (Hoofdstuk II). Hierin wordt een overzicht gegeven van de studies die deze variabelen hebben onderzocht en hieruit komt naar voren dat de vmPFC een kerngebied is voor empathie en agressie. De meeste studies hebben de Dorsolaterale Prefrontale Cortex (DLPFC) gestimuleerd, mede omdat dit gebied wat minder diep in het brein ligt en hierdoor makkelijker te bereiken is met tDCS, maar ook omdat dit gebied vaker is gestimuleerd in voorgaande onderzoeken en ook de daarop volgende studies kiezen voor deze aanpak. Het blijkt echter dat door het stimuleren van de DLPFC, voornamelijk bij verslaving, er vooral veel verandering teweeg wordt gebracht in de vmPFC. Door de bevindingen vanuit dit review onderzoek en vanuit de eerder onderliggende theorieën die de vmPFC als cruciale kern beschrijven is in dit het onderzoek dat dragend is voor dit proefschrift gekozen voor het moduleren van de vmPFC.

In Hoofdstuk III is het studieprotocol beschreven van de studie naar het effect van HD-tDCS in het verhogen van empathische vermogens en verlagen van agressief gedrag. Dit studie is gerandomiseerd (willekeurige indeling van patiënten in groepen), gecontroleerd (er is ook een controlegroep meegenomen om te vergelijken) en dubbelblind (zowel onderzoeker als patiënt weten niet in welke conditie ze zitten). In dit protocol wordt beschreven dat de studie afgenomen wordt in twee forensische verslavingsklinieken in Poortugaal, Rotterdam. De Forensische Verslavingskliniek (FVA) en de Afdeling

Forensische Zorg (AFZ), beide met net een ander beveiligings- en integratieniveau. Beide instellingen bieden zorg voor verslaafde justitiabelen. Na een zorgvuldige screening op medicatie en neurologische stoornissen werden de patiënten willekeurig toegewezen aan dan wel de actieve tDCS behandeling of de sham (controle) conditie. Omdat het belangrijk is te zien wat de tDCS voor effecten teweegbrengt, werden de resultaten vergeleken door middel van een pretest (baseline sessie voor de behandeling) en een posttest (na de behandeling). In de pretest werden verschillende metingen gedaan, waaronder zelfrapportagevragenlijsten, een EEG-taak in rusttoestand, passieve empathie taak tijdens EEG (afbeeldingen van agressieve en neutrale situaties), een empathietaak met het beoordelen van de afbeeldingen uit de passieve empathie taak en ten slotte een agressie taak (de Point Subtraction Aggression Paradigm; PSAP, Chereck, 1992). De pretest werd altijd op de vrijdag uitgevoerd voordat de week begon waarin de patiënten de interventie met tDCS ondergingen. Tijdens de daaropvolgende week ontvingen de deelnemers gedurende vijf opeenvolgende dagen twee keer per dag 20 minuten HD-tDCS-stimulatie of sham stimulatie, eindigend op een vrijdag. De uitgebreide beschrijving hiervan is te lezen in Hoofdstuk III. De primaire uitkomstmaten van deze studie waren de empathietaken, de agressietaak en de elektrofysiologische maten (de ERP). De vragenlijsten dienden als secundaire uitkomstmaten, maar waren van zeker belang op de effectiviteit van de behandeling. Het doel van het gebruik van HD-tDCS was het verhogen van empathische vermogens en het reduceren van agressief gedrag van forensische patiënten.

In Hoofdstuk IV worden de resultaten van de hierboven beschreven studie besproken. De resultaten zijn veelbelovend op het verlagen van agressie omdat is aangetoond dat voor zowel de agressietaak (de PSAP) en als voor de zelf-gerapporteerde agressie (RPQ) een afname te zien was voor de tDCS groep in vergelijking met de controlegroep. Voor zowel de PSAP als de RPQ ging het om een afname in reactieve agressie. De afname van agressieve reacties op de PSAP wordt gezien als een indicatie voor het meten van reactieve agressie en de afname in reactieve agressie antwoorden was ook te zien op de vragenlijst. Op elektrofysiologische niveau werd er een toename gevonden in de LPP-amplitude (dus een toename in de emotionele verwerking) na de tDCS interventie in vergelijking met de controlegroep. Er werden geen effecten gevonden op de empathietaak, zelf-gerapporteerde empathie en voor de P3 amplitude. De huidige bevindingen zijn de eerste die laten zien dat tDCS effectief is in het moduleren van elektrofysiologische reacties die betrokken zijn bij emotionele verwerking en daarmee het reduceren van agressie in verslaafde justitiabelen. De resultaten tonen daarmee het potentieel aan van tDCS als interventie in de forensische geestelijke gezondheidszorg.



Uit onderzoeken is gebleken dat de activiteit van het brein in ruststaat kenmerkend kan zijn om onderliggende stoornissen in kaart te brengen en de karakteristieke veranderingen in breinfuncties in patiënten kan aantonen. Gebaseerd op deze onderzoeken en onderliggende theorieën is er tijdens de studie ook een EEG-taak opgenomen die de activiteit meet in rust toestand. In Hoofdstuk V wordt er dieper ingegaan op de elektrofysiologische activiteit door het analyseren van de EEG-taak in rust toestand. In deze taak moesten de patiënten één minuut hun ogen dichtdoen en vervolgens één minuut hun ogen opendoen, deze condities wisselden elkaar af gedurende vier minuten. De data van deze EEG-taak is toen verder geanalyseerd en vervolgens is gekeken wat het effect van de HD-tDCS behandeling is op drie verschillende methoden van het bekijken van breinactiviteit tijdens rust.

Allereerst is er gekeken naar het effect van de tDCS behandeling op drie verschillende frequentiebanden. Zoals eerder genoemd duidt elk van deze frequentiebanden een verschillend moment aan in de brein oscillaties en kan worden gelinkt aan verschillend gedrag. Gebaseerd op eerdere onderzoeken wordt er in dit proefschrift onderzoek gedaan naar de activiteit van theta (4-7 Hz), alpha (8-13 Hz) en beta (14-30 Hz), omdat deze in relatie staan met agressieve gedragingen of kenmerkend zijn voor de hersenactiviteit van gewelddadige patiënten. Vanuit de literatuur blijkt dat gewelddadige individuen een verhoogde activiteit in de theta laten zien en een verlaagde activiteit in de alpha. Dit is in vergelijken met individuen zonder agressieve trekken of een antisociale stoornis afwijkend. Echter hebben we voor de eerste onderzoeksvraag geen effect gevonden van de tDCS behandeling op de power (activiteit) in deze frequentiebanden.

Een andere manier om naar corticale activiteit te kijken is het bestuderen van de asymmetrische activiteit, het blijkt dat vooral de linkerhersenhalft gerelateerd is aan agressieve gedragingen en het beïnvloeden van de activiteit in de rechterhersenhalft kan leiden tot een reductie van agressie. We hebben geen effect gevonden van de tDCS interventie van pretest tot posttest op de asymmetrische activiteit in alpha, in beta vonden we een verband, maar ook dit bleek niet over de behandelperiode te zijn veranderd.

De laatste methode die wordt beschreven in Hoofdstuk V is het analyseren van de synchronisatie tussen elektrodes. Op deze manier kan er gekeken worden naar de communicatie tussen hersengebieden. Elektrodes zijn verbonden aan bepaalde gebieden in het brein en aangezien het frontale gedeelte van het brein het kerngebied is voor agressie is er in deze studie is voor de frontale elektrodes gekozen. Door middel van een analyse is gekeken of de tDCS behandeling de communicatie tussen de elektrodes heeft versterkt. In de alpha en beta frequentiebanden is er verhoogde synchronisatie, of communicatie, gevonden na de tDCS behandeling. Hierdoor kan er gesuggereerd worden dat de activiteit in het frontale gebied van het brein verhoogd kan worden door de tDCS behandeling.

De resultaten tonen hiermee aan dat tDCS kan worden gebruikt om de activiteit te moduleren in het brein van gewelddadige patiënten en hopelijk hiermee de cognitieve vaardigheden te kunnen versterken.

In Hoofdstuk VI worden de resultaten van alle studies beschreven en wordt er een conclusie gegeven alsmede implicaties voor het gebruik van tDCS in de forensische praktijk.

De bevindingen in dit proefschrift hebben kunnen aantonen dat tDCS een effectieve interventie zou kunnen zijn voor het verlagen van agressie in forensische patiënten. Voordat het gebruik van tDCS als behandelmethode in de forensische praktijk kan worden geïmplementeerd zijn er wel nog een paar stappen nodig. Allereerst is er meer onderzoek nodig in verschillende instellingen in zowel Nederland als internationale klinieken om te kunnen bepalen of de gevonden effecten uit deze ‘proof of concept studie’ kunnen worden gerepliceerd. Ten tweede is het van belang om verschillende soorten gewelddadige patiënten te onderzoeken om te kijken of dezelfde effecten kunnen worden behaald en hierdoor de generaliseerbaarheid van neuromodulatie als behandeling te kunnen toetsen. Ten derde zou het gebruiken van neuro-imaging technieken (fMRI) samen met EEG-tDCS kunnen bijdragen aan het in kaart brengen van de activatie van specifieke gebieden in het brein. Op deze manier kan specifiek worden bekeken naar de breingebieden die in relatie staan met de tekorten of afwijkende hersenactiviteit en zo kan optimaal worden bepaald welke gebieden er moeten worden gemoduleerd. Ten vierde is het van cruciaal belang om per onderzoek en doelgroep te bepalen welk tDCS protocol het beste past om de gewenste resultaten te kunnen bereiken. Het bepalen van de sterkte van de stimulatie, locatie van de elektrodes, hoelang een stimulatie moet duren, hoe vaak dit moet worden herhaald en hoelang het effect aanhoudt (follow-up). Tenslotte, zou de samenwerking van verschillende onderzoekers nationaal en internationaal zou hierin transparant moeten zijn. De communicatie en betrokkenheid van meerdere disciplines, zowel in biologisch (neuro) als in psychosociaal in wetenschappelijk onderzoek zou moeten worden gestimuleerd. Op deze manier zouden deze velden samen een doorslaggevende rol kunnen spelen in het verwezenlijken van het gemeenschappelijke doel om neuromodulatie te implementeren in de (forensische) zorg. Het gebruiken van neuromodulatie in de forensische praktijk staat nog in de kinderschoenen, maar dit proefschrift is een goede ‘proof of concept’ studie in het aantonen van de effectiviteit van tDCS als behandelmethode in het reduceren van agressief gedrag en het moduleren van elektrofysiologische activiteit in het brein van forensische patiënten.



Concluderend, de studies beschreven in dit proefschrift tonen aan dat HD-tDCS een effectieve behandeling kan zijn als aanvulling op de al bestaande behandelingen in de forensische praktijk. Het is non-invasief, heeft geen bijwerkingen en is gebruiksvriendelijk na een goede training. Onze resultaten toonden aan dat meerdere sessies van HD-tDCS gericht op de vmPFC resulteerden in verminderde agressie. Bovendien resulteerde deze modulatie ook in verhoogde activiteit in het brein na het zien van afbeeldingen met agressieve scènes. Dit duidt op een verhoogde activatie voor- en emotionele evaluatie van scènes met slachtoffers van agressie (i.e. empathische verwerking). Verder heeft dit proefschrift ook aangetoond dat tDCS een invloed heeft op de corticale activiteit in rust en heeft het de samenwerking in het frontale gebied van het brein vergroot. Deze resultaten leiden tot een beter begrip van agressie door gewelddadige individuen en deze neurobiologische inzichten kunnen in de toekomst bijdragen aan het verminderen van recidive en een beter begrip van het agressieve brein. Hierdoor kan dit ondersteuning bieden voor de effectiviteit van alternatieve aanvullende behandelinterventies binnen de forensische verslavingszorg.





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ABOUT THE AUTHOR

CARMEN-SILVA SERGIU



Carmen-Silva Sergiou was born in Paralimni, Cyprus on 12-05-1991. After completing her secondary (pre-university; Gymnasium/VWO) education at het Willem de Zwijger College in Bussum, the Netherlands she studied Psychology at de Vrije Universiteit Amsterdam. She obtained her bachelor degree in Cognitive and Neuropsychology. Already in her Bachelor Carmen became very passionate about the forensic field. She started to do volunteering work in the clinic of Long-term Forensic Care of Inforsa two times a week. In addition, she started as a research assistant for a professor at the Vrije Universiteit to collect data in multiple forensic institutions. After completing her bachelor, she started the Clinical Forensic Psychology master at the University of Amsterdam. During her internship period she went to Washington State University in Pullman, Washington to work at a lab where they do research on narcissism. After her graduation, Carmen developed her passion for the forensic field even more and started to work as a forensic psychologist at the P.I. Veenhuizen in Drenthe. During this period, she simultaneously worked as a research assistant for the Erasmus University Rotterdam. She obtained data in nine forensic institutions all over the Netherlands.

After doing this for a few months she got the opportunity to start with her PhD in clinical forensic psychology at the Erasmus University Rotterdam. During her PhD Carmen was a tutor in psychology courses of the bachelor but also helped in coordinating the course Forensic Psychology for the master students, where she always did the international classes. Furthermore, Carmen was a member of the Postgraduate School for Research and Education and Experimental Psychopathology (EPP) for three years. During her PhD she presented at multiple international conferences and helped voluntarily as a translator and research committee in the Aftermath Foundation for Psychopathy Victims. In the last part of her PhD trajectory, she had the opportunity to collaborate with a professor at Harvard Medical School to write the last chapter of her dissertation and went for a three-month work visit to Boston. She received two travel grants to achieve this goal. Currently, Carmen is working as a post-doctoral researcher on the Virtual Burglary Project for the Max Planck Institute for the study of Crime, Security and Law at Universiteit Leiden.

List of publications

Publications that form the basis of this dissertation:

*Sergiou, C. S., Santarnecchi, E., Romanella, S. M., Wieser, M. J., Franken, I. H. A., Rassin, E. G. C., & van Dongen, J. D. M. (2021). tDCS targeting the Ventromedial Prefrontal Cortex reduces reactive aggression and modulates electrophysiological responses in a forensic population. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*. <https://doi.org/https://doi.org/10.1016/j.bpsc.2021.05.007>

*Sergiou, C.S., Woods, A., Franken, I.H.A., & van Dongen, J.D.M. (2020). Transcranial direct current stimulation (tDCS) as an intervention to improve empathic abilities and reduce violent behavior in forensic offenders: study protocol for a randomized controlled trial. *Trials*. (2020) 21:263 <https://doi.org/10.1186/s13063-020-4074-0>.

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Other international publications

*Ekthiari H., Zangen, A., Del Felice, A., Shahbabaie, A., Gourdriaan, A., **Sergiou C.S.**, et al. (2019). Transcranial Electrical and Magnetic Stimulation (tES and TMS) for Addiction Medicine: A consensus paper on the present state of the science and the road ahead. *Neuroscience & Biobehavioral Reviews*.

*Slotboom, J., Hoppenbrouwers, S.S., In 't Hout, W., **Sergiou, C.S.**, Van der Stigchel, S. & Theeuwes, J. (2016). Visual attention in violent offenders: susceptibility to distraction. *Psychiatry Research*, Vol.251

*Hoppenbrouwers, S.S., Van der Stigchel, S., Sergiou C.S., & Theeuwes, J. (2016). Top down attention and selection history in psychopathy: evidence from a community sample. *Journal of Abnormal Psychology*, Vol.124, No.4,

*Lui, J., **Sergiou, C.S.**, Barry, C. (2016). Interventions for Improving Affective Abilities in Adolescents: An Integrative Review Across Community and Clinical Samples of Adolescents. *Adolescent Research Review Clinical Samples of Adolescents*.

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*Dongen, J. van, **Sergiou, C.**, Franken, I. (2019). Middelengebruik en geweld. ontwikkeling en validatie van een testbatterij. Wetenschappelijk Onderzoeken Documentatiecentrum. Ministerie van Justitie en Veiligheid.

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